



UNIVERSITEIT VAN AMSTERDAM

Data-Driven Decision Making in Operations Research

NOTES BY

Hengyi Li

LECTURER

Dr. A. V. den Boer

INTRODUCTION. In this course, we will study data-driven decision problems: optimisation problems for which the objective function (i.e., the relation between decision and outcome) is unknown upfront, and has to be learned from accumulating data. These problems have an intrinsic tension between statistical goals and optimisation goals: learning how the system behaves (the statistical goal) is accelerated by experimenting with different actions, while for making good decisions (the optimisation goal), one would like to limit experimentation and use estimated optimal decisions. We will study this exploration-exploitation trade-off for so-called *multi-armed bandit problems*, the paradigmatic framework for dynamic optimisation problems with incomplete information. We will discuss standard building blocks of the state-of-the-art theory, and we will discuss applications such as dynamic pricing and assortment optimisation problems.

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Bandits and Concentration

SECTION 1.1

| Introduction

Operations Research (OR) is concerned with optimising decisions in business or societal contexts such as: optimising selling prices; bids for online advertisement auctions; how much inventory to order; route optimisation for package delivery; which vaccination strategy minimises completion time, and so on.

OR approach: translate real-life problem into mathematical optimisation problem (“modelling”, is an art and a craft); determine (approximate) optimal solution (research often focuses on this), and implement solution in real-life context (usually the hard part, and requires “convincing” stakeholders).

In a *static* problem there is just a single decision to take; in a *dynamic* problem there are multiple, sequential decisions to take.

The mathematical optimisation problem usually depends on *unknown* quantities (parameters, distributions): sales or demand as function of price; probability of winning the auction as function of your bid; travel time as function of selected route, and so on.

Static: collect data, estimate unknowns, optimise decision, done. Dynamic: implemented decisions often generate *new data*: prices generate new sales data; bids generate data on success/failure to win the auction; routes generate realised travel time, and so on.

In conclusion, new data $\Rightarrow$ re-estimate the unknown quantities $\Rightarrow$ re-solve optimisation problem.

SECTION 1.2

| Stochastic Bandits

We consider a decision maker (DM) who, at each discrete time period $t \in \mathbb{N}$, selects a (possibly random) action A_t from a nonempty action space $\mathcal{A} \neq \emptyset$. Subsequently, and before selecting action A_{t+1} , the DM receives a reward X_t . For all actions $a \in \mathcal{A}$ and time periods $t \in \mathbb{N}$, let $X_{t,a}$ be the *random reward* that the DM would receive if he selected action a at time t . Then $X_t = X_{t,A_t}$, for all t . We assume that $\{X_{t,a} : t \in \mathbb{N}\}$ are *i.i.d.* random variables, taking values in a set $\mathcal{X} \subset \mathbb{R}$ that is independent of a , and with probability distribution ν_a that is *unknown* to the DM. The expectation of $X_{t,a}$ is denoted by μ_a , and is assumed to be finite for all a . We write $\nu = (\nu_a : a \in \mathcal{A})$ and $\mu = (\mu_a : a \in \mathcal{A})$.

It is worth emphasising that the DM *only* observes the reward corresponding to his action at time t , i.e., X_{t,A_t} ; he does not observe what he would have earned if he would have used a different action, i.e., $X_{t,a}$ for all $a \neq A_t$.

The actions are also referred to as *arms*, which refers to the analogy of a slot machine or bandit that gives a random reward after pulling its arm. If the bandit has multiple arms (a *multi-armed bandit*), each with an unknown and possibly different reward distribution, then a gambler faces the problem of having to learn which arm has the largest expected reward—analogous to the sequential decision problem with uncertainty described above. If $\mathcal{A}$ is a finite set, the decision problem is called a *finite-armed multi-armed bandit problem*, or *finite-armed bandit*; if $\mathcal{A} = [0, 1]^d$ for some $d \in \mathbb{N}$, the decision problem is called a *continuum-armed bandit*; if $\mathcal{A}$ is a collection of combinatorial structures, we speak of a *combinatorial bandit*; and so on. In this chapter we assume that $\mathcal{A} = \{1, \dots, K\}$ is a finite set, with $K \in \mathbb{N}$ the number of actions/arms.

The actions of the DM are required to be *non-anticipatory*, meaning that each action A_t may depend on the past actions and rewards $A_1, X_1, \dots, A_{t-1}, X_{t-1}$, but not on the future A_{t+1}, X_{t+1} . More precisely, we assume that the DM's actions are described by a *policy*: a collection of probability distributions

$$\pi = (\pi(\cdot \mid h) : h \in H)$$

on the measurable space $(\mathcal{A}, 2^{\mathcal{A}})$, where

$$H = \bigcup_{t \in \mathbb{N}} (\mathcal{A} \times \mathcal{X})^{t-1}$$

is the collection of all possible action-reward sequences (*histories* $a_1, x_1, \dots, a_{t-1}, x_{t-1}$), such that, conditionally on $A_1, X_1, \dots, A_{t-1}, X_{t-1} = h$, action A_t has distribution $\pi(\cdot \mid h)$, for all $h \in H$ and all $t \in \mathbb{N}$. The set of histories H contains an empty sequence $\emptyset$, and $\pi(\cdot \mid \emptyset)$ is the probability distribution of the first action A_1 . Thus, the policy describes the probability distributions of the actions, for each possibly observed history:

$$\mathbb{P}(A_t = a \mid (A_1, X_1, \dots, A_{t-1}, X_{t-1}) = h) = \pi(\{a\} \mid h),$$

for all $a \in \mathcal{A}$ and all $h \in H$. Since $\mathcal{A}$ is in this chapter assumed to be finite, we simply write $\pi(a \mid h) = \pi(\{a\} \mid h)$ for the conditional probability that $A_t = a$.

If all $\pi(h)$ are point-masses, i.e., for each history $h \in H$ there is an action $a(h)$ such that $\pi(a(h) \mid h) = 1$, then π is called a *deterministic* or *non-random* policy. In that case, π can be construed as a map from histories $h \in H$ to actions $a(h) \in \mathcal{A}$; there is therefore a one-to-one

correspondence between the space of deterministic policies and $\mathcal{A}^H$, the space of functions $H \rightarrow \mathcal{A}$. A policy that is not deterministic is called *random* or *randomised*. We let Π denote the collection of all policies.

It is convenient to define $H_t := (A_1, X_1, \dots, A_{t-1}, X_{t-1})$, the observed actions and rewards (the *history*) after $t \in \mathbb{N}$ time periods, with $H_0 := \emptyset$ and $H_\infty := (A_t, X_t)_{t \in \mathbb{N}}$. To emphasise that the distribution of H_∞ depends on the policy π and the reward distributions ν , we write $\mathbb{P}_\nu^\pi$ for the probability distribution of H_∞ , and $\mathbb{E}_\nu^\pi$ for the corresponding expectation operator. For example, if the rewards have discrete support $\mathcal{X}$ and probability mass function $p_a(x) := \nu_a(\{x\})$, then, for all $n \in \mathbb{N}$ and all histories $(a_1, x_1, \dots, a_n, x_n) \in H$,

$$\mathbb{P}_\nu^\pi(H_n = (a_1, x_1, \dots, a_n, x_n)) = \prod_{t=1}^n \pi(a_t \mid a_1, x_1, \dots, a_{t-1}, x_{t-1}) p_{a_t}(x_t),$$

and, for all functions f on the histories of length n ,

$$\mathbb{E}_\nu^\pi[f(H_n)] = \sum_{(a_1, x_1, \dots, a_n, x_n) \in H} f(a_1, x_1, \dots, a_n, x_n) \mathbb{P}_\nu^\pi(H_n = (a_1, x_1, \dots, a_n, x_n)).$$

If π and ν are clear from the context, we sometimes omit the sub- and superscript ν, π and simply write $\mathbb{P}$.

The goal of the DM is to make good decisions, that is, to use a *good policy*. There are several ways to measure the quality or performance of a policy. Let

$$\mu^* := \max\{\mu_a : a \in \mathcal{A}\}$$

be the highest attainable expected reward, and let a^* be an action such that $\mu^* = \mu_{a^*}$. Such action is referred to as a *best arm* or an *optimal action*. If the DM would know which action is optimal, he would simply play this action in all time periods in order to maximise his expected reward. However, since the reward distributions ν_a and the expected rewards μ_a are unknown, the DM might frequently use sub-optimal actions and experience a loss in reward compared to μ^* . The total expected loss induced by a policy π after $n \in \mathbb{N}$ time periods is called the *regret*, and is defined by

$$R_\nu^\pi(n) := n\mu^* - \mathbb{E}_\nu^\pi \left[\sum_{t=1}^n X_t \right]. \quad (1.1)$$

The first term on the right-hand side of (1.1) is the optimal expected reward that the DM could have obtained in n time periods if he had known the reward distributions, namely, by selecting an optimal action in all time periods. Observe that μ^* depends on ν , although that is not made explicit in the notation. The second term on the right-hand side of (1.1) is the actual expected reward that the DM obtains. The difference can be interpreted as the costs of not knowing the reward distributions, or the *costs of uncertainty*. Regret is always non-negative, and the lower the regret, the better the policy.

There are many alternative ways to measure the performance of a policy. For example, instead of the expectation of $n\mu^* - \sum_{t=1}^n X_t$, one could try to minimise a quantile, or the *conditional value-at-risk*. Another alternative is to assign weight γ^t to losses in period t , for some discount factor $\gamma \in (0, 1)$, and minimise the discounted loss

$$\mathbb{E}_\nu^\pi \left[\sum_{t=1}^n \gamma^t (\mu^* - X_t) \right].$$

In this expression it is natural to take $n = \infty$, in which case one minimises the *infinite-horizon discounted loss*.

Yet another class of performance measures consider only what happens in the last time period. For example, one could be interested in minimising the expected loss in the last time period, $\mathbb{E}_v^\pi[\mu^* - X_t]$, or in minimising the probability of selecting a sub-optimal action in the last time period, $\mathbb{P}_v^\pi(\mu_{A_n} < \mu^*)$. In these cases, the losses incurred in the first $n - 1$ periods are irrelevant; all that counts is making a good decision at time n . A characteristic feature of multi-armed bandit problems is that *all losses count*. The goal of the DM is not to obtain a good action in one particular time period, but to do well in all time periods. Furthermore, the DM is not (primarily) interested in obtaining high-quality estimates of the distributions v_a or the expectations μ_a ; his goal is to take high-quality decisions.

In these lecture notes we will focus on designing policies that minimise the regret, as defined in (1.1). In addition, we will also consider the *worst-case regret*, defined as

$$R^\pi(n) := \sup_{v \in \mathcal{V}} R_v^\pi(n) \quad (1.2)$$

Here $\mathcal{V}$ denotes a class of vectors of reward probability distributions v that is known to the DM. The set $\mathcal{V}$ allows the DM to model restrictions, properties, or knowledge about the reward distributions, that can be helpful in designing a good policy. Typically, $\mathcal{V}$ contains assumptions on the support, moments, or tail behaviour of the reward distributions, or consists of a family of distributions. For example, $\mathcal{V}$ could consist of all vectors of Bernoulli distributions of length K , or of all vectors (of length K) of probability distributions with support $[0, \infty)$ and finite variance, and so on. The DM does not have to impose assumptions (other than that the rewards have finite expectation; otherwise the regret is not meaningful). However, if distributional information on the rewards is available, it often turns out to be beneficial to use that information in the design and analysis of a policy.

Both performance measures (1.1) and (1.2) have advantages and limitations. If, for example, the DM knows that his policy π satisfies $R^\pi(n) < R^{\pi'}(n)$, where π' is some alternative policy, then he might infer that π is “better than” π' . However, it could be that $R_v^{\pi'}(n) > R_v^\pi(n)$ for just a few $v \in \mathcal{V}$ that are unlikely (but not impossible) to occur, while $R_v^{\pi'}(n) < R_v^\pi(n)$ for all other v . In this case, it is not clear if a DM would necessarily prefer π over π' . Moreover, one could argue that the only performance measure that matters is the regret for the “true” $v \in \mathcal{V}$, and that therefore one should search for a policy that minimises (1.1). The problem with this argument is of course that the “true” v is unknown. What if $R_v^{\pi'}(n) > R_v^\pi(n)$ but $R_{v'}^{\pi'}(n) < R_{v'}^\pi(n)$, for some v and v' ? Should the DM use policy π or policy π' ? What the “right” performance criterion is, to a certain extent, subjective, and for this reason we analyse both performance measures (1.1) and (1.2).

EXAMPLE 1.1. Let $\mathcal{B}(p)$ denote a Bernoulli distribution with parameter $p \in [0, 1]$, i.e.,

$$\mathbb{P}(\emptyset) = 0, \quad \mathbb{P}(\{0\}) = 1 - p, \quad \mathbb{P}(\{1\}) = p, \quad \mathbb{P}(\{0, 1\}) = 1.$$

And class of all Bernoulli-distributed reward *vectors*:

$$\mathcal{V} = \{(\mathcal{B}(\mu_1), \dots, (\mathcal{B}(\mu_K)) : (\mu_1, \dots, \mu_K) \in [0, 1]^K\}$$

In this case: distribution vector v characterised by its mean vector μ . Instance-dependent regret $R_\mu^\pi(n)$ depends on $\mu = (\mu_1, \dots, \mu_K)$. Instance-independent regret

$$R^\pi(n) = \sum_{\mu \in [0, 1]^K} R_\mu^\pi(n).$$

SECTION 1.3

| Concentration of Measure

Ideally we could find a policy π^* such that

$$R_v^{\pi^*}(n) = \inf_{\pi \in \Pi} R_v^\pi(n), \quad \forall v \in \mathcal{V},$$

or, weaker, a policy π^* such that

$$\sup_{v \in \mathcal{V}} R_v^{\pi^*}(n) = \inf_{\pi \in \Pi} \sup_{v \in \mathcal{V}} R_v^\pi(n)$$

for all n , or for a given n , which is intractable except in simple cases. We try to find a policy π^* such that $R_v^{\pi^*}(n) = \inf_{\pi \in \Pi} R_v^\pi(n)$ for the “true” $v \in \mathcal{V}$ only. We search for asymptotically optimal policies: policies π such that $R_v^\pi(n)$ grows as slowly as possible in n , for all $v \in \mathcal{V}$ or $R^\pi(n)$ grows as slowly as possible in n (in practice n might be large).

To prove asymptotic optimality of a policy, we need to find an upper bound on the regret of π as function of n , and find a lower bound on the regret of any policy. Regret upper bounds crucially depend on *concentration inequalities*.

Throughout this section, let Z be a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, with expectation $\mu \in \mathbb{R}$ and finite variance. Let $n \in \mathbb{N}$, and let $Z_1, \dots, Z_n$ be *i.i.d.* random variables distributed as Z . Let $\epsilon > 0$. The observation

$$\mathbb{E}[|Z|] \geq \mathbb{E}[|Z| \mathbb{1}_{|Z| \geq \epsilon}] \geq \mathbb{E}[\epsilon \mathbb{1}_{|Z| \geq \epsilon}] = \epsilon \mathbb{P}(|Z| \geq \epsilon)$$

implies *Markov's inequality*:

$$\mathbb{P}(|Z| \geq \epsilon) \leq \frac{\mathbb{E}[|Z|]}{\epsilon}.$$

Furthermore, the observation

$$\mathbb{P}(|Z - \mathbb{E}[Z]| \geq \epsilon) = \mathbb{P}\left((Z - \mathbb{E}[Z])^2 \geq \epsilon^2\right)$$

together with *Markov's inequality* yields *Chebyshev's inequality*:

$$\mathbb{P}(|Z - \mathbb{E}[Z]| \geq \epsilon) \leq \frac{\text{Var}(Z)}{\epsilon^2}.$$

Applying *Chebyshev's inequality* to the sample average of $Z_1, \dots, Z_n$ implies

$$\mathbb{P}\left(\left|\frac{1}{n} \sum_{i=1}^n Z_i - \mu\right| \geq \epsilon\right) \leq \frac{\text{Var}(Z)}{n\epsilon^2}, \quad \forall \epsilon > 0.$$

Thus, the probability that the sample average of $Z_1, \dots, Z_n$ lies at least ϵ away from μ is bounded by a constant times n^{-1} . All that this result requires is that the variance of Z is finite. This rate is far from tight:

$$\mathbb{P}(\mathcal{N}(0, 1) > x) = \int_x^\infty \frac{x}{y} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy \leq \frac{1}{x\sqrt{2\pi}} \int_x^\infty y e^{-\frac{y^2}{2}} dy = \frac{1}{x\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

for all $x > 0$. Thus, if $Z \sim \mathcal{N}(\mu, \sigma^2)$, then $\bar{Z}_n \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$ and

$$\mathbb{P}(\bar{Z}_n > \mu + \epsilon) = \mathbb{P}\left(\mathcal{N}(0, 1) > \epsilon \sqrt{\frac{n}{\sigma^2}}\right) \leq \frac{\sigma}{\sqrt{2\pi}\epsilon^2 n} e^{-\frac{n\epsilon^2}{2\sigma^2}}$$

exponential decay in n . Much sharper concentration inequalities (than Chebyshev) are possible, and ideally we derive such inequalities without using explicit formula for the CDF.

For a random variable Z , its *moment-generating function* (MGF) is

$$M_Z(t) := \mathbb{E}[\exp(tZ)]$$

and its *cumulant-generating function* is

$$\psi_Z(t) := \log M_Z(t).$$

EXAMPLE 1.2. (i) If $Z \sim \mathcal{N}(\mu, \sigma^2)$, then $M_Z(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$ for all $t \in \mathbb{R}$.

(ii) If $Z \sim \text{Exp}(\lambda)$, then $M_Z(t) = \frac{\lambda}{\lambda - t}$ for all $t < \lambda$.

(iii) If $Z \sim \text{Cauchy}(0, 1)$, i.e., Z has pdf $(\pi(1 + x^2))^{-1}$ for all $x \in \mathbb{R}$, then $M_Z(t)$ only defined for $t = 0$ ($M_Z(0) = 1$).

LEMMA 1.3.

Let $I := \{t \in \mathbb{R} : M_Z(t) < \infty\}$ denote the domain of M_Z .

(i) I is a (possibly degenerate) interval containing zero.

(ii) M_Z is infinitely often differentiable on the interior of I , with derivatives equal to

$$\frac{d^k}{dt^k} M_Z(t) = \mathbb{E}[Z^k \exp(tZ)], \quad \forall k \in \mathbb{N}.$$

The moment-generating function can be used to derive concentration inequalities. A well-known technique is the so-called *Chernoff method*. Let $\epsilon \geq 0$, and let $t > 0$ be in the domain of M_Z . By *Markov's inequality*,

$$\mathbb{P}(Z - \mu \geq \epsilon) = \mathbb{P}(\exp(tZ) \geq \exp(t(\mu + \epsilon))) \leq \frac{M_Z(t)}{\exp(t(\mu + \epsilon))},$$

so that, by optimising over all such t , we obtain

$$\mathbb{P}(Z - \mu \geq \epsilon) \leq \inf_{t>0} \frac{M_Z(t)}{\exp(t(\mu + \epsilon))}.$$

To solve the right-hand side in terms of t , we need to have an upper bound on the MGF of Z .

EXAMPLE 1.4. If $Z \sim \mathcal{N}(\mu, \sigma^2)$, then $M_Z(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right)$, and

$$\mathbb{P}(Z \geq \mu + \epsilon) \leq \inf_{t>0} \frac{M_Z(t)}{\exp(t(\mu + \epsilon))} = \inf_{t>0} \exp\left(\frac{\sigma^2 t^2}{2} - t\epsilon\right) = \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right)$$

with $t = \frac{\epsilon}{\sigma^2}$.

DEFINITION 1.5 (Subgaussianity).

Z is called *s-subgaussian*, for $s > 0$, if its MGF is defined for all $t \in \mathbb{R}$, and is dominated by the MGF of a $\mathcal{N}(\mu, s^2)$ random variable:

$$M_Z(t) \leq \exp\left(\mu t + \frac{s^2 t^2}{2}\right), \quad \forall t \in \mathbb{R}.$$

If we apply the Chernoff method to subgaussian random variables, we obtain the following concentration inequality.

LEMMA 1.6.

If Z is s -subgaussian then, for all $\epsilon > 0$,

$$\begin{aligned}\mathbb{P}(Z \geq \mu + \epsilon) &\leq \exp\left(-\frac{\epsilon^2}{2s^2}\right) \\ \mathbb{P}(Z \leq \mu - \epsilon) &\leq \exp\left(-\frac{\epsilon^2}{2s^2}\right) \\ \mathbb{P}(|Z - \mu| \geq \epsilon) &\leq 2 \exp\left(-\frac{\epsilon^2}{2s^2}\right).\end{aligned}$$

If $Z, Z_1, \dots, Z_n$ are i.i.d. s -subgaussian and $\bar{Z}_n := n^{-1} \sum_{i=1}^n Z_i$ is the sample average, then for all t ,

$$\begin{aligned}M_{\bar{Z}_n}(t) &= \mathbb{E}\left[\exp\left(t \cdot \frac{1}{n} \sum_{i=1}^n Z_i\right)\right] = \mathbb{E}\left[\prod_{i=1}^n \exp\left(t \cdot \frac{1}{n} Z_i\right)\right] = \prod_{i=1}^n \mathbb{E}\left[\exp\left(t \cdot \frac{1}{n} Z_i\right)\right] \\ &\leq \prod_{i=1}^n \exp\left(\mu\left(\frac{t}{n}\right) + \frac{1}{2}s^2\left(\frac{t}{n}\right)^2\right) = \exp\left(\mu t + \frac{s^2 t^2}{2n}\right).\end{aligned}$$

Hence $\bar{Z}_n$ is $\frac{s}{\sqrt{n}}$ -subgaussian, and thus

$$\begin{aligned}\mathbb{P}(\bar{Z}_n - \mu \geq \epsilon) &\leq \exp\left(-\frac{\epsilon^2}{2\left(\frac{s}{\sqrt{n}}\right)^2}\right) = \exp\left(-\frac{\epsilon^2 n}{2s^2}\right) \\ \mathbb{P}(\bar{Z}_n - \mu \leq -\epsilon) &\leq \exp\left(-\frac{\epsilon^2 n}{2s^2}\right), \\ \mathbb{P}(|\bar{Z}_n - \mu| \geq \epsilon) &\leq 2 \exp\left(-\frac{\epsilon^2 n}{2s^2}\right).\end{aligned}$$

Simplifying assumption: rewards are 1-subgaussian (scale rewards if necessary), subgaussian-factor is thus known.

EXAMPLE 1.7 (Bounded Random Variable). If Z is bounded with support $[a, b]$, $a < b$, then Z is $\frac{1}{2}(b - a)$ -subgaussian and

$$\mathbb{P}(\bar{Z}_n - \mu \geq \epsilon) \leq \exp\left(-\frac{2\epsilon^2 n}{(b - a)^2}\right),$$

which is the *Hoeffding's inequality*.

Proof. For the first part, it suffices to prove that $\psi_X(\lambda) \leq \frac{\lambda^2(b-a)^2}{4}$. By Taylor's theorem, for some λ' between 0 and λ ,

$$\psi_X(\lambda) = \psi_X(0) + \psi'_X(\lambda')\lambda + \psi''_X(\lambda')\frac{\lambda^2}{2}.$$

To bound the last term, introduce the distribution P_λ for $\lambda \in \mathbb{R}$ arbitrary: $P_\lambda(dz) = e^{-\psi_X(\lambda)} e^{\lambda z} P(dz)$. Show that $\psi''_X(\lambda) = \mathbb{V}[Z]$, where $Z \sim P_\lambda$. Now, since $Z \in [a, b]$ with probability one, argue (without relying on $\mathbb{E}[Z]$) that $\mathbb{V}[Z] \leq \frac{(b-a)^2}{4}$. $\square$

Remark. Without assumption such as subgaussianity, tails can be very 'thick' i.e., $\mathbb{P}(Z > x) = x^{-1-\epsilon}$, small $\epsilon > 0$, has finite mean but very thick tail.

Exercises

EXERCISE 1.1. Consider a two-armed Bernoulli bandit.

- (i) Give a policy that minimises the worst-case or instance-independent regret after $n = 1$ periods over all $(\mu_1, \mu_2) \in [0, 1]^2$, and prove that this is the case.
- (ii) Similar question for $n = 2$. If you want, try $n = 3$.

SOLUTION. (i) Given that $n = 1$ and we have no prior information about μ_1 and μ_2 , to minimise the regret, we can give a policy that we choose each arm with equal probability, i.e., $\pi(A_1 = 1) = \pi(A_1 = 2) = \frac{1}{2}$.

Let A_1 denote the arm chosen in the first period. Since we choose each arm with probability $\frac{1}{2}$, The expected reward is then:

$$\mathbb{E}^\pi[X_{A_1}] = \frac{1}{2}\mu_1 + \frac{1}{2}\mu_2,$$

where $(\mu_1, \mu_2) \in [0, 1]^2$. Then the regret is

$$R^\pi(1) = \mu^* - \mathbb{E}^\pi[X_{A_1}] = \mu^* - \left(\frac{1}{2}\mu_1 + \frac{1}{2}\mu_2\right),$$

where $\mu^* = \max(\mu_1, \mu_2)$ is the reward of the optimal arm. And this regret can be rewritten as

$$R^\pi(1) = \frac{1}{2}|\mu_1 - \mu_2|,$$

where the worst-case regret has maximum $\frac{1}{2}$.

To prove that the policy is indeed the best one, consider any alternative policy π^* such that $\pi^*(A_1 = 1) = p$ and $\pi^*(A_1 = 2) = 1 - p$ with $p \in [0, 1]$. The expected reward under this policy would be:

$$\mathbb{E}^{\pi^*}[X_{A_1}] = p\mu_1 + (1 - p)\mu_2.$$

The regret for this policy is

$$R^{\pi^*}(1) = \mu^* - \mathbb{E}^{\pi^*}[X_{A_1}] = \max(\mu_1, \mu_2) - (p\mu_1 + (1 - p)\mu_2).$$

And the worst-case regret for this policy is

$$\max_{(\mu_1, \mu_2) \in [0, 1]^2} R^{\pi^*}(1) = \max_{(\mu_1, \mu_2) \in [0, 1]^2} \{(1 - p)(\mu_1 - \mu_2), p(\mu_2 - \mu_1)\} = \max_{(\mu_1, \mu_2) \in [0, 1]^2} (p, 1 - p)|\mu_1 - \mu_2|,$$

where the minimum possible worst-case regret is achieved when $p = \frac{1}{2}$.

- (ii) In the first round, choose each arm with equal probability: $\pi(A_1 = 1) = \pi(A_1 = 2) = \frac{1}{2}$. In the second round, choose the arm with the higher observed reward. If the rewards are equal (a tie), choose either arm with equal probability.

Let A_1 be the arm chosen in the first round. The expected reward is:

$$\mathbb{E}^\pi[X_{A_1}] = \frac{1}{2}\mu_1 + \frac{1}{2}\mu_2,$$

where $(\mu_1, \mu_2) \in [0, 1]^2$. Since X_{A_1} follows a Bernoulli distribution, it can either be 0 or 1. If $X_{A_1} = 1$, we choose the same arm again in the second round; if $X_{A_1} = 0$, we choose the other arm in the second round. Therefore, the expected reward for the second round can be calculated as:

$$\begin{aligned} \mathbb{E}^\pi[X_{A_2}] &= \frac{1}{2}\mathbb{E}^\pi[X_{A_2} \mid A_1 = 1] + \frac{1}{2}\mathbb{E}^\pi[X_{A_2} \mid A_1 = 2] \\ &= \frac{1}{2}(\mu_1\mathbb{P}(X_{A_1} = 1) + \mu_2\mathbb{P}(X_{A_1} = 0)) + \frac{1}{2}(\mu_2\mathbb{P}(X_{A_1} = 1) + \mu_1\mathbb{P}(X_{A_1} = 0)) \\ &= \frac{1}{2}(\mu_1^2 + \mu_2(1 - \mu_1)) + \frac{1}{2}(\mu_2^2 + \mu_1(1 - \mu_2)) = \frac{1}{2}(\mu_1^2 + \mu_2^2 + \mu_1 + \mu_2 - 2\mu_1\mu_2). \end{aligned}$$

Now, the total expected reward over two rounds is

$$\mathbb{E}^\pi[X_{A_1}] + \mathbb{E}^\pi[X_{A_2}] = \frac{1}{2}\mu_1 + \frac{1}{2}\mu_2 + \frac{1}{2}(\mu_1^2 + \mu_2^2 + \mu_1 + \mu_2 - 2\mu_1\mu_2) = \mu_1 + \mu_2 - \frac{1}{2}(\mu_1 - \mu_2)^2.$$

The regret after 2 rounds is then:

$$\begin{aligned} R^\pi(2) &= 2\mu^* - \left(\mu_1 + \mu_2 - \frac{1}{2}(\mu_1 - \mu_2)^2\right) \\ &= 2\max(\mu_1, \mu_2) - \left(\mu_1 + \mu_2 - \frac{1}{2}(\mu_1 - \mu_2)^2\right) = |\mu_1 - \mu_2| + \frac{1}{2}(\mu_1 - \mu_2)^2, \end{aligned}$$

where the worst-case regret has maximum $\frac{3}{2}$. This policy equally balances the exploration and exploitation phases. Any policy π^* that does not choose each arm with equal probability in the first round could have a regret higher than $\frac{3}{2}$, because choosing one arm deterministically in the first round would result in no (or less) information about the other arm, leading to potentially higher regret.

(iii) **First Round.** Choose each arm with equal probability: $\pi(A_1 = 1) = \pi(A_1 = 2) = \frac{1}{2}$.

Second Round. Choose the arm with the higher observed reward from the first round. If the rewards are tied, choose either arm with equal probability.

Third Round. Choose the arm that has performed better over the first two rounds using tie-break rule.

The first and second rounds are the same as before. The expected reward for the third round can be calculated as:

For this calculation, write

$$\begin{aligned} Q_1 &:= \frac{1}{2}(\mathbb{E}^\pi[X_{A_3} \mid A_1 = 1, A_2 = 1] + \mathbb{E}^\pi[X_{A_3} \mid A_1 = 1, A_2 = 2]) \\ &\quad + \frac{1}{2}(\mathbb{E}^\pi[X_{A_3} \mid A_1 = 2, A_2 = 2] + \mathbb{E}^\pi[X_{A_3} \mid A_1 = 2, A_2 = 1]) \end{aligned}$$

Then

$$\begin{aligned} \mathbb{E}^\pi[X_{A_3}] &= Q_1 = \frac{1}{2}(\mu_1^2 + \mu_2(1 - \mu_1)) + \frac{1}{2}(\mu_2^2 + \mu_1(1 - \mu_2)). \end{aligned}$$

Then the total expected reward is

$$\frac{1}{2}\mu_1 + \frac{1}{2}\mu_2 + \left(\mu_1^2 + \mu_2^2 + \mu_1 + \mu_2 - 2\mu_1\mu_2\right) = \frac{3}{2}(\mu_1 + \mu_2) - (\mu_1 - \mu_2)^2.$$

The regret after 3 rounds is

$$R^\pi(3) = 3\mu^* - \left(\frac{3}{2}(\mu_1 + \mu_2) - (\mu_1 - \mu_2)^2\right) = \frac{3}{2}|\mu_1 - \mu_2| + (\mu_1 - \mu_2)^2,$$

where the worst-case regret has maximum $\frac{5}{2}$.

□

The ETC and the UCB Policies

SECTION 2.1

| The Explore-Then-Commit Algorithm

ETC is characterised by the number of times it explores each arm, denoted by a natural number m . Because there are k actions, the algorithm will explore for mk rounds before choosing a best action for the remaining rounds.

DEFINITION 2.1.

For $a \in \mathcal{A}$ and $t \in \mathbb{N}$, after round t , define *the number of times action a has been played*

$$T_a(t) := \sum_{s=1}^t \mathbb{1}_{\{A_s=a\}},$$

and *average reward received from arm a*

$$\hat{\mu}_a(t) := \frac{1}{T_a(t)} \sum_{s=1}^t X_s \mathbb{1}_{\{A_s=a\}},$$

which is only defined if $T_a(t) > 0$. If $T_a(t) = 0$, define $\hat{\mu}_a(t) := \infty$.

Explore. Let $A_t = a$ for $t = 1 + (a-1)m, \dots, am$.

Commit. For $t = mk + 1, \dots, n$, choose

$$A_t \in \arg \max_{a \in \mathcal{A}} \hat{\mu}_a(mk).$$

In case of ties, choose the smallest index.

Remark. Informal/Implicit description of policy. It does not require continuous updating of estimates during the commit phase. Don't forget to specify tie-break rule.

Regret of the ETC Policy

Write $R(n), \mathbb{E}, \mu^*$ instead of $R_v^\pi(n), \mathbb{E}_v^\pi, \mu^*(v)$ in (1.1). For all $a \in \mathcal{A}$, write *optimality gap* $\Delta_a = \mu^* - \mu_a$. Observe

$$\begin{aligned} R(n) &= \sum_{t=1}^n \mathbb{E}[\mu^* - X_t] = \sum_{t=1}^n \sum_{a=1}^k \mathbb{E}[\mu^* - X_t \mid A_t = a] \mathbb{P}(A_t = a) \\ &= \sum_{a=1}^k (\mu^* - \mu_a) \sum_{t=1}^n \mathbb{P}(A_t = a) = \sum_{a=1}^k \Delta_a \mathbb{E}[T_a(n)]. \end{aligned}$$

Need to bound $\mathbb{E}[T_a(n)]$ for suboptimal arms. Assume *w.l.o.g.* that the first arm is optimal, which means that $\mu_1 = \mu^* = \max_i \mu_i$ (relabel if necessary). Let $a \in \{2, \dots, k\}$ and $1 \leq m \leq \frac{n}{k}$. In the first mk rounds, the policy is deterministic, choosing each action exactly m times. Subsequently it chooses a single action maximising the average reward during exploration. Thus:

$$\begin{aligned} \mathbb{E}[T_a(n)] &= m + (n - mk) \mathbb{P}(A_{mk+1} = a) \\ &\leq m + (n - mk) \mathbb{P}\left(\hat{\mu}_a(mk) \geq \max_{j \neq a} \hat{\mu}_j(mk)\right) \leq m + (n - mk) \mathbb{P}(\hat{\mu}_a(mk) \geq \hat{\mu}_1(mk)), \end{aligned}$$

where

$$\begin{aligned} \mathbb{P}(\hat{\mu}_a(mk) \geq \hat{\mu}_1(mk)) &= \mathbb{P}\left(\underbrace{\hat{\mu}_a(mk) - \hat{\mu}_1(mk) - \mu_a + \mu_1}_{\sqrt{\frac{2}{m}}\text{-subgaussian}} \geq \Delta_a\right) \\ &\leq \exp\left(-\frac{\Delta_a^2}{2\left(\sqrt{\frac{2}{m}}\right)^2}\right) = \exp\left(-\frac{1}{4}m\Delta_a^2\right). \end{aligned}$$

Therefore, for all $1 \leq m \leq \frac{n}{k}$,

$$R(n) = \sum_{a=1}^k \Delta_a \mathbb{E}[T_a(n)] \leq \sum_{a=1}^k \Delta_a \left\{ m + (n - mk) \exp\left(-\frac{1}{4}m\Delta_a^2\right) \right\}.$$

How to Choose m ?

Consider $k = 2$, arm 1 again optimal, suppose $\Delta_2 > 0$,

$$R(n) \leq \Delta_2 \left\{ m + n \exp\left(-\frac{1}{4}m\Delta_2^2\right) \right\}.$$

Minimise *w.r.t.* m :

$$\frac{d\{m + n \exp(-\frac{1}{4}m\Delta_2^2)\}}{dm} = 1 - \frac{n\Delta_2^2}{4} \exp\left(-\frac{1}{4}m\Delta_2^2\right).$$

Choose

$$m^* = \max \left\{ 1, \left\lceil \frac{4}{\Delta_2^2} \log\left(\frac{n\Delta_2^2}{4}\right) \right\rceil \right\}.$$

Since $\lceil x \rceil \leq 1 + x$, we have, if $\frac{n\Delta_2^2}{4} \geq 1$,

$$R(n) \leq \Delta_2 + \frac{4}{\Delta_2} \log \left(\frac{n\Delta_2^2}{4} \right) + \frac{4}{\Delta_2} = \Delta_2 + \frac{4}{\Delta_2} \left(\log \frac{n\Delta_2^2}{4} + 1 \right),$$

and, if $\frac{n\Delta_2^2}{4} < 1$,

$$R(n) \leq \Delta_2 \left\{ 1 + n \exp \left(-\frac{1}{4} \Delta_2^2 \right) \right\} \leq \Delta_2 + \frac{4}{\Delta_2}.$$

With $\log^+(x) := \max\{0, \log(x)\}$,

$$R(n) \leq \min \left\{ n\Delta_2, \Delta_2 + \frac{4}{\Delta_2} \left(1 + \log^+ \left(\frac{n\Delta_2^2}{4} \right) \right) \right\}.$$

Observe

$$\begin{aligned} \frac{d}{d\Delta_2} \left[\frac{4}{\Delta_2} \left(1 + \log^+ \left(\frac{n\Delta_2^2}{4} \right) \right) \right] &= -\frac{4}{\Delta_2^2} (1 + \log^+ (\dots)) + \frac{4}{\Delta_2} \cdot \frac{2}{\Delta_2} \mathbb{1}_{\left\{ \frac{n\Delta_2^2}{4} > 1 \right\}} \\ &< \frac{4}{\Delta_2} \left(1 - \log^+ \left(\frac{n\Delta_2^2}{4} \right) \right) < 0 \end{aligned}$$

if $\frac{n\Delta_2^4}{4} < e$. If $\Delta_2 \leq 2\sqrt{\frac{e}{n}}$, then $R(n) \leq \Delta_2 n \leq 2\sqrt{en}$. If $\Delta_2 > 2\sqrt{\frac{e}{n}}$, then $\frac{n\Delta_2^4}{4} > e$, and thus

$$R(n) \leq \Delta_2 + \frac{4}{2\sqrt{\frac{e}{n}}} \cdot 2 \leq \Delta_2 + 3\sqrt{n}$$

which is almost instance-independent. If Δ_2 is bounded, e.g., by 1, then

$$R(n) \leq 1 + 3\sqrt{n},$$

which is the instance-independent upper bound.

Remark. Without bound on Δ_2 there is no true instance-independent bound.

EXAMPLE 2.2. ETC with 2 arms, 1-subgaussian rewards, and $\Delta_2 > 0$, has

$$R(n) \leq C \cdot \left(\Delta_2 + \frac{\log n}{\Delta_2} \right)$$

for all n large enough and some $C > 0$, and

$$R(n) \leq \Delta_2 + 3\sqrt{n}.$$

Remark. The choice of m depends on the unknown optimality gap Δ_2 , so not really useful.

Choose m Independent of Δ_2

$$R(n) \leq \Delta_2 m + n \cdot \Delta_2 \exp \left(-\frac{1}{4} m \Delta_2^2 \right).$$

Suppose again $\Delta_2 \leq 1$. Since

$$\frac{d}{d\Delta_2} \Delta_2 \exp \left(-\frac{1}{4} m \Delta_2^2 \right) = \left(1 - \frac{m \Delta_2^2}{2} \right) \exp \left(-\frac{1}{4} m \Delta_2^2 \right),$$

then $\Delta_2 \exp\left(-\frac{1}{4}m\Delta_2^2\right)$ is maximised at $\Delta_2 = \sqrt{\frac{2}{m}}$, and thus, for all Δ_2 ,

$$R(n) \leq m + n \cdot \sqrt{\frac{2}{m}} \exp\left(-\frac{1}{2}\right).$$

Optimise *w.r.t.* m : choose $m = (2e)^{-\frac{1}{3}}n^{\frac{2}{3}}$ yields

$$R(n) \leq \left(2^{-\frac{1}{3}} + 2^{\frac{2}{3}}\right)n^{-\frac{1}{3}} \cdot n^{\frac{2}{3}}.$$

Remark. This is a worse instance-independent bound than $n^{\frac{1}{2}}$. A lot of exploration, $m \sim n^{\frac{2}{3}}$, is needed.

This rate $n^{\frac{2}{3}}$ is tight. Suppose both arms are Gaussian with variance 1. Write $\Delta = \Delta_2$.

$$R(n) = m\Delta + (n - 2m)\Delta\mathbb{P}(\hat{\mu}_2(m) \geq \hat{\mu}_1(m)) = m\Delta + (n - 2m)\Delta\mathbb{P}\left(\mathcal{N}(0, 1) \geq \sqrt{\frac{m}{2}}\right)$$

Suppose $m \leq 2n^{\frac{2}{3}}$. Let $\Delta = \sqrt{\frac{2}{m}}$. Then $\Delta \geq n^{-\frac{1}{3}}$, and

$$R(n) = m\Delta(1 - 2\mathbb{P}(\mathcal{N}(0, 1) \geq 1)) + n\Delta\mathbb{P}(\mathcal{N}(0, 1) \geq 1) \geq m\Delta \cdot 0.68 + n\Delta \cdot 0.15 \geq 0.15n^{\frac{2}{3}}.$$

Suppose $m > 2n^{\frac{2}{3}}$. Then $R(n) \geq 2n^{\frac{2}{3}}\Delta$, hence

$$R(n) \geq 0.15 \cdot \min(1, \Delta)n^{\frac{2}{3}}.$$

The $n^{\frac{2}{3}}$ -rate cannot be improved without additional assumptions. On the other hand, one wonders if $n^{\frac{1}{2}}$ is possible with other policies.

SECTION 2.2

| Other Policies

ETC with Adaptive Commitment Time

Explore all arms equally often until some condition is met (i.e., m is data-dependent). If correctly done, on a two-armed 1-subgaussian bandit with $\mu^* = \mu_1$,

$$R_\nu(n) \leq C \left(\Delta_2 + \frac{\log n}{\Delta_2} \right)$$

for some $C > 0$ independent of ν , and

$$R(n) \leq \Delta_2 + C' \sqrt{n \log n}$$

for some $C' > 0$.

 ϵ -Greedy

Let $\epsilon_t \in [0, 1]$, for all $t \in \mathbb{N}$. Let Z_t be Bernoulli with parameter ϵ_t . If $Z_t = 1$, choose A_t uniformly at random. If $Z_t = 0$, choose $A_t \in \arg \max_a \hat{\mu}_a(t)$ (with some tie-break rule). If ϵ_t chosen correctly $\sim \frac{1}{t}$, on a two-armed 1-subgaussian bandit with $\mu^* = \mu_1$,

$$R_\nu(n) \leq C \left(\Delta_2 + \frac{\log n}{\Delta_2} \right)$$

for some $C > 0$ independent of ν .

Adaptive Elimination

Operates in *phases*. Arms are *active* or *eliminated*. In each phase, use all *active* arms equally often. After phase ℓ , eliminate all arms with average reward *too low*. Input: sequence $(m_\ell : \ell \in \mathbb{N})$ of integers. $A_1 := \{1, \dots, k\}$ (active arms). For all $\ell \in \mathbb{N}$: choose each arm $i \in A_\ell$ exactly m_ℓ times, let $\hat{\mu}_{i,\ell}$ be average reward of arm i during phase ℓ , and set

$$A_{\ell+1} = \left\{ i \in A_\ell : \hat{\mu}_{i,\ell} \geq \max_{j \in A_\ell} \hat{\mu}_{j,\ell} - 2^{-\ell} \right\}.$$

On a k -armed 1-subgaussian bandit:

$$R_\nu(n) \leq C \left(\sum_{i: \Delta_i \geq 0} \Delta_i + \frac{\log n}{\Delta_i} \right)$$

for some $C > 0$ independent of ν .

Follow-the-Leader

Also called *greedy* or *myopic*. Always choose the empirically best arm (with some tie-break rule). Not too good.

For different policies we obtain bounds,

$$R_v(n) \leq C \left(\sum_{i: \Delta_i \geq 0} \Delta_i + \frac{\log n}{\Delta_i} \right)$$

for some $C > 0$ independent of v , and

$$R(n) \leq \sum_i \Delta_i + C' \sqrt{nk \log k}$$

for some $C' > 0$. $\log n$ grows slowly. Worst-case is much worse: not just a higher *constant*, but a *higher order*.

Anytime Algorithms

Policies that do not require knowledge of n are called *anytime algorithms*. *Doubling trick*: run the algorithm with different, well-chosen, values of n . Suppose the worst-case regret $R(n) \leq C\sqrt{n}$, for all n , but n is unknown. Run the algorithm $1, 2, 4, 8, 16, \dots, 2^m$ times, with $m = \lceil \log_2(n+1) \rceil - 1$.

$$R(n) \leq \sum_{i=0}^m R(2^i) \leq \sum_{i=0}^m C\sqrt{2^i} = C \frac{\sqrt{2}^{m+1} - 1}{\sqrt{2} - 1} \leq C \frac{\sqrt{2}^{\log_2(n+1)+1} - 1}{\sqrt{2} - 1} \leq C \frac{\sqrt{2}}{\sqrt{2} - 1} \cdot \sqrt{n+1}.$$

$\lceil x \rceil \leq x + 1$.

Similar reasoning if $R(n) \leq C \log(n+1)$.

SECTION 2.3

| The Upper Confidence Bound Algorithm

If $Z_1, \dots, Z_n$ are 1-subgaussian, *i.i.d.*, with mean μ , and *empirical mean*

$$\hat{\mu} := \frac{1}{n} \sum_{i=1}^n Z_i,$$

then

$$\mathbb{P}(\hat{\mu} \leq \mu - \epsilon) \leq \exp\left(-\frac{\epsilon^2 n}{2}\right), \quad \forall \epsilon \geq 0.$$

Let $\delta \in (0, 1)$. Choose ϵ such that $\exp\left(-\frac{\epsilon^2 n}{2}\right) = \delta$. Then

$$\mathbb{P}\left(\mu \geq \hat{\mu} + \sqrt{\frac{2 \log(\frac{1}{\delta})}{n}}\right) \leq \delta, \quad \forall \delta \in (0, 1).$$

Hence, for all $a \in A$, $\delta \in (0, 1)$, and $t \in \mathbb{N}$, we can define

$$\text{UCB}_a(t, \delta) := \begin{cases} \hat{\mu}_a(t) + \sqrt{\frac{2 \log(\frac{1}{\delta})}{T_a(t)}} & T_a(t) \geq 1 \\ \infty & T_a(t) = 0 \end{cases},$$

where $\sqrt{\frac{2 \log(\frac{1}{\delta})}{T_a(t)}}$ is the *exploration bonus*.

UCB Policy(n, δ) and the Regret

Use each action once: $A_t = t$ for $t = 1, \dots, k$. For all $t > k$, choose

$$A_t \in \arg \max_{a \in A} \text{UCB}_a(t-1, \delta),$$

where the choice of δ is an input parameter (should not be too large).

THEOREM 2.3.

Let π be $\text{UCB}(n, \delta)$ with $\delta = \frac{1}{n^2}$. There is a $C > 0$, independent of v and n , such that the regret:

$$R(n) \leq C \cdot \sum_{a: \Delta_a > 0} \left(\Delta_a + \frac{\log n}{\Delta_a} \right).$$

Proof. As before,

$$R(n) = \sum_{a=1}^k \Delta_a \mathbb{E}[T_a(n)].$$

W.l.o.g., assume arm 1 is optimal: $\mu_1 = \mu^*$. Let $a \in \{2, \dots, k\}$ be a suboptimal arm: $\mu_a \leq \mu_1$. Let $u_a \in \mathbb{N}$ be some *large number*. Define the *good event*:

$$G_a := \{\mu_1 < \text{UCB}_1(t, \delta) \forall t = 1, \dots, n\} \cap \left\{ \hat{\mu}_{a, u_a} + \sqrt{\frac{2 \log \frac{1}{\delta}}{u_a}} < \mu_1 \right\},$$

where the UCB of arm 1 is never below μ_1 and the u_a -th UCB of arm a is below μ_1 . Think of G_a as a high-probability event, if u_a is sufficiently large. By definition of the UCB,

$$\mathbb{P}(\mu_1 \geq \text{UCB}_1(t, \delta) \text{ for some } t = 1, \dots, n) \leq \sum_{s=1}^n \mathbb{P}\left(\mu_1 \geq \hat{\mu}_{1,s} + \sqrt{\frac{2 \log \frac{1}{\delta}}{s}}\right) \leq n\delta.$$

Also

$$\begin{aligned} \mathbb{P}\left(\hat{\mu}_{a,u_a} + \sqrt{\frac{2 \log \frac{1}{\delta}}{u_a}} \geq \mu_1\right) &= \mathbb{P}\left(\hat{\mu}_{a,u_a} - \mu_a \geq \Delta_a - \sqrt{\frac{2 \log \frac{1}{\delta}}{u_a}}\right) \leq \mathbb{P}\left(\hat{\mu}_{a,u_a} - \mu_a \geq \frac{1}{2}\Delta_a\right) \\ &\leq \exp\left(-\frac{u_a \Delta_a^2}{8}\right) \leq \delta \end{aligned}$$

if $u_a = \lceil 8\Delta_a^{-2} \log \frac{1}{\delta} \rceil$. Hence, G_a is a high-probability event if u_a is sufficiently large, and $\mathbb{P}(G_a^c) \leq (n+1)\delta$.

LEMMA 2.4.

On G_a , $T_a(n) \leq u_a$.

Proof. Suppose $T_a(n) > u_a$ on G_a . Let $t \in \{1, \dots, n\}$ such that $T_a(t-1) = u_a$ and $A_t = a$.

$$\text{UCB}_a(t-1) = \hat{\mu}_a(t-1) + \sqrt{\frac{2 \log(\frac{1}{\delta})}{T_a(t-1)}} = \hat{\mu}_{a,u_a} + \sqrt{\frac{2 \log(\frac{1}{\delta})}{u_a}} < \mu_1 < \text{UCB}_1(t-1),$$

$T_a(t-1) = u_a$; Definition of G_a .

which is a contradiction. $\square$

Putting everything together,

$$\mathbb{E}[T_a(n)] = \mathbb{E}[T_a(n)\mathbf{1}_{G_a}] + \mathbb{E}[T_a(n)\mathbf{1}_{G_a^c}] \leq u_a \mathbb{P}(G_a) + n \mathbb{P}(G_a^c) \leq \left\lceil \frac{8 \log \frac{1}{\delta}}{\Delta_a^2} \right\rceil \cdot 1 + n \cdot (n+1)\delta.$$

This holds for all suboptimal arms. Choose $\delta = n^{-2}$. Then, using $\lceil x \rceil \leq x + 1$ and $n \cdot (n+1)\delta \leq 2$, by summing over all suboptimal arms,

$$R(n) = \sum_{a=1}^k \Delta_a \mathbb{E}[T_a(n)] \leq \sum_{a:\Delta_a > 0} \left(\frac{16 \log n}{\Delta_a} + 3\Delta_a \right).$$

$\square$

Instance-Dependent Bound

Again of the form

$$R(n) \leq C \sum_{a:\Delta_a > 0} \left(\Delta_a + \frac{\log n}{\Delta_a} \right)$$

for some universal $C > 0$.

Instance-Independent Bound

Let $\Delta > 0$.

$$R(n) = \sum_{a:\Delta_a > 0} \Delta_a \mathbb{E}[T_a(n)]$$

$$\leq \sum_{a:\Delta_a < \Delta} \Delta \mathbb{E}[T_a(n)] + \sum_{a:\Delta_a \geq \Delta} \left(3\Delta_a + \frac{16 \log n}{\Delta_a} \right) \leq \Delta n + \sum_{a=1}^k 3\Delta_a + \frac{16k \log n}{\Delta}$$

since $\sum_a \mathbb{E}[T_a(n)] = n$. Minimise *w.r.t.* Δ implies $\Delta = \sqrt{\frac{16k \log n}{n}}$, gives *instance-independent bound*:

$$R(n) \leq 8\sqrt{nk \log n} + 3 \sum_{a=1}^k \Delta_a.$$

Therefore,

$$R_v^\pi(n) = O\left(\sum_{a:\Delta_a > 0} \left\{ \Delta_a + \frac{\log n}{\Delta_a} \right\}\right), \quad R^\pi(n) = O\left(\sqrt{nk \log n}\right).$$

Remark. (i) Improves upon the $n^{\frac{2}{3}}$ of ETC.

(ii) No tuning parameter that depends on unknown instances.

(iii) Worst-case regret $\tilde{O}(\sqrt{nk})$, up to logarithmic terms.

(iv) By refined analysis, the $\sqrt{\log n}$ term can be removed.

(v) No algorithm can have worst-case regret lower than $O(\sqrt{nk})$. The right amount of optimism is (close to) asymptotically optimal.

Some Extensions

Exercises

EXERCISE 2.1 (Adaptive Commitment Times). Suppose that ETC interacts with a two-armed 1-subgaussian bandit $\nu \in \mathcal{E}$ with means $\mu_1, \mu_2 \in \mathbb{R}$ and $\Delta_\nu = |\mu_1 - \mu_2|$.

- (i) Find a choice of m that only depends on the horizon n and not Δ such that there exists a constant $C > 0$ such that for any n and for any $\nu \in \mathcal{E}$, the regret $R_n(\nu)$ of Algorithm 1 is bounded by

$$R_n(\nu) \leq (\Delta_\nu + C)n^{\frac{2}{3}}.$$

Furthermore, show that there is no $C > 0$ such that for any problem instance ν and $n \geq 1$, $R_n(\nu) \leq \Delta_\nu + Cn^{\frac{2}{3}}$ holds.

- (ii) Now suppose the commitment time is allowed to be data dependent, which means the algorithm explores each arm alternately until some condition is met and then commits to a single arm for the remainder. Design a condition such that the regret of the resulting algorithm can be bounded by

$$R_n(\nu) \leq \Delta_\nu + \frac{C \log n}{\Delta_\nu}, \quad (2.1)$$

where C is a universal constant. Your condition should only depend on the observed rewards and the time horizon. It should not depend on μ_1, μ_2 or Δ_ν .

- (iii) Show that any algorithm for which (2.1) holds also satisfies

$$R_n(\nu) \leq \Delta_\nu + C\sqrt{n \log(n)}$$

for any $n \geq 1$ and $\nu \in \mathcal{E}$ and a suitably chosen universal constant.

- (iv) As for (ii), but now the objective is to design a condition such that for any $n \geq 1$ and $\nu \in \mathcal{E}$, the regret of the resulting algorithm is bounded by

$$R_n(\nu) \leq \Delta_\nu + \frac{C \log \max\{e, n\Delta_\nu^2\}}{\Delta_\nu}. \quad (2.2)$$

- (v) Show that any algorithm for which (2.2) holds also satisfies that for any $n \geq 1$ and $\nu \in \mathcal{E}$, $R_n(\nu) \leq \Delta_\nu + C\sqrt{n}$ for suitably chosen universal constant $C > 0$.

SOLUTION. (i) By (6.4), we have

$$R_n \leq m\Delta + n\Delta \exp\left(-\frac{m\Delta^2}{4}\right),$$

and by differentiation:

$$\frac{d\left(n\Delta \exp\left(-\frac{m\Delta^2}{4}\right)\right)}{d\Delta} = n \exp\left(-\frac{m\Delta^2}{4}\right) + n\Delta \exp\left(-\frac{m\Delta^2}{4}\right)\left(-\frac{m\Delta}{2}\right) = n \exp\left(-\frac{m\Delta^2}{4}\right)\left(1 - \frac{m\Delta^2}{2}\right),$$

we can substitute Δ in the second term by $\Delta^* = \sqrt{\frac{2}{m}}$ and obtain

$$R_n \leq m\Delta + n\sqrt{\frac{2}{m}} \exp\left(-\frac{m\left(\sqrt{\frac{2}{m}}\right)^2}{4}\right) = m\Delta + n\sqrt{\frac{2}{m}} \exp\left(-\frac{1}{2}\right).$$

Now let $m = \left\lceil n^{\frac{2}{3}} \right\rceil$ which only depends on n and we have

$$R_n \leq \left(n^{\frac{2}{3}} + 1\right)\Delta + n\sqrt{\frac{2}{n^{\frac{2}{3}}}} \exp\left(-\frac{1}{2}\right) = \left(\Delta + \sqrt{2} \exp\left(-\frac{1}{2}\right)\right)n^{\frac{2}{3}} + \Delta.$$

So for any n and any $v \in \mathcal{E}$, there exists a constant C such that

$$Cn^{\frac{2}{3}} \geq \sqrt{2} \exp\left(-\frac{1}{2}\right) + \Delta_v,$$

and the regret is bounded by

$$R_n(v) \leq (\Delta_v + C)n^{\frac{2}{3}}.$$

Assume there exists $C_0 > 0$ such that for any problem instance v and $n \geq 1$, we have

$$R_n(v) \leq \Delta_v + C_0 n^{\frac{2}{3}}.$$

By the definition of the ETC algorithm, we know the lower bound for the regret: $R_n(v) \geq m\Delta_v$. Thus, for any problem instance, we have

$$m\Delta_v \leq \Delta_v + C_0 n^{\frac{2}{3}} \Rightarrow m \leq 1 + \frac{C_0 n^{\frac{2}{3}}}{\Delta_v}.$$

Let $\Delta_v \rightarrow \infty$, then we have $m \leq 1$. But it is possible that after selecting both arms for 0 or 1 time, we still select the suboptimal arm, and in this case, $R_n(v) = (n - m)\Delta_v$. Then if we choose n large enough, we would have

$$(n - m)\Delta_v > (\Delta_v + C_0)n^{\frac{2}{3}},$$

which is contradictory. Hence, we show that there does not exist a constant C such that for any problem instance the regret is bounded by $(\Delta_v + C)n^{\frac{2}{3}}$.

(ii) Now we design the stopping condition as an event

$$G = \left\{ \exists m \in \left\{1, 2, \dots, \left\lfloor \frac{n}{2} \right\rfloor\right\} : |\hat{\mu}_1(2m) - \hat{\mu}_2(2m)| > 2\sqrt{\frac{2\log(\frac{n}{\delta})}{m}} \right\}.$$

In the algorithm we keep exploring each arm alternately until two averages' difference exceeds $2\sqrt{\frac{2\log(\frac{n}{\delta})}{m}}$, then select the arm with the higher average reward, otherwise, if their difference does not exceed, we would keep exploring until the end. In the case that G happens, we denote

$$m^* = \min \left\{ m \in \left\{1, 2, \dots, \left\lfloor \frac{n}{2} \right\rfloor\right\} : |\hat{\mu}_1(2m) - \hat{\mu}_2(2m)| > 2\sqrt{\frac{2\log(\frac{n}{\delta})}{m}} \right\}$$

as the stopping exploring time, and let $m^* = \frac{n}{2}$ if G does not happen. Also, we define another event

$$G_0 = \left\{ |\hat{\mu}_i(2m) - \mu_i| \leq \sqrt{\frac{2\log(\frac{n}{\delta})}{m}}, i \in \{1, 2\}, m \in \left\{1, 2, \dots, \left\lfloor \frac{n}{2} \right\rfloor\right\} \right\}.$$

Since $\hat{\mu}_i(2m)$ is a $\frac{1}{\sqrt{m}}$ -subgaussian, the probability $\mathbb{P}(G_0)$ has a lower bound as

$$\begin{aligned} \mathbb{P}(G_0) &= 1 - \mathbb{P}(G_0^c) = 1 - \mathbb{P}\left(\exists i \in \{1, 2\}, m \in \left\{1, 2, \dots, \left\lfloor \frac{n}{2} \right\rfloor\right\} : |\hat{\mu}_i(2m) - \mu_i| \geq \sqrt{\frac{2\log(\frac{n}{\delta})}{m}}\right) \\ &\geq 1 - \sum_{i=1}^2 \sum_{m=1}^{\left\lfloor \frac{n}{2} \right\rfloor} \mathbb{P}\left(|\hat{\mu}_i(2m) - \mu_i| \geq \sqrt{\frac{2\log(\frac{n}{\delta})}{m}}\right) \geq 1 - \sum_{i=1}^2 \sum_{m=1}^{\left\lfloor \frac{n}{2} \right\rfloor} 2 \exp\left(-\frac{\left(\sqrt{\frac{2\log(\frac{n}{\delta})}{m}}\right)^2 m}{2}\right) \end{aligned}$$

$$\geq 1 - 2 \cdot \frac{n}{2} \cdot \frac{2\delta}{n} = 1 - 2\delta.$$

Given that G_0 happens, if G also happens and we select arm 1 after $2m^*$ rounds, i.e., $\hat{\mu}_1(2m^*) > \hat{\mu}_2(2m^*) + 2\sqrt{\frac{2\log(\frac{n}{\delta})}{m}}$, arm 1 is actually the optimal arm, since

$$\mu_1 \geq \hat{\mu}_1(2m^*) - \sqrt{\frac{2\log(\frac{n}{\delta})}{m}} > \hat{\mu}_2(2m^*) + 2\sqrt{\frac{2\log(\frac{n}{\delta})}{m}} - \sqrt{\frac{2\log(\frac{n}{\delta})}{m}} = \hat{\mu}_2(2m^*) + \sqrt{\frac{2\log(\frac{n}{\delta})}{m}} \geq \mu_2.$$

Thus, it implies conditional on G_0 , the regret $R_n = \Delta E(m^* | G_0)$. *W.l.o.g.*, we assume arm 1 is optimal. Thus, we can derive the regret's bound as

$$\begin{aligned} R_n &= \Delta E(T_2(n)) = \Delta(E(T_2(n | G_0))\mathbb{P}(G_0) + E(T_2(n | G_0^c))\mathbb{P}(G_0^c)) \\ &\leq \Delta(E(T_2(n | G_0)) + (n - m) \cdot 2\delta) \leq \Delta(E(m^* | G_0) + 2n\delta). \end{aligned}$$

For $m \in \{1, 2, \dots, m^* - 1\}$, given that G_0 happens, we have

$$\begin{aligned} 2\sqrt{\frac{2\log(\frac{n}{\delta})}{m}} &\geq |\hat{\mu}_1(2m) - \hat{\mu}_2(2m)| \geq \hat{\mu}_1(2m) - \hat{\mu}_2(2m) \\ &\geq \left(\mu_1 - \sqrt{\frac{2\log(\frac{n}{\delta})}{m}}\right) - \left(\mu_2 + \sqrt{\frac{2\log(\frac{n}{\delta})}{m}}\right) = \Delta - 2\sqrt{\frac{2\log(\frac{n}{\delta})}{m}}, \end{aligned}$$

and by rearranging terms, we obtain

$$m \leq \frac{32\log \frac{n}{\delta}}{\Delta^2}.$$

Then the conditional expectation is bounded by $E(m^* | G_0) \leq \frac{32\log \frac{n}{\delta}}{\Delta^2} + 1$, and by taking $\delta = \frac{1}{n}$, the regret's bound is given as

$$R_n \leq \Delta \left(\left(\frac{32\log(\frac{n}{\frac{1}{n}})}{\Delta^2} + 1 \right) + 2n \cdot \frac{1}{n} \right) = 3\Delta + \frac{64\log(n)}{\Delta}.$$

Hence, for any $n \geq 1$ and any $v \in \mathcal{E}$, by choosing C large enough the regret of the algorithm is bounded by

$$R_n(v) \leq \Delta_v + \frac{C \log(n)}{\Delta_v}.$$

- (iii) Since the regret can never be larger than $n\Delta$ and we have $R_n(v) \leq \Delta_v + \frac{C \log(n)}{\Delta_v}$, if $\Delta_v \leq \sqrt{\frac{\log(n)}{n}}$, we have

$$R_n(v) \leq n\Delta_v \leq n\sqrt{\frac{\log(n)}{n}} = \sqrt{n \log(n)},$$

and if $\Delta_v \geq \sqrt{\frac{\log(n)}{n}}$, the regret is bounded by

$$R_n(v) \leq \Delta_v + \frac{C \log(n)}{\Delta_v} \leq \Delta_v + \frac{C \log(n)}{\sqrt{\frac{\log(n)}{n}}} = \Delta_v + C\sqrt{n \log(n)}.$$

So for any $n \geq 1$ and $v \in \mathcal{E}$, by taking $C' = \max\{1, C\}$, we have

$$R_n(v) \leq \Delta_v + C'\sqrt{n \log(n)}.$$

- (iv) Similarly, now we replace the condition in (ii) by

$$G = \left\{ \exists m \in \left\{ 1, 2, \dots, \left\lfloor \frac{n}{2} \right\rfloor \right\} : |\hat{\mu}_1(2m) - \hat{\mu}_2(2m)| > 2\sqrt{\frac{2\log(\frac{n}{m\delta})}{m}} \right\},$$

and the stopping exploring time

$$m^* = \min \left\{ m \in \left\{ 1, 2, \dots, \left\lfloor \frac{n}{2} \right\rfloor \right\} : |\hat{\mu}_1(2m) - \hat{\mu}_2(2m)| > 2\sqrt{\frac{2\log(\frac{n}{m\delta})}{m}} \right\}.$$

Again, we need to redefine another event

$$G_0 = \left\{ |\hat{\mu}_i(2m) - \mu_i| \leq \sqrt{\frac{2\log(\frac{n}{m\delta})}{m}}, i \in \{1, 2\}, m \in \left\{ 1, 2, \dots, \left\lfloor \frac{n}{2} \right\rfloor \right\} \right\}.$$

For G^c we have

$$\begin{aligned} \mathbb{P}(G_0^c) &= \mathbb{P}\left(\exists i \in \{1, 2\}, m \in \left\{ 1, 2, \dots, \left\lfloor \frac{n}{2} \right\rfloor \right\} : |\hat{\mu}_i(2m) - \mu_i| \geq \sqrt{\frac{2\log(\frac{n}{m\delta})}{m}}\right) \\ &\leq \sum_{i=1}^2 \sum_{m=1}^{\left\lfloor \frac{n}{2} \right\rfloor} \mathbb{P}\left(|\hat{\mu}_i(2m) - \mu_i| \geq \sqrt{\frac{2\log(\frac{n}{m\delta})}{m}}\right) \leq \sum_{i=1}^2 \sum_{m=1}^{\left\lfloor \frac{n}{2} \right\rfloor} 2 \exp\left(-\frac{(\sqrt{\frac{2\log(\frac{n}{m\delta})}{m}})^2 m}{2}\right) \\ &= 2 \sum_{m=1}^{\left\lfloor \frac{n}{2} \right\rfloor} \frac{2m\delta}{n} \leq n\delta, \end{aligned}$$

and from (b) we know given that G_0 happens, during commitment times(if exists) the algorithm always select the optimal arm, so the regret is given as

$$R_n \leq \Delta(E(m^* | G_0)\mathbb{P}(G_0) + (n - m)\mathbb{P}(G_0^c)) \leq \Delta(E(m^* | G_0) + n^2\delta).$$

Also, for $m \in \{1, 2, \dots, m^* - 1\}$, if G_0 happens,

$$\begin{aligned} 2\sqrt{\frac{2\log(\frac{n}{m\delta})}{m}} &\geq |\hat{\mu}_1(2m) - \hat{\mu}_2(2m)| \geq \hat{\mu}_1(2m) - \hat{\mu}_2(2m) \\ &\geq \left(\mu_1 - \sqrt{\frac{2\log(\frac{n}{m\delta})}{m}}\right) - \left(\mu_2 + \sqrt{\frac{2\log(\frac{n}{m\delta})}{m}}\right) = \Delta - 2\sqrt{\frac{2\log(\frac{n}{m\delta})}{m}}, \end{aligned}$$

then we set $\delta = \frac{1}{n^2}$ and obtain that

$$m^* \leq \frac{C \log(\max\{e, n\Delta^2\})}{\Delta^2}$$

for a constant $C > 0$. Thus, the regret is bounded by

$$R_n \leq \Delta(E(m^* | G_0) + n^2\delta) \leq \Delta\left(\frac{C \log(\max\{e, n\Delta^2\})}{\Delta^2} + n^2 \frac{1}{n^2}\right) = \frac{C \log(\max\{e, n\Delta^2\})}{\Delta} + \Delta.$$

(v) Suppose an algorithm satisfies

$$R_n(v) \leq \Delta_v + \frac{C \log(\max\{e, n\Delta^2\})}{\Delta_v}.$$

By differentiation, we have

$$\frac{d\left(\frac{\log(n\Delta^2)}{\Delta}\right)}{d\Delta} = \frac{2 - \log(n\Delta^2)}{\Delta^2},$$

so $\frac{\log(n\Delta^2)}{\Delta}$ is maximised at $\Delta = \frac{e}{\sqrt{n}}$ such that $n\Delta_v^2 = e^2 > e$ and the maximum equals $\frac{2\sqrt{n}}{e}$. If $\Delta_v \leq \frac{1}{\sqrt{n}}$, then we have

$$R_n(v) \leq n\Delta_v \leq n \cdot \frac{1}{\sqrt{n}} = \sqrt{n}.$$

And if $\Delta_v > \frac{1}{\sqrt{n}}$, we have

$$R_n(v) \leq \Delta_v + \frac{C \log(\max\{e, n\Delta_v^2\})}{\Delta_v} \leq \Delta_v + C \cdot \frac{2\sqrt{n}}{e} = \Delta_v + \frac{2C}{e} \sqrt{n}.$$

Now let $C' = \max\{1, \frac{2C}{e}\}$, we know for any $n \geq 1$ and $v \in \mathcal{E}$, the regret is bounded by

$$R_n(v) \leq \Delta_v + C' \sqrt{n}.$$

□

EXERCISE 2.2 (Elimination Algorithm). A simple way to generalise the ETC policy to multiple arms and overcome the problem of tuning the commitment time is to use an elimination algorithm. The algorithm operates in phases and maintains an active set of arms that could be optimal. In the ℓ -th phase, the algorithm aims to eliminate from the active set all arms i for which $\Delta_i \geq 2^{-\ell}$.

Algorithm 1 Phased elimination for finite-armed bandits

Input: k and sequence $(m_\ell)_\ell$

$A_1 = \{1, 2, \dots, k\}$

for $\ell = 1, 2, 3, \dots$ **do**

 Choose each arm $i \in A_\ell$ exactly m_ℓ times

 Let $\hat{\mu}_{i,\ell}$ be the average reward for arm i from this phase only

 Update active set:

$$A_{\ell+1} = \left\{ i : \hat{\mu}_{i,\ell} + 2^{-\ell} \geq \max_{j \in A_\ell} \hat{\mu}_{j,\ell} \right\}$$

end for

Without loss of generality, assume that arm 1 is an optimal arm. You may assume that the horizon n is known.

(i) Show that for any $\ell \geq 1$,

$$\mathbb{P}(1 \notin A_{\ell+1}, 1 \in A_\ell) \leq k \exp\left(-\frac{m_\ell 2^{-2\ell}}{4}\right).$$

(ii) Show that if $i \in [k]$ and $\ell \geq 1$ are such that $\Delta_i \geq 2^{-\ell}$, then

$$\mathbb{P}(i \in A_{\ell+1}, 1 \in A_\ell, i \in A_\ell) \leq \exp\left(-\frac{m_\ell (\Delta_i - 2^{-\ell})^2}{4}\right).$$

(iii) Let $\ell_i = \min\left\{\ell \geq 1 : 2^{-\ell} \leq \frac{\Delta_i}{2}\right\}$. Choose m_ℓ in such a way that

$$\mathbb{P}(\exists \ell : 1 \notin A_\ell) \leq \frac{1}{n} \quad \text{and} \quad \mathbb{P}(i \in A_{\ell_i+1}) \leq \frac{1}{n}.$$

(iv) Show that your algorithm has regret at most

$$R_n \leq C \sum_{i: \Delta_i > 0} \left(\Delta_i + \frac{1}{\Delta_i} \log(n) \right),$$

where $C > 0$ is a carefully chosen universal constant.

(v) Modify your choice of m_ℓ and show that the regret of the resulting algorithm satisfies

$$R_n \leq C \sum_{i: \Delta_i > 0} \left(\Delta_i + \frac{1}{\Delta_i} \log \max \{e, n\Delta_i^2\} \right).$$

(vi) Show that with an appropriate universal constant $C' > 0$, the regret satisfies

$$R_n \leq \sum_i \Delta_i + C' \sqrt{nk \log(k)}.$$

SOLUTION. (i) By the definition of the algorithm, we have

$$A_{\ell+1} = \left\{ i : \hat{\mu}_{i,\ell} + 2^{-\ell} \geq \max_{j \in A_\ell} \hat{\mu}_{j,\ell} \right\}.$$

Now we are supposed to choose the arm 1 m_ℓ times, i.e., $1 \in A_\ell$, getting the average reward $\hat{\mu}_{1,\ell}$. To make sure the arm 1 isn't the optimal arm for $\ell + 1$, i.e., $1 \notin A_{\ell+1}$, there must exist at least one $j \in A_\ell \setminus \{1\}$ such that $\hat{\mu}_{1,\ell} + 2^{-\ell} < \hat{\mu}_{j,\ell}$, or we can write

$$\begin{aligned} & \mathbb{P}(1 \notin A_{\ell+1}, 1 \in A_\ell) \\ & \leq \mathbb{P}\left(1 \in A_\ell, \exists j \in A_\ell \setminus \{1\} : \hat{\mu}_{1,\ell} + 2^{-\ell} \leq \hat{\mu}_{j,\ell}\right) = \mathbb{P}\left(1 \in A_\ell, \exists j \in A_\ell \setminus \{1\} : \hat{\mu}_{j,\ell} - \hat{\mu}_{1,\ell} \geq 2^{-\ell}\right). \end{aligned}$$

For $\sqrt{\frac{1}{m_\ell}}$ -subgaussian random variables $\hat{\mu}_{j,\ell}$ and $\hat{\mu}_{1,\ell}$, by Lemma 5.4 (c), $\hat{\mu}_{j,\ell} - \hat{\mu}_{1,\ell}$ is $\sqrt{\frac{2}{m_\ell}}$ -subgaussian, then by Theorem 5.3, we have

$$\mathbb{P}(1 \notin A_{\ell+1}, 1 \in A_\ell) \leq \exp\left(-\frac{m_\ell 2^{-2\ell}}{4}\right) \leq (k-1) \exp\left(-\frac{m_\ell 2^{-2\ell}}{4}\right) \leq k \exp\left(-\frac{m_\ell 2^{-2\ell}}{4}\right).$$

take the union bound.

(ii) Now we require the arm i to be the optimal arm in $A_{\ell+1}$, note that we have assumed the arm 1 is the optimal arm in A_ℓ and $\max_{j \in A_\ell} \hat{\mu}_{j,\ell} = \mu^* = \hat{\mu}_{1,\ell}$, we need the equation $\hat{\mu}_{i,\ell} + 2^{-\ell} \geq \hat{\mu}_{1,\ell}$ satisfies. Similarly, we have

$$\begin{aligned} & \mathbb{P}(i \in A_{\ell+1}, 1 \in A_\ell, i \in A_\ell) \\ & \leq \mathbb{P}(i \in A_\ell, 1 \in A_\ell, \hat{\mu}_{i,\ell} + 2^{-\ell} \geq \hat{\mu}_{1,\ell}) = \mathbb{P}(i \in A_\ell, 1 \in A_\ell, \hat{\mu}_{i,\ell} - \hat{\mu}_{1,\ell} \geq -2^{-\ell}) \\ & = \mathbb{P}(i \in A_\ell, 1 \in A_\ell, \hat{\mu}_{i,\ell} - \hat{\mu}_{1,\ell} - \mu_i + \mu_1 \geq \Delta_i - 2^{-\ell}) \leq \exp\left(-\frac{m_\ell (\Delta_i - 2^{-\ell})^2}{4}\right) \end{aligned}$$

optimality gap $\Delta_i = \mu^* - \mu_i = \mu_1 - \mu_i$.

by Theorem 5.3, where $\hat{\mu}_{i,\ell} - \hat{\mu}_{1,\ell} - \mu_i + \mu_1$ is again $\sqrt{\frac{2}{m_\ell}}$ -subgaussian.

(iii) Choose

$$m_\ell = 2^{4+2\ell} \log\left(\frac{\ell}{\delta}\right),$$

where

$$\delta = \frac{1}{n} \cdot \frac{1}{1 + \frac{k\pi^2}{6}}$$

is a constant between (0, 1). From part (i), we have

$$\begin{aligned}\mathbb{P}(\exists \ell : 1 \notin A_\ell) &\leq \sum_{\ell=1}^{\infty} \mathbb{P}(1 \notin A_{\ell+1}, 1 \in A_\ell) \leq k \exp\left(-\frac{m_\ell 2^{-2\ell}}{4}\right) = k \exp\left(-\frac{2^{4+2\ell} \log\left(\frac{\ell}{\delta}\right) 2^{-2\ell}}{4}\right) \\ &= k \exp\left(-4 \log\left(\frac{\ell}{\delta}\right)\right) = k \delta^4 \left(\frac{1}{\ell}\right)^4 \leq k \delta \sum_{\ell=1}^{\infty} \frac{1}{\ell^2} = \frac{k\pi^2}{6} \delta = \frac{k\pi^2}{6} \cdot \frac{1}{n} \cdot \frac{1}{1 + \frac{k\pi^2}{6}} \\ &= \frac{1}{n} \cdot \frac{\frac{k\pi^2}{6}}{1 + \frac{k\pi^2}{6}} \leq \frac{1}{n}.\end{aligned}$$

$$\sum_{\ell=1}^{\infty} \frac{1}{\ell^2} = \frac{\pi^2}{6}.$$

And by part (ii), we have

$$\begin{aligned}\mathbb{P}(i \in A_{\ell_i+1}) &\leq \mathbb{P}(i \in A_{\ell_i+1}, 1 \in A_{\ell_i}, i \in A_{\ell_i}) + \mathbb{P}(1 \notin A_{\ell_i}) \leq \exp\left(-\frac{m_\ell (\Delta_i - 2^{-\ell_i})^2}{4}\right) + \frac{k\pi^2}{6} \delta \\ &\leq \exp\left(-\frac{m_\ell 2^{-2\ell_i}}{16}\right) + \frac{k\pi^2}{6} \delta \leq \delta \left(1 + \frac{k\pi^2}{6}\right) \\ &= \frac{1}{n} \cdot \frac{1}{1 + \frac{k\pi^2}{6}} \left(1 + \frac{k\pi^2}{6}\right) = \frac{1}{n}.\end{aligned}$$

$$\ell_i = \min \left\{ \ell \geq 1 : 2^{-\ell} \leq \frac{\Delta_i}{2} \right\}; \exp\left(-\frac{m_\ell 2^{-2\ell}}{4}\right) \leq \frac{k\pi^2}{6} \delta.$$

Therefore, we have proved

$$\mathbb{P}(\exists \ell : 1 \notin A_\ell) \leq \frac{1}{n} \quad \text{and} \quad \mathbb{P}(i \in A_{\ell_i+1}) \leq \frac{1}{n}.$$

- (iv) When $n < k$, the total number of pulls is less than the number of arms, meaning each arm is tried at most once. The regret in this case is trivially bounded by $R_n \leq \sum_{i: \Delta_i > 0} \Delta_i$, which is below the desired bound provided $C > 1$.

When $n \geq k$, if there are no suboptimal arms, the statement is trivial because the regret is zero. Otherwise, let i be a suboptimal arm with suboptimality gap Δ_i . As before, we let

$$\ell_i = \min \left\{ \ell \geq 1 : 2^{-\ell} \leq \frac{\Delta_i}{2} \right\},$$

then we have $2^{2\ell_i} \leq \frac{16}{\Delta_i^2}$, and the number of pulls $m_\ell \geq m_1 \geq 1$ for all $\ell \geq 1$. And we can calculate the expected number of times the suboptimal arm is pulled:

$$\mathbb{E}[T_i(n)] \leq n \mathbb{P}(i \in A_{\ell_i+1}) + \sum_{\ell=1}^{\ell_i \wedge n} m_\ell \leq 1 + \sum_{\ell=1}^{\ell_i \wedge n} 2^{4+2\ell} \log\left(\frac{\ell}{\delta}\right).$$

$$\mathbb{P}(i \in A_{\ell_i+1}) \leq \frac{1}{n}.$$

Note that for the geometric series, we have

$$\sum_{\ell=1}^L 2^{4+2\ell} = 2^4 \sum_{\ell=1}^L 2^{2\ell} \approx 2^4 \cdot \frac{4(4^L - 1)}{3} \leq \frac{2^6}{3} \cdot 2^{2L}. \quad (2.3)$$

And the logarithmic term

$$\log\left(\frac{\ell}{\delta}\right) \leq \log\left(\frac{\ell_i}{\delta}\right) = \log\left(\frac{\ell_i}{\frac{1}{n} \cdot \frac{1}{1 + \frac{k\pi^2}{6}}}\right) = \log\left(n \ell_i \left(\frac{1}{1 + \frac{k\pi^2}{6}}\right)\right) \leq \tilde{C} \log(nk), \quad (2.4)$$

where $\tilde{C}$ is a suitably large constant. Combine (2.3) and (2.4) with the previous result, we have

$$\mathbb{E}[T_i(n)] \leq 1 + C \cdot 2^{2\ell_i} \log(nk) \leq 1 + \frac{16C}{\Delta_i^2} \log(nk),$$

$$2^{2\ell_i} \leq \frac{16}{\Delta_i^2}.$$

where C is a suitably large constant. Therefore, by Lemma 4.5, the total regret is:

$$R_n = \sum_{i:\Delta_i>0} R_i = \sum_{i:\Delta_i>0} \Delta_i \mathbb{E}[T_i(n)] \leq \sum_{i:\Delta_i>0} \Delta_i \left(1 + \frac{16C}{\Delta_i^2} \log(nk) \right) \leq C \sum_{i:\Delta_i>0} \left(\Delta_i + \frac{1}{\Delta_i} \log(n) \right),$$

$$\log(nk) \leq 2 \log(n).$$

where $C > 0$ is a carefully chosen universal constant.

(v) The modified choice of m_ℓ is:

$$m_\ell = C \cdot 2^{2^\ell} \log \max \left(e, kn2^{-2^\ell} \right),$$

where C is a suitably large constant. Then we have two probability bounds:

$$\mathbb{P}(1 \notin A_\ell) \leq \frac{2^{2^\ell}}{n}, \quad \mathbb{P}(i \in A_{\ell+1}) \leq \frac{2^{2^\ell}}{n}.$$

And we can calculate the expected number of times the suboptimal arm is pulled:

$$\begin{aligned} \mathbb{E}[T_i(n)] &\leq n\mathbb{P}(i \in A_{\ell+1}) + \sum_{\ell=1}^{\ell_i \wedge n} m_\ell \\ &\leq 2^{2^\ell} + \sum_{\ell=1}^{\ell_i \wedge n} C \cdot 2^{2^\ell} \log \max \left(e, kn2^{-2^\ell} \right) \leq \frac{16}{\Delta_i^2} + C \sum_{\ell=1}^{\ell_i} 2^{2^\ell} \log \max \left(e, kn2^{-2^\ell} \right). \end{aligned}$$

We note that 2^{2^ℓ} grows very quickly and dominates the logarithm. Let

$$f(x) = 2^{2^x} \log \max \left(e, kn2^{-2^x} \right).$$

We can calculate the integral:

$$\int_1^{\ell_i} 2^{2^x} \log \max \left(e, kn2^{-2^x} \right) dx = \frac{1}{\ln 2} \int_2^{2^{\ell_i}} \frac{2^u}{u} \log \max \left(e, knu^{-2} \right) du.$$

substitute $u = 2^x$.

We have

$$\log \max \left(e, knu^{-2} \right) = \begin{cases} \log(knu^{-2}) = \log(kn) - 2 \log(u), & u \leq \sqrt{\frac{kn}{e}} \\ \log(e) = 1, & u > \sqrt{\frac{kn}{e}} \end{cases}.$$

When $u \leq \sqrt{\frac{kn}{e}}$, the integral is

$$\begin{aligned} &\frac{1}{\ln 2} \int_2^{\sqrt{\frac{kn}{e}}} \frac{2^u}{u} (\log(kn) - 2 \log(u)) du \\ &= \frac{1}{\ln 2} \left(\log(kn) \int_2^{\sqrt{\frac{kn}{e}}} \frac{2^u}{u} du - 2 \int_2^{\sqrt{\frac{kn}{e}}} \frac{2^u \log(u)}{u} du \right) \approx \frac{1}{\ln 2} \left(\log(kn) \frac{2^{\sqrt{\frac{kn}{e}}}}{\sqrt{\frac{kn}{e}}} - 2 \log \sqrt{\frac{kn}{e}} \frac{2^{\sqrt{\frac{kn}{e}}}}{\sqrt{\frac{kn}{e}}} \right). \end{aligned}$$

When $u > \sqrt{\frac{kn}{e}}$, the integral is

$$\frac{1}{\ln 2} \int_2^{\sqrt{\frac{kn}{e}}} \frac{2^u}{u} du = \frac{\text{Ei} \left(\ln 2 \sqrt{\frac{kn}{e}} \right) - \text{Ei}(2 \ln 2)}{\ln 2} \approx \frac{2^{2^{\ell_i}}}{2^{\ell_i}}.$$

Combining both parts, we get an upper bound:

$$\sum_{\ell=1}^{\ell_i} 2^{2^\ell} \log \max \left(e, kn2^{-2^\ell} \right) \leq \int_1^{\ell_i} 2^{2^x} \log \max \left(e, kn2^{-2^x} \right) dx \leq \frac{C}{\Delta_i^2} \log \max(e, n\Delta_i^2).$$

Therefore, by Lemma 4.5, the total regret is:

$$\begin{aligned} R_n &= \sum_{i:\Delta_i>0} \Delta_i \mathbb{E}[T_i(n)] \leq \sum_{i:\Delta_i>0} \Delta_i \left(\frac{16}{\Delta_i^2} + C \sum_{\ell=1}^{\ell_i} 2^{2\ell} \log \max(e, kn2^{-2\ell}) \right) \\ &\leq \sum_{i:\Delta_i>0} \Delta_i \left(\frac{16}{\Delta_i^2} + C \int_1^{\ell_i} 2^{2x} \log \max(e, kn2^{-2x}) dx \right) \leq C \sum_{i:\Delta_i>0} \left(\Delta_i + \frac{1}{\Delta_i} \log \max\{e, n\Delta_i^2\} \right). \end{aligned}$$

(vi) From part (v), let $\Delta_i = \sqrt{\frac{k}{n}}$, by Lemma 4.5, we have the total regret:

$$R_n = \sum_{i:\Delta_i>0} \Delta_i \mathbb{E}[T_i(n)] \leq n\Delta + \sum_{i:\Delta_i>0} \frac{C'}{\Delta_i} \log \max(e, nk\Delta_i^2).$$

We notice the function $g(x) = \frac{1}{x} \log \max(e, nkx^2)$ is monotonically increasing for $x > \sqrt{\frac{kn}{e}}$, to get an upper bound, we consider the integral:

$$\begin{aligned} \int_{\sqrt{\frac{k}{n}}}^{\infty} \frac{1}{x} \log \max(e, nkx^2) dx &= \int_{\sqrt{\frac{k}{n}}}^{\infty} \frac{1}{x} \log(nkx^2) dx \\ &= \int_{\sqrt{\frac{k}{n}}}^{\infty} \frac{1}{x} (\log(nk) + 2\log(x)) dx \approx \log(nk) \log\left(\frac{n}{k}\right) + \log^2\left(\frac{n}{k}\right), \end{aligned}$$

which grow slower than the term $\sqrt{nk \log(k)}$. Therefore, we have

$$R_n \leq n\Delta + \sum_{i:\Delta_i>0} \frac{C'}{\Delta_i} \log \max(e, nk\Delta_i^2) = n\Delta + \int_{\sqrt{\frac{k}{n}}}^{\infty} \frac{1}{x} \log \max(e, nkx^2) dx \leq \sum_i \Delta_i + C' \sqrt{nk \log(k)},$$

where C' is an appropriate universal constant. □

EXERCISE 2.3 (Empirical Comparison). (i) Implement [Algorithm 2](#).

Algorithm 2 UCB(δ)

Input: k and δ
for $t \in 1, \dots, n$ **do**
 Choose action $A_t = \arg \max_i UCB_i(t-1, \delta)$
 Observe reward X_t and update upper confidence bounds
end for

- (ii) Reproduce [Figure 2.1](#).
- (iii) Explain the shape of the curves for ETC. In particular, when $m = 50$, we see a bump, a dip and then a linear asymptote as Δ grows. Why does the curve look like this?
- (iv) Design an experiment to determine the practical effect of the choice of δ .
- (v) Explain your results.

SOLUTION. (i) With the same settings for *Experiment 7.1* in BA, we implement mainly the UCB algorithm with parameter $\delta = \frac{1}{n^2}$ with the following code:

```
import numpy as np
import matplotlib.pyplot as plt
```

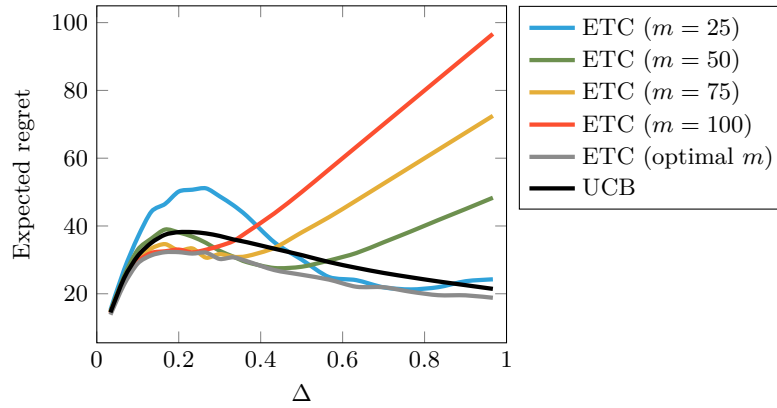


FIGURE 2.1. Experiment showing universality of UCB relative to fixed instances of ETC

```

class Bandit:
    def __init__(self, num_arms):
        self.num_arms = num_arms
        self.means = np.zeros(num_arms)
        self.means[0] = 0

    def pull_arm(self, arm):
        return np.random.normal(self.means[arm], 1)

class UCB:
    def __init__(self, num_arms, delta):
        self.num_arms = num_arms
        self.counts = np.zeros(num_arms)
        self.values = np.zeros(num_arms)
        self.time = 0
        self.delta = delta

    def select_arm(self):
        self.time += 1
        exploration_bonus = np.sqrt(2 * np.log(1/self.delta) /
        ↪ (self.counts + 1e-5))
        return np.argmax(self.values + exploration_bonus)

    def update(self, arm, reward):
        self.counts[arm] += 1
        self.values[arm] += (reward - self.values[arm]) / self.counts[arm]

class ETC:
    def __init__(self, num_arms, m):
        self.num_arms = num_arms
        self.values = np.zeros(num_arms)
        self.m = m
        self.explore_counts = np.zeros(num_arms)
        self.total_counts = 0

    def select_arm(self):

```

```

        if self.total_counts < self.num_arms * self.m:
            return np.argmin(self.explore_counts)
        else:
            return np.argmax(self.values)

    def update(self, arm, reward):
        self.explore_counts[arm] += 1
        self.total_counts += 1
        self.values[arm] += (reward - self.values[arm]) /
        ↪ self.explore_counts[arm] if self.explore_counts[arm] <= self.m
        ↪ else 0

    def simulate(bandit, algorithm, num_rounds):
        rewards = np.zeros(num_rounds)
        for t in range(num_rounds):
            arm = algorithm.select_arm()
            reward = bandit.pull_arm(arm)
            rewards[t] = reward
            algorithm.update(arm, reward)
        return np.max(bandit.means) * num_rounds - np.sum(rewards)

```

The main methods and results would be exhibited in (ii).

(ii) Implementing the following code:

```

class ETC:
    def __init__(self, num_arms, m):
        self.num_arms = num_arms
        self.values = np.zeros(num_arms)
        self.m = m
        self.explore_counts = np.zeros(num_arms)
        self.total_counts = 0

    def select_arm(self):
        if self.total_counts < self.num_arms * self.m:
            return np.argmin(self.explore_counts)
        else:
            return np.argmax(self.values)

    def update(self, arm, reward):
        self.explore_counts[arm] += 1
        self.total_counts += 1
        self.values[arm] += (reward - self.values[arm]) /
        ↪ self.explore_counts[arm] if self.explore_counts[arm] <= self.m
        ↪ else 0

from scipy.interpolate import make_interp_spline
from numpy.polynomial import Polynomial

def main():
    num_arms = 2
    num_rounds = 1000
    num_simulations = 100000

```

```

delta = 1 / (num_rounds)**2

m_values = [25, 50, 75, 100]

Deltas = np.linspace(0.01, 1, 50)

ucb_regrets = np.zeros((len(Deltas),))
etc_regrets = np.zeros((len(Deltas), len(m_values)))
etc_regrets_optimal = np.zeros((len(Deltas),))

for i, Delta in enumerate(Deltas):
    m_optimal = max(1, np.ceil(4 / Delta**2 * np.log(num_rounds *
        ↪ Delta**2 / 4)))

    bandit = Bandit(num_arms)
    bandit.means[1] = -Delta

    ucb_total_regret = 0
    etc_total_regret = np.zeros((len(m_values),))
    etc_total_regret_optimal = 0

    for _ in range(num_simulations):
        ucb_algorithm = UCB(num_arms, delta)
        ucb_regret = simulate(bandit, ucb_algorithm, num_rounds)
        ucb_total_regret += ucb_regret

        for j, m in enumerate(m_values):
            etc_algorithm = ETC(num_arms, m)
            etc_regret = simulate(bandit, etc_algorithm, num_rounds)
            etc_total_regret[j] += etc_regret

        etc_algorithm_optimal = ETC(num_arms, m_optimal)
        etc_regret_optimal = simulate(bandit, etc_algorithm_optimal,
            ↪ num_rounds)
        etc_total_regret_optimal += etc_regret_optimal

    ucb_regrets[i] = ucb_total_regret / num_simulations

    for j in range(len(m_values)):
        etc_regrets[i][j] = etc_total_regret[j] / num_simulations

    etc_regrets_optimal[i] = etc_total_regret_optimal /
        ↪ num_simulations

new_Deltas = np.linspace(min(Deltas), max(Deltas), 300)

spline_ucb_regrets = make_interp_spline(Deltas, ucb_regrets)
smooth_ucb_regrets = spline_ucb_regrets(new_Deltas)
degree = 5
ucb_polynomial = Polynomial.fit(new_Deltas, smooth_ucb_regrets,
    ↪ degree)
smooth_ucb_regrets_fit = ucb_polynomial(new_Deltas)

```

```

plt.plot(new_Deltas, smooth_ucb_regrets_fit, label='UCB',
↪ linestyle='--')

for j, m in enumerate(m_values):
    spline_etc_regrets = make_interp_spline(Deltas, etc_regrets[:, j])
    smooth_etc_regrets = spline_etc_regrets(new_Deltas)
    etc_polynomial = Polynomial.fit(new_Deltas, smooth_etc_regrets,
↪ degree)
    smooth_etc_regrets_fit = etc_polynomial(new_Deltas)
    plt.plot(new_Deltas, smooth_etc_regrets_fit, label=f'ETC (m={m})',
↪ linestyle='-')

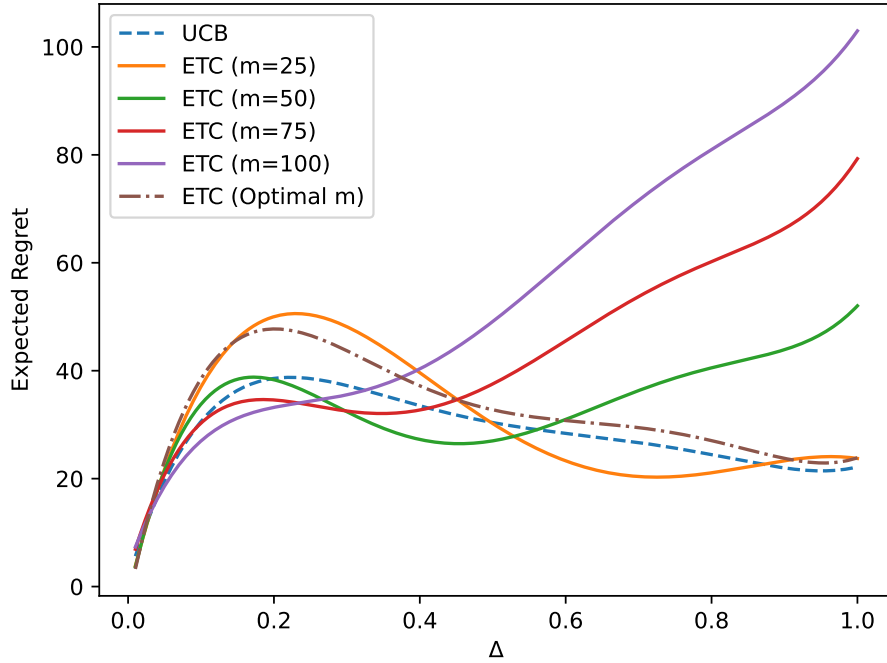
spline_etc_regrets_optimal = make_interp_spline(Deltas,
↪ etc_regrets_optimal)
smooth_etc_regrets_optimal = spline_etc_regrets_optimal(new_Deltas)
etc_optimal_polynomial = Polynomial.fit(new_Deltas,
↪ smooth_etc_regrets_optimal, degree)
smooth_etc_regrets_optimal_fit = etc_optimal_polynomial(new_Deltas)
plt.plot(new_Deltas, smooth_etc_regrets_optimal_fit, label='ETC
↪ (Optimal m)', linestyle='-.')

plt.xlabel(r'$\Delta$')
plt.ylabel('Expected Regret')
plt.legend()
plt.show()

if __name__ == "__main__":
    main()

```

Combining with classes in (i) and the above, we reproduce the [Figure 2.1](#) as follows:



- (iii) When $m = 50$, initially, the algorithm starts with a relatively high level of exploration. This means it explores different arms more aggressively to gather information about their potential rewards. As a result, the regret initially increases because the algorithm chooses the suboptimal arm many times during exploring time but may still choose the suboptimal arm after exploring because of non-significant differences.

As Δ grows, the difference in means between the optimal arm and the suboptimal arm increases. This larger difference encourages the algorithm to exploit the arm with the higher mean more frequently, which means it is more possible for the algorithm to find the optimal arm after exploring. Consequently, the regret starts to decrease, leading to the observed dip in the curve. However, as Δ continues to increase, the arms become more distinguishable, and the algorithm can always exploit the arm with the highest mean after exploring. Then the commitment phase leads to a linear asymptote ($\approx (n - 50)\Delta$) as the regret. In this phase, the regret's slope remains relatively constant because the algorithm consistently chooses the optimal arm, minimising the regret over time.

- (iv) To test the practical effect of different choices of δ , we design an experiment with the same set-up as the *Experiment 7.1* but let $\Delta \in [0, 3]$ instead, and we would implement ucb algorithms with parameter

$$\delta \in \left\{ \frac{1}{\sqrt{n}}, \frac{1}{n}, \frac{1}{n^2}, \frac{1}{n^3}, \frac{1}{n^4} \right\}$$

respectively. Implementing the following code:

```
from scipy.interpolate import make_interp_spline
from numpy.polynomial import Polynomial

def main():
    num_arms = 2
    num_rounds = 1000
    num_simulations = 100
```

```

delta_values = [1 / np.sqrt(num_rounds), 1 / num_rounds, 1 /
    ↪ (num_rounds)**2, 1 / (num_rounds)**3, 1 / (num_rounds)**4]

Deltas = np.linspace(0.01, 3, 50)

ucb_regrets = np.zeros((len(Deltas), len(delta_values)))

for i, Delta in enumerate(Deltas):
    bandit = Bandit(num_arms)
    bandit.means[1] = -Delta

    ucb_total_regret = np.zeros((len(delta_values),))

    for _ in range(num_simulations):
        for j, delta in enumerate(delta_values):
            ucb_algorithm = UCB(num_arms, delta)
            ucb_regret = simulate(bandit, ucb_algorithm, num_rounds)
            ucb_total_regret[j] += ucb_regret

    for j in range(len(delta_values)):
        ucb_regrets[i][j] = ucb_total_regret[j] / num_simulations

new_Deltas = np.linspace(min(Deltas), max(Deltas), 300)

degree = 5

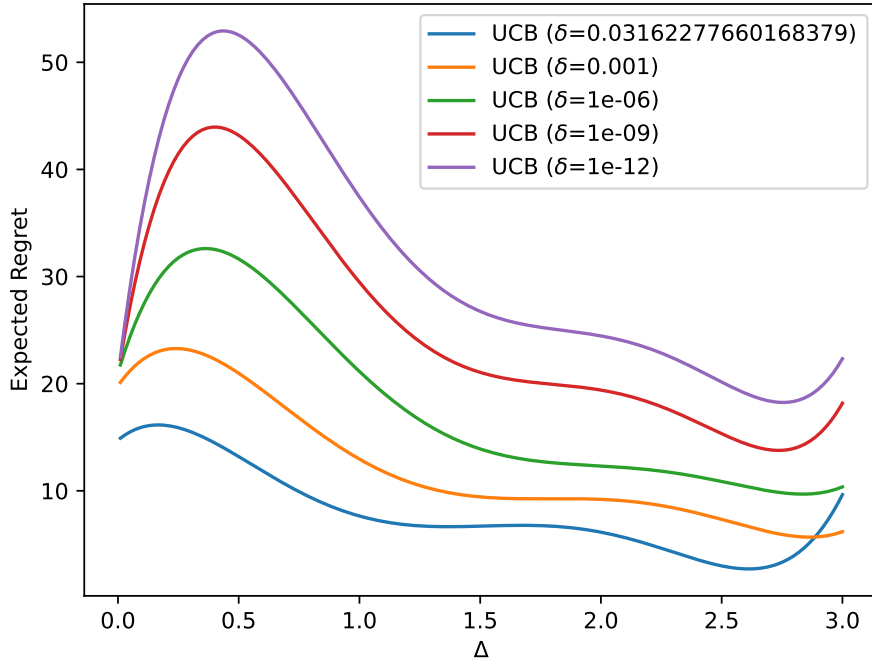
for j, delta in enumerate(delta_values):
    spline_ucb_regrets = make_interp_spline(Deltas, ucb_regrets[:, j])
    smooth_ucb_regrets = spline_ucb_regrets(new_Deltas)
    ucb_polynomial = Polynomial.fit(new_Deltas, smooth_ucb_regrets,
    ↪ degree)
    smooth_ucb_regrets_fit = ucb_polynomial(new_Deltas)
    plt.plot(new_Deltas, smooth_ucb_regrets_fit, label=f'UCB
    ↪ ($\delta$={delta})', linestyle='-')

plt.xlabel(r'$\Delta$')
plt.ylabel('Expected Regret')
plt.legend()
plt.show()

if __name__ == "__main__":
    main()

```

Combining with classes in (i) and the above, we would reproduce *Figure 2.1* with respect to five values of δ as follows:



- (v) From the graph in (iv) we can find that as the parameter δ grows, for almost any Δ , the expected regret increases, which means the UCB would perform worse. The reason might be that as δ becomes smaller, the exploration bonus part in $ucb(t, \delta)$ would become very large so that the difference between empirical means becomes more indistinguishable, which means the algorithm would explore the suboptimal arms more times and leads to a higher regret.

At the same time, notice that if the parameter δ is not small enough, once the optimal arm drops below its true mean and the confidence interval fails, then the algorithm would stop playing the optimal arm and suffer linear regret because the exploration bonus is small. In this experiment, we do not observe this, and the reason may be that the variance for the optimal arm is low, which means that it is hard for the optimal arm to drop too much, so we observe that $\delta = \frac{1}{\sqrt{n}}$ performs better than $\delta = \frac{1}{n}$.

In summary, to choose an appropriate δ , we should notice that δ should be small enough to ensure optimism with high probability, but not so small that suboptimal arms are explored excessively.

□

Adversarial Bandits, the EXP3 Policy, and Regret Lower Bounds

SECTION 3.1

Adversarial Bandit Environments

Nature chooses the rewards *adversarially*, i.e., to make your life difficult. Idea: if we get a regret bound, this will hedge against the *worst case* that may occur. ‘Nature will simply set all rewards to zero’. But then *all policies are optimal* and have regret zero. We look at worst-case *regret*, not worst-case revenue.

Adversarial setting is a zero-sum game between you and Nature. You move first: announce the policy. Then Nature moves: determines rewards $x_{t,a}$. Then the game is played (outcome is not fixed, because actions may be randomised). Nature is *oblivious*: does not respond to your particular actions. Nature tries to *maximise* your regret, you try to *minimise* your regret.

Randomising actions will be essential. Suppose $A = \{1, 2\}$, $T = 1$, $x_{t,a} \in \{0, 1\}$ (lower-case because not-random). If $A_1 = 1$, nature sets $x_{1,1} = 0$, $x_{1,2} = 1$, regret 1; if $A_1 = 2$, nature sets $x_{1,1} = 1$, $x_{1,2} = 0$, regret 1. Whatever *fixed policy* you play, you endure regret 1. If $A_1 = 1$ w.p. $\frac{1}{2}$ and $A_1 = 2$ w.p. $\frac{1}{2}$, then *randomised policy* halves the regret.

$x_{1,1}$	$x_{1,2}$	Best reward	My reward	Regret
0	0	0	0	0
0	1	1	0.5	0.5
1	0	1	0.5	0.5
1	1	1	1	0

Setting

Action set $A = \{1, \dots, k\}$, $k \geq 2$. For all $t = 1, \dots, n$, select action A_t , observe reward $X_t = x_{t,A_t}$. Here $x_{t,a} \in [0, 1]$ for all t and a . Write

$$v = (x_{t,a} : t = 1, \dots, n, a \in \mathcal{A}) \in [0, 1]^{nk}.$$

Oblivious opponent: v may depend on your policy π , but not on the data $(A_1, X_1, \dots)$. Policy $\pi = (\pi_t(\cdot \mid h))$: collection of distributions on $\mathcal{A}$, $\forall h = (a_1, x_1, \dots, a_{t-1}, x_{t-1})$.

$$\mathbb{P}(A_t = a \mid (A_1, X_1, \dots, A_{t-1}, X_{t-1}) = h) = \pi_t(a \mid h).$$

Only source of randomness is the policy.

Performance Metric

The best decision may be *time-dependent*. Suggests performance metric:

$$\max_{v=(x_{t,a}) \in [0,1]^{nk}} \sum_{t=1}^n \left\{ \max_{a \in \mathcal{A}} x_{t,a} - \mathbb{E}_v^\pi[x_{t,A_t}] \right\}.$$

However, this gives Nature *much power*, i.e., one may not get sublinear regret.

EXAMPLE 3.1. Fix π . For all t , let α_t be an action that is selected with smallest probability, i.e., α_t minimises $\mathbb{P}^\pi(A_t = a)$ w.r.t. $a \in \mathcal{A}$. This depends on $\alpha_1, \dots, \alpha_{t-1}$ and $x_{1,a}, \dots, x_{t-1,a}$, for all $a \in \mathcal{A}$, but not on the actual data. Set $x_{t,\alpha_t} = 1$ and $x_{t,a} = 0$ for all $a \neq \alpha_t$. Then

$$\sum_{t=1}^n \left\{ \max_{a \in \mathcal{A}} x_{t,a} - \mathbb{E}_v^\pi[x_{t,A_t}] \right\} \geq \sum_{t=1}^n 1 \cdot \mathbb{P}^\pi(A_t \neq \alpha_t) \geq n \left(1 - \frac{1}{k} \right).$$

linear regret.

Alternative Performance Metric

Loss w.r.t. the *best-fixed-action-in-hindsight*,

$$R_v^\pi(n) := \left\{ \max_{a \in \mathcal{A}} \sum_{t=1}^n x_{t,a} \right\} - \mathbb{E}_v^\pi \left[\sum_{t=1}^n x_{t,A_t} \right], \quad R^\pi(n) := \sup_{v \in [0,1]^{nk}} R_v^\pi(n).$$

instance-dependent; instance-independent.

If you actually would have selected the best-action-in-hindsight upfront, Nature would have chosen different rewards. In fact, if you always play a fixed action $\tilde{a}$, then $x_{t,a} := \mathbb{1}_{\{a \neq \tilde{a}\}}$ for all t gives linear regret. So, it is a performance measure, but not without (interpretive) problems.

EXAMPLE 3.2. Let π be a policy, $\tilde{v}_1, \dots, \tilde{v}_k$ distributions on $[0, 1]$, and $\tilde{v} = (\tilde{v}_1, \dots, \tilde{v}_k)$. Let $X_{t,a}$ i.i.d. according to $\tilde{v}_a$, for all $t = 1, \dots, n$ and all $a \in \mathcal{A}$.

$$\begin{aligned} \text{Stochastic-}R_v^\pi(n) &= \max_{a \in \mathcal{A}} \mathbb{E}_{\tilde{v}}^\pi \left[\sum_{t=1}^n X_{t,a} - X_{t,A_t} \right] \leq \mathbb{E}_{\tilde{v}}^\pi \left[\max_{a \in \mathcal{A}} \sum_{t=1}^n \{X_{t,a} - X_{t,A_t}\} \right] \\ &= \text{Adversarial-}R_{\{X_{t,a}\}}^\pi(n) \leq \text{Adversarial-}R^\pi(n). \end{aligned}$$

The expectation of $X_{t,a}$ is independent of t .

If $Z_1, \dots, Z_k$ are random variables, then

$$Z_a \leq \max_a Z_a, \quad a.s., \Rightarrow \mathbb{E}[Z_a] \leq \mathbb{E} \left[\max_a Z_a \right], \Rightarrow \max_a \mathbb{E}[Z_a] \leq \mathbb{E} \left[\max_a Z_a \right].$$

Take $\sup_{\tilde{\nu}} \text{Stochastic-}R_{\tilde{\nu}}^{\pi}(n)$. Adversarial Multi-Armed Bandit (MAB) problem is *at least as difficult as* stochastic MAB, because Nature can always choose *i.i.d.* rewards according to some distributions.

Construction of a Policy

Idea. You need to randomise, and you want to *follow the leader*, i.e., select best-performing action.

Solution. Choose each arm with probability proportional to total reward up to that moment.

SECTION 3.2

| The EXP3 Algorithm

The **exponential** weighting for **exploration** and **exploitation** policy tuning parameter $\eta > 0$. For all t , choose $A_t = a$ with probability

$$P_{t,a} := \frac{\exp(\eta \hat{S}_{t-1,a})}{\sum_{i=1}^k \exp(\eta \hat{S}_{t-1,i})},$$

where $\hat{S}_{t-1,a}$ is an estimate of $\sum_{s=1}^{t-1} x_{s,a}$, the total reward gained from arm a up to time $t-1$, with $\hat{S}_{0,a} := 0$. If $\eta \approx 0$, then uniformly selected action implies exploration; if $\eta \rightarrow \infty$, we pick the best implies exploitation.

To estimate $\sum_{s=1}^t x_{s,a}$, we estimate $x_{t,a}$ for all actions, also the ones did not play. Let $y_{t,a} := 1 - x_{t,a}$, loss at time t , action a . Let

$$P_{t,a} := \mathbb{P}_v^\pi(A_t = a \mid \underbrace{A_1, X_1, \dots, A_{t-1}, X_{t-1}}_{=: H_{t-1}}) = \pi(a \mid H_{t-1}).$$

Suppose π is such that $P_{t,a} > 0$ a.s., for all t, a . Estimate $y_{t,a}$ by

$$\hat{Y}_{t,a} := \frac{(1 - X_t) \mathbb{1}_{\{A_t=a\}}}{P_{t,a}} = \begin{cases} \frac{y_{t,a}}{P_{t,a}}, & A_t = a \\ 0, & A_t \neq a \end{cases},$$

and $\hat{X}_{t,a} := 1 - \hat{Y}_{t,a}$. Compute this estimate after observing the reward in period t . Bias:

$$\mathbb{E}_v^\pi[\hat{Y}_{t,a} \mid H_{t-1}] = \frac{y_{t,a}}{P_{t,a}} \mathbb{P}_v^\pi(A_t = a \mid H_{t-1}) + 0 \cdot \mathbb{P}_v^\pi(A_t \neq a \mid H_{t-1}) = y_{t,a} \quad a.s.$$

Hence

$$\mathbb{E}_v^\pi[\hat{Y}_{t,a}] = \mathbb{E}_v^\pi[\mathbb{E}_v^\pi[\hat{Y}_{t,a} \mid H_{t-1}]] = y_{t,a}, \quad \mathbb{E}_v^\pi[\hat{X}_{t,a}] = 1 - y_{t,a} = x_{t,a},$$

unbiased estimator.

where $\hat{X}_{t,a}$ and $\hat{Y}_{t,a}$ of course do depend on X_t , but we don't put $\hat{X}_{t,A_t} = X_t = x_{t,A_t}$. We don't estimate rewards directly:

$$\hat{X}_{t,a} := \frac{X_t \mathbb{1}_{\{A_t=a\}}}{P_{t,a}},$$

because this estimator is unbounded from above, whereas $\hat{X}_{t,a} \leq 1$ a.s. If $P_{t,a}$ is small, the (conditional) variance of the estimates can be big:

$$\begin{aligned} \mathbb{E}_v^\pi \left[\left(\hat{X}_{t,a} - x_{t,a} \right)^2 \mid H_{t-1} \right] &= \mathbb{E}_v^\pi \left[\left(\hat{X}_{t,a} - y_{t,a} \right)^2 \mid H_{t-1} \right] \\ &= \left(\frac{y_{t,a}}{P_{t,a}} - y_{t,a} \right)^2 P_{t,a} + (0 - y_{t,a})^2 (1 - P_{t,a}) \\ &= y_{t,a}^2 \cdot \frac{(1 - P_{t,a})^2 + (1 - P_{t,a}) P_{t,a}}{P_{t,a}} = y_{t,a}^2 \cdot \frac{1 - P_{t,a}}{P_{t,a}} \quad a.s. \end{aligned}$$

If $P_{t,a} \approx 0$, the regret (before expectation) will have large variance. A solution is to estimate $y_{t,a}$ by

$$\hat{Y}_{t,a} := \frac{(1 - X_t) \mathbb{1}_{\{A_t=a\}}}{P_{t,a} + \gamma}$$

for some $\gamma > 0$, where we introduce bias to reduce the variance.

Regret Upper Bound

Let $\hat{S}_{t,a} = \sum_{s=1}^t \hat{X}_{s,a}$ for all t, a . Let $\eta > 0$, and, for all t , choose $A_t = a$ with probability

$$P_{t,a} := \frac{\exp(\eta \hat{S}_{t-1,a})}{\sum_{i=1}^k \exp(\eta \hat{S}_{t-1,i})}.$$

Observe $P_{t,a}$ is really the probability of $A_t = a$ conditionally on H_{t-1} , where $P_{t,a}$ depends on $\{\hat{S}_{t-1,a} : a \in A\}$, which recursively depends on $\{P_{s,a} : s < t, a \in A\}$.

THEOREM 3.3.

Let π be $\text{Exp3}(\eta)$. Then, for all $v \in [0, 1]^{nk}$, the regret

$$R_v^\pi(n) \leq \frac{\log k}{\eta} + \eta nk.$$

In particular, $\eta = \sqrt{\frac{\log k}{nk}}$ yields $R^\pi(n) \leq 2\sqrt{nk \log k}$.

Proof. Let $v \in [0, 1]^{nk}$. The regret is given by

$$R_v^\pi(n) = \max_{a \in \mathcal{A}} \left\{ \sum_{t=1}^n x_{t,a} - \sum_{t=1}^n \mathbb{E}_v^\pi[X_t] \right\}.$$

Observe that $x_{t,a} = \mathbb{E}_v^\pi[\hat{X}_{t,a}]$ and

$$\mathbb{E}_v^\pi[X_t] = \mathbb{E}_v^\pi \left[\mathbb{E}_v^\pi \left[\sum_{a=1}^k x_{t,a} \mathbb{1}_{\{A_t=a\}} \mid H_{t-1} \right] \right] = \mathbb{E}_v^\pi \left[\sum_{a=1}^k x_{t,a} P_{t,a} \right].$$

So that

$$R_v^\pi(n) = \max_{a \in \mathcal{A}} \left\{ \mathbb{E}_v^\pi \left[\sum_{t=1}^n \left\{ \hat{X}_{t,a} - \sum_{a=1}^k x_{t,a} P_{t,a} \right\} \right] \right\}.$$

For all $a \in A$,

$$\begin{aligned} \exp \left(\eta \sum_{t=1}^n \hat{X}_{t,a} \right) &= \exp(\eta \hat{S}_{n,a}) \leq \sum_{i=1}^k \exp(\eta \hat{S}_{n,i}) \\ &= \frac{\sum_{i=1}^k \exp(\eta \hat{S}_{n,i})}{\sum_{i=1}^k \exp(\eta \hat{S}_{n-1,i})} \cdot \frac{\sum_{i=1}^k \exp(\eta \hat{S}_{n-1,i})}{\sum_{i=1}^k \exp(\eta \hat{S}_{n-2,i})} \cdots \frac{\sum_{i=1}^k \exp(\eta \hat{S}_{1,i})}{\sum_{i=1}^k \exp(\eta \hat{S}_{0,i})} \cdot k \\ &= k \cdot \prod_{t=1}^n \sum_{i=1}^k \frac{\exp(\eta \hat{S}_{t-1,i})}{\sum_{i=1}^k \exp(\eta \hat{S}_{t-1,i})} \cdot \exp(\eta \hat{X}_{t,i}) = k \cdot \prod_{t=1}^n \sum_{i=1}^k P_{t,i} \exp(\eta \hat{X}_{t,i}) \\ &\leq k \cdot \prod_{t=1}^n \sum_{i=1}^k P_{t,i} \{1 + \eta \hat{X}_{t,i} + \eta^2 \hat{X}_{t,i}^2\} \leq k \cdot \prod_{t=1}^n \exp \left(\sum_{i=1}^k P_{t,i} \{ \eta \hat{X}_{t,i} + \eta^2 \hat{X}_{t,i}^2 \} \right). \end{aligned}$$

$\exp(z) \leq 1 + z + z^2$ if $z \leq 1$; $\sum_{i=1}^k P_{t,i} = 1$ and $1 + x \leq \exp(x)$.

Take log, divide by η :

$$\sum_{t=1}^n \hat{X}_{t,a} \leq \frac{\log k}{\eta} + \sum_{t=1}^n \sum_{i=1}^k P_{t,i} \{ \hat{X}_{t,i} + \eta \hat{X}_{t,i}^2 \} \quad a.s.$$

$$\Rightarrow \sum_{t=1}^n \hat{X}_{t,a} - \sum_{t=1}^n \sum_{i=1}^k P_{t,i} \hat{X}_{t,i} \leq \frac{\log k}{\eta} + \sum_{t=1}^n \sum_{i=1}^k P_{t,i} \eta \hat{X}_{t,i}^2 \quad a.s.$$

Now

$$\begin{aligned} \mathbb{E}_v^\pi [P_{t,i} \hat{X}_{t,i}^2 \mid H_{t-1}] &= P_{t,i} \cdot \left\{ P_{t,i} \left(\frac{1 - y_{t,i}}{P_{t,i}} \right)^2 + (1 - P_{t,i}) \cdot (1 - 0)^2 \right\} \\ &= (P_{t,i} - y_{t,i})^2 + (1 - P_{t,i}) \cdot P_{t,i} \leq 1 \quad a.s., \end{aligned}$$

and thus

$$R_v^\pi(n) = \max_{a \in A} \left\{ \mathbb{E}_v^\pi \left[\sum_{t=1}^n \left\{ \hat{X}_{t,a} - \sum_{a=1}^k x_{t,a} P_{t,a} \right\} \right] \right\} \leq \frac{\log k}{\eta} + \eta n k.$$

Choose $\eta = \sqrt{\frac{\log k}{nk}} \leq 1$ for all $n \geq 1, k \geq 2$. Then

$$\sup_v R_v^\pi(n) \leq 2\sqrt{nk \log k}.$$

□

Remark. (i) Close to $\sqrt{nk}$ upper bound, asymptotically near-optimal. Same as stochastic bandit, $\sqrt{\log k}$ can be removed by refined analysis.

(ii) Observe the exploration-exploitation trade-off in η , where η depends on n : doubling trick, or properly chosen η_t .

(iii) Policy also works for stochastic bandit.

(iv) Variance of

$$\max_a \left\{ \sum_{t=1}^n (x_{t,a} - X_t) \right\}$$

can be high. Remedy:

$$\hat{Y}_{t,a} := \frac{(1 - X_t) \mathbb{1}_{\{A_t=a\}}}{P_{t,a} + \gamma}$$

for some $\gamma > 0$.

(v) Benchmark of best-fixed-action-in-hindsight is not always informative. E.g. if $\mu_a(t)$ is slowly moving, you want to track the $\mu(t)$ processes.

SECTION 3.3

| Minimax Lower Bound by Vogel

Select the Highest-Mean Bernoulli Sample

Let $\mathfrak{B}$ be the class of Bernoulli probability distributions. Suppose you observe two *i.i.d.* Bernoulli samples:

$$\mathbf{Z} = (Z_1, \dots, Z_n) \quad \text{and} \quad \mathbf{Z}' = (Z'_1, \dots, Z'_n)$$

from Bernoulli random variables Z, Z' with *unknown* mean $p := \mathbb{E}[Z] \in [0, 1]$ and $q := \mathbb{E}[Z'] \in [0, 1]$. Based on $\mathbf{Z}, \mathbf{Z}'$, determine which of Z, Z' has the highest mean. Decisions are encoded by a *decision rule*:

$$d : \{0, 1\}^n \times \{0, 1\}^n \rightarrow [0, 1]$$

where $d(z, z')$ denotes the probability of selecting Z , and $1 - d(z, z')$ denotes the probability of selecting Z' , given data $z, z' \in \{0, 1\}^n$. The probability $P(d; p, q)$ of selecting Z , as function of d, p, q , is

For this calculation, write

$$Q_1 := \sum_{z \in \{0, 1\}^n} \sum_{z' \in \{0, 1\}^n} d(z, z') \cdot \mathbb{P}(\mathbf{Z} = z, \mathbf{Z}' = z')$$

Then

$$P(d; p, q) :$$

$$= Q_1 = \sum_{z \in \{0, 1\}^n} \sum_{z' \in \{0, 1\}^n} d(z, z') \cdot \prod_{i,j=1}^n p^{z_i} (1-p)^{1-z_i} q^{z'_j} (1-q)^{1-z'_j}.$$

The *loss* or *regret* of d , given p, q is

$$L(d; p, q) := (q - p)^+ P(d; p, q) + (p - q)^+ (1 - P(d; p, q)).$$

Let $\mathcal{D}_n$ be the class of all decision rules. A rule $d' \in \mathcal{D}_n$ is *minimax optimal* if

$$\sup_{(p,q) \in [0,1]^2} L(d'; p, q) = \inf_{d \in \mathcal{D}_n} \sup_{(p,q) \in [0,1]^2} L(d; p, q).$$

The natural decision rule $\mathfrak{d}_n$ is defined by

$$\mathfrak{d}_n(z, z') := \begin{cases} 1 & \text{if } \sum_{i=1}^n z_i > \sum_{i=1}^n z'_i \\ \frac{1}{2} & \text{if } \sum_{i=1}^n z_i = \sum_{i=1}^n z'_i \\ 0 & \text{if } \sum_{i=1}^n z_i < \sum_{i=1}^n z'_i \end{cases}$$

LEMMA 3.4 (Vogel, 1960).

The natural decision rule is *minimax optimal*.

Proof. (i) We can restrict to symmetric rules. Let $d \in \mathcal{D}_n$. Define its symmetrisation

$$d^\sigma(z, z') := \frac{1}{2} d(z, z') + \frac{1}{2} (1 - d(z', z)), \quad \forall z, z' \in \{0, 1\}^n.$$

Then

For this calculation, write

$$\mathcal{Q}_1 := \sum_{z \in \{0,1\}^n} \sum_{z' \in \{0,1\}^n} \left\{ \frac{1}{2}d(z, z') + \frac{1}{2}(1 - d(z', z)) \right\} \cdot \mathbb{P}(\mathbf{Z} = z, \mathbf{Z}' = z')$$

Then

$$\begin{aligned} P(d^\sigma; p, q) &= \sum_{z \in \{0,1\}^n} \sum_{z' \in \{0,1\}^n} d^\sigma(z, z') \cdot \mathbb{P}(\mathbf{Z} = z, \mathbf{Z}' = z') \\ &= \mathcal{Q}_1 = \frac{1}{2}P(d; p, q) + \frac{1}{2}(1 - P(d; q, p)) \end{aligned}$$

by interchanging z and z' . And thus

For this calculation, write

$\mathcal{Q}_1$

$$:= (q - p)^+ \left\{ \frac{1}{2}P(d; p, q) + \frac{1}{2}(1 - P(d; q, p)) \right\} + (p - q)^+ \left\{ \frac{1}{2}P(d; q, p) + \frac{1}{2}(1 - P(d; p, q)) \right\}$$

Then

$$\begin{aligned} L(d^\sigma; p, q) &= (q - p)^+ P(d^\sigma; p, q) + (p - q)^+ (1 - P(d^\sigma; p, q)) \\ &= \mathcal{Q}_1 = \frac{1}{2}L(d; p, q) + \frac{1}{2}L(d; q, p), \end{aligned}$$

and therefore

$$\begin{aligned} \sup_{p, q} L(d^\sigma; p, q) &= \sup_{p, q} \left\{ \frac{1}{2}L(d; p, q) + \frac{1}{2}L(d; q, p) \right\} \\ &\leq \sup_{p, q} \frac{1}{2}L(d; p, q) + \sup_{p, q} \frac{1}{2}L(d; q, p) = \sup_{p, q} L(d; p, q). \end{aligned}$$

Worst-case loss of a rule cannot increase by using its symmetrisation. We can restrict to symmetric rules: $d(z, z') = 1 - d(z', z)$ for all z, z' .

(ii) Total losses are sufficient statistics.

For this calculation, write

$$\mathcal{Q}_1 := \sum_{z \in \{0,1\}^n} \sum_{z' \in \{0,1\}^n} d(z, z') \prod_{i=1}^n p^{z_i} (1 - p)^{1 - z_i} q^{z'_i} (1 - q)^{1 - z'_i}$$

Then

$$\begin{aligned} P(d; p, q) &= \sum_{z \in \{0,1\}^n} \sum_{z' \in \{0,1\}^n} d(z, z') \cdot \mathbb{P}(\mathbf{Z} = z, \mathbf{Z}' = z') \\ &= \mathcal{Q}_1 = \sum_{k, l=0}^n p^k (1 - p)^{n-k} q^l (1 - q)^{n-l} \sum_{(z, z') \in \mathcal{Z}_{k, l}} d(z, z') \end{aligned}$$

where

$$\mathcal{Z}_{k, l} := \left\{ (z, z') \in \{0, 1\}^n \times \{0, 1\}^n : \sum_{i=1}^n z_i = k, \sum_{i=1}^n z'_i = l \right\}.$$

We can replace each $d(z, z')$ by its average

$$|\mathcal{Z}_{k, l}|^{-1} \sum_{(z, z') \in \mathcal{Z}_{k, l}} d(z, z')$$

over $\mathcal{Z}_{k,l}$, without affecting $P(d; p, q)$ or $L(d; p, q)$. Thus, a minimax rule is *w.l.o.g.* a function of the total losses only:

$$d(z, z') \equiv f\left(\sum_i z_i, \sum_i z'_i\right)$$

for some $f : \{0, 1, \dots, n\}^2 \rightarrow [0, 1]$,

$$P(d; p, q) = \sum_{k,l=0}^n p^k (1-p)^{n-k} q^l (1-q)^{n-l} \cdot f(k, l) \cdot \binom{n}{k} \binom{n}{l}.$$

(iii) Determine optimal f . Let

$$d(z, z') = f\left(\sum_i z_i, \sum_i z'_i\right)$$

be symmetric: $f(l, k) = 1 - f(k, l)$. Then $L(d; p, q)$ can be written as:

For this calculation, write

$$\begin{aligned} Q_1 &:= (q-p)^+ \sum_{k,l=0}^n p^k (1-p)^{n-k} q^l (1-q)^{n-l} \binom{n}{k} \binom{n}{l} f(k, l) \\ &\quad + (p-q)^+ \sum_{k,l=0}^n p^k (1-p)^{n-k} q^l (1-q)^{n-l} \binom{n}{k} \binom{n}{l} (1 - f(k, l)) \end{aligned}$$

$$\begin{aligned} Q_2 &:= \sum_{k=0}^n \pi_{k,k} \{(q-p)^+ f(k, k) + (p-q)^+ (1 - f(k, k))\} \\ &\quad + \sum_{k>l} \{\pi_{k,l} \{(q-p)^+ f(k, l) + (p-q)^+ (1 - f(k, l))\} \\ &\quad + \pi_{l,k} \{(q-p)^+ f(l, k) + (p-q)^+ (1 - f(l, k))\}\} \end{aligned}$$

Then

$$\begin{aligned} L(d; p, q) &= (q-p)^+ P(d; p, q) + (p-q)^+ (1 - P(d; p, q)) \\ &= Q_1 = Q_2. \end{aligned}$$

Now use symmetry of f : $f(k, k) = \frac{1}{2}$, $f(k, l) = 1 - f(l, k)$.

$$\begin{aligned} L(d; p, q) &= \frac{(q-p)^+ + (p-q)^+}{2} + \sum_{k=0}^n \pi_{k,k} \\ &\quad + (q-p)^+ \sum_{k>l} \{\pi_{k,l} f(k, l) + \pi_{l,k} (1 - f(k, l))\} \end{aligned} \tag{3.1}$$

$$+ (p-q)^+ \sum_{k>l} \{\pi_{k,l} (1 - f(k, l)) + \pi_{l,k} f(k, l)\}, \tag{3.2}$$

where term (3.1) smallest if $f(k, l) = 1$ for all $k > l$; term (3.2) smallest if $f(k, l) = 0$ for all $k < l$. Thus $\mathfrak{d}_n$ is optimal, for all p, q ,

$$\sup_{p,q} L(d; p, q) \geq \sup_{p,q} L(\mathfrak{d}_n; p, q), \quad \forall d \in \mathcal{D}_n.$$

Which *weights* $\pi_{\cdot, \cdot}$ are highest? This determines optimal f . Recall

$$\pi_{k,l} := \binom{n}{k} p^k (1-p)^{n-k} \binom{n}{l} q^l (1-q)^{n-l}.$$

If $q > p$ and $k > l$ then

$$\frac{\pi_{k,l}}{\pi_{l,k}} = \left(\frac{p}{q} \cdot \frac{1-q}{1-p}\right)^{k-l} < 1$$

and thus $\pi_{k,l} < \pi_{l,k}$. Similarly, if $p > q$ and $k > l$ then $\pi_{k,l} > \pi_{l,k}$. Thus, for each fixed $n \in \mathbb{N}$, the natural decision rule $\mathfrak{d}_n$ is minimax optimal.

Now suppose you have to select Z or Z' for each $n = 1, \dots, N$, some $N \in \mathbb{N}$. Then the *sequential* natural decision rule, that selects Z at stage n with probability $\mathfrak{d}_n(Z_1, \dots, Z_n, Z'_1, \dots, Z'_n)$, $\forall n = 1, \dots, N$, is minimax optimal:

$$\inf_{(d_n)_{1 \leq n \leq N}} \sup_{(p,q) \in [0,1]^2} \sum_{n=1}^N L(d_n; p, q) = \sup_{(p,q) \in [0,1]^2} \sum_{n=1}^N L(\mathcal{D}_n; p, q).$$

□

THEOREM 3.5 (Vogel, 1960).

Suppose that $K = 2$. Then, for all policies $\pi \in \Pi$ and all $T \in \mathbb{N}$, the regret

$$\sup_{v \in \mathfrak{B}^2} R_v^\pi(T) \geq 0.1873\sqrt{T} - 30.$$

Proof. (i) We can assume that each A_t is determined based on *all observations*

$$X_{1,1}, \dots, X_{1,t-1}, X_{2,1}, \dots, X_{2,t-1}$$

instead of $(X_{s,A_s} : 1 \leq s \leq t-1)$, which can only make the lower bound smaller.

(ii) Relate to the *sequential natural decision rule*.

For this calculation, write

$$\mathcal{Q}_1 := \sup_{0 \leq \mu_1, \mu_2 \leq 1} \sum_{t=2}^T \left\{ (\mu_1 - \mu_2)^+ \mathbb{P}_{(\mu_1, \mu_2)}^\pi(A_t = 2) + (\mu_2 - \mu_1)^+ \mathbb{P}_{(\mu_1, \mu_2)}^\pi(A_t = 1) \right\}$$

Then

$$\begin{aligned} \sup_{v \in \mathfrak{B}^2} R_v^\pi(T) &= \sup_{0 \leq \mu_1, \mu_2 \leq 1} \sum_{t=1}^T \left\{ \max\{\mu_1, \mu_2\} - \mathbb{E}_{(\mu_1, \mu_2)}^\pi[X_t] \right\} \\ &= \mathcal{Q}_1 = \sup_{0 \leq \mu_1, \mu_2 \leq 1} \sum_{t=2}^T L(d_{t-1}^\pi; \mu_1, \mu_2), \end{aligned}$$

where for all $t = 2, \dots, T$ and $z, z' \in \{0, 1\}^{t-1}$,

$$d_{t-1}^\pi(z, z') := \mathbb{P}_{(\mu_1, \mu_2)}^\pi(A_t = 1 \mid (X_{1,1}, \dots, X_{t-1,1}) = z; (X_{1,2}, \dots, X_{t-1,2}) = z').$$

Since the *sequential natural decision rule* is minimax optimal,

For this calculation, write

$$\mathcal{Q}_1 := \sup_{(\mu_1, \mu_2) \in [0,1]^2} \sum_{t=2}^T L(\mathfrak{d}_{t-1}; \mu_1, \mu_2)$$

Then

$$\begin{aligned} \sup_{v \in \mathfrak{B}^2} R_v^\pi(T) &\geq \mathcal{Q}_1 \geq \sup_{(\mu_1, \mu_2) \in [0,1]^2} \sum_{t=1}^{T-1} \left\{ (\mu_1 - \mu_2)^+ \mathbb{P}(\hat{\mu}_{2,t} > \hat{\mu}_{1,t}) + (\mu_2 - \mu_1)^+ \mathbb{P}(\hat{\mu}_{1,t} > \hat{\mu}_{2,t}) \right\}. \end{aligned}$$

(iii) Apply *Berry-Esseen theorem*. Let $\mu_1, \mu_2 \in (0, 1)$, write

$$\sigma_{\mu_1, \mu_2} := \sqrt{\mu_1(1 - \mu_1) + \mu_2(1 - \mu_2)}.$$

Then:

For this calculation, write

$$\begin{aligned} \mathcal{Q}_1 &:= \sum_{t=1}^{T-1} (\mu_1 - \mu_2)^+ P\left(\frac{1}{\sqrt{t}} \sum_{s=1}^t X_{2,s} - X_{1,s} - \mu_2 + \mu_1 > \sqrt{t} \frac{\mu_1 - \mu_2}{\sigma_{\mu_1, \mu_2}}\right) \\ &\quad + \sum_{t=1}^{T-1} (\mu_2 - \mu_1)^+ P\left(\frac{1}{\sqrt{t}} \sum_{s=1}^t X_{1,s} - X_{2,s} - \mu_1 + \mu_2 > \sqrt{t} \frac{\mu_2 - \mu_1}{\sigma_{\mu_1, \mu_2}}\right) \end{aligned}$$

Then

$$\begin{aligned} &\sum_{t=1}^{T-1} \{(\mu_1 - \mu_2)^+ P(\hat{\mu}_{2,t} > \hat{\mu}_{1,t}) + (\mu_2 - \mu_1)^+ P(\hat{\mu}_{1,t} > \hat{\mu}_{2,t})\} \\ &= \mathcal{Q}_1 \geq \sum_{t=1}^{T-1} |\mu_1 - \mu_2| \left[\Phi\left(-\sqrt{t} \frac{|\mu_1 - \mu_2|}{\sigma_{\mu_1, \mu_2}}\right) - |\mu_1 - \mu_2| \cdot C \cdot \frac{\rho_{\mu_1, \mu_2}}{\sigma_{\mu_1, \mu_2}^3 \sqrt{t}} \right], \end{aligned}$$

using

$$1 - \Phi(z) \equiv \Phi(-z), \quad \rho_{\mu_1, \mu_2} := \mathbb{E}[|X_{2,1} - X_{1,1} - \mu_2 + \mu_1|^3], \quad C = 0.4748,$$

and the *Berry-Esseen theorem* (Shevtsova, 2011): if Y_s i.i.d. $\sim Y$,

$$\sup_{x \in \mathbb{R}} \left| P\left(\frac{1}{\sqrt{t}} \sum_{s=1}^t \frac{Y_s - \mathbb{E}[Y]}{\sigma_Y} \leq x\right) - \Phi(x) \right| \leq C \cdot \frac{\mathbb{E}[|Y - \mathbb{E}[Y]|^3]}{\sigma_Y^3 \sqrt{t}}.$$

(iv) Choose μ_1, μ_2 . Write $\delta := |\mu_1 - \mu_2|$ and $\sigma := \sigma_{\mu_1, \mu_2}$.

$$\begin{aligned} \sum_{t=1}^{T-1} \delta \Phi\left(-\sqrt{t} \frac{\delta}{\sigma}\right) &\geq \int_1^T \delta \Phi\left(-\sqrt{t} \frac{\delta}{\sigma}\right) dt \\ &= \frac{2\sigma^2}{\delta} \int_0^{\frac{\sqrt{T}\delta}{\sigma}} y \Phi(-y) dy - \int_0^1 \delta \Phi\left(-\sqrt{t} \frac{\delta}{\sigma}\right) dt \geq 2\sigma\sqrt{T} \cdot \underbrace{\frac{1}{c} \int_0^c y \Phi(-y) dy}_{\text{we want this to be large}} - \delta. \end{aligned}$$

$$y = \frac{\sqrt{t}\delta}{\sigma}, dt = \frac{2\sigma^2}{\delta^2} y dy; c := \frac{\sqrt{T}\delta}{\sigma}.$$

Let

$$c^* := \arg \max \left\{ x^{-1} \int_0^x y \Phi(-y) dy : x > 0 \right\} \approx 1.2467.$$

Choose

$$\mu_1 = \frac{1}{2} + \frac{\delta}{2}, \mu_2 = \frac{1}{2} - \frac{\delta}{2}, \delta := \frac{c^*}{\sqrt{(c^*)^2 + 2T}} \in \left(0, \frac{1}{2}\right).$$

Then

$$\sigma_{\mu_1, \mu_2} = \sqrt{\frac{1 - \delta^2}{2}}, \quad c = \frac{\sqrt{T}\delta}{\sigma_{\mu_1, \mu_2}} = c^*,$$

and therefore

$$2\sigma\sqrt{T} \cdot \frac{1}{c} \int_0^c y \Phi(-y) dy \geq 2\sigma_{\mu_1, \mu_2} \cdot \sqrt{T} \cdot 0.132447 \geq 0.1873\sqrt{T} - 0.1457T^{-1/2}$$

using

$$\sigma_{\mu_1, \mu_2} \geq \sqrt{\frac{1}{2}} - \sqrt{\frac{1}{2}}\delta^2, \quad -\delta^2 \geq \frac{(c^*)^2}{2T}.$$

It follows that

$$\sup_{v \in \mathbb{B}^2} R_v^\pi(T) \geq 0.1873\sqrt{T} - c_T,$$

where, with the choice of μ_1, μ_2 , and δ , the remainder term satisfies

$$c_T := \sum_{t=1}^{T-1} \delta \cdot C \cdot \frac{\rho_{\mu_1, \mu_2}}{\sigma_{\mu_1, \mu_2}^3 \sqrt{t}} + \delta + 0.1457T^{-\frac{1}{2}} \leq 30.$$

□

Remark. (i) The proof assumes that you observe all rewards. In this case, the natural decision rule is minimax optimal.

(ii) Berry-Esseen relates Bernoulli probabilities to a Gaussian integral. Worst case attained at $\mu_1, \mu_2 = \frac{1}{2} \pm \delta$, for $\delta \sim T^{-\frac{1}{2}}$.

More generally:

(iii) The minimax rate $\sqrt{T}$ cannot be improved, even in a simple 2-armed Bernoulli bandit.

(iv) (Refinements of) UCB have the asymptotically optimal dependence on T .

(v) The constant 0.1873 can be improved to 0.307 (Bather 1985) and perhaps to 0.317.

(vi) The technique of Vogel does not extend to k arms.

SECTION 3.4

| Kullback-Leibler Divergence

Let $\emptyset \neq S \subset \mathbb{R}$ be *discrete* set: $S = \{s_1, \dots\}$. Let $\mathcal{P}(S)$ be the collection of all probability distributions on $(S, 2^S)$. For $p \in \mathcal{P}(S)$ write $p(x) := p(\{x\})$ for the *probability mass function*. If $f : S \rightarrow \mathbb{R}$, write

$$\mathbb{E}_{X \sim p}[f(X)] := \sum_{x \in S} p(x)f(x).$$

If $p, q \in \mathcal{P}(S)$ and $p(x) = 0$ whenever $q(x) = 0$, write $p \ll q$: p is *absolutely continuous w.r.t. q* . For all $p, q \in \mathcal{P}(S)$ such that $p \ll q$, define the *Kullback–Leibler divergence*:

$$\text{KL}(p, q) := \sum_{x \in S: p(x) > 0} p(x) \log \frac{p(x)}{q(x)}.$$

Convention: $0 \log \frac{0}{x} := 0$ for all $x \geq 0$, and $\text{KL}(p, q) := \infty$ if $p \not\ll q$. Then $\text{KL}(p, q)$ defined for all $p, q \in \mathcal{P}(S)$, and

$$\text{KL}(p, q) = \sum_{x \in S} p(x) \log \frac{p(x)}{q(x)}.$$

EXAMPLE 3.6. If p and q Bernoulli distributions on $S = \{0, 1\}$ then

$$\text{KL}(p, q) = p(0) \log \frac{p(0)}{q(0)} + p(1) \log \frac{p(1)}{q(1)}.$$

If p and q geometrically distributed with parameters λ, μ then

$$\text{KL}(p, q) = \sum_{k=1}^{\infty} \lambda(1-\lambda)^{k-1} \log \frac{\lambda(1-\lambda)^{k-1}}{\mu(1-\mu)^{k-1}}.$$

We focus on probability mass functions (S discrete), but you can also define KL divergence for probability densities (replace $\sum$ by $\int$) or general probability distributions:

$$\text{KL}(p, q) = \int \log \left(\frac{dp}{dq} \right) dp.$$

Remark. (i) KL divergence is also called *relative entropy* from q to p . Notations: $\text{KL}(p, q)$, $D_{\text{KL}}(P \parallel Q)$, $D(P, Q)$.

(ii) You can think of KL-divergence as an expected log-likelihood ratio.

(iii) $\text{KL}(p, q)$ is not symmetric and does not satisfy the triangle inequality. So $\text{KL}(p, q)$ is not a proper distance measure between probability distributions.

(iv) There are many other divergences (e.g. by replacing $\log$ by some f).

LEMMA 3.7 (Gibbs' Inequality).

$\text{KL}(p, q) \geq 0$ for all $p, q \in \mathcal{P}(S)$.

Proof. Let $p, q \in \mathcal{P}(S)$ and let $f(y) := y \log y$, for $y > 0$. The case $p \not\ll q$ is trivial; assume $p \ll q$. Since f is convex, Jensen's inequality implies

$$\text{KL}(p, q) = \sum_{x \in S: p(x) > 0} q(x) \cdot \frac{p(x)}{q(x)} \cdot \log \left(\frac{p(x)}{q(x)} \right) = \sum_{x \in S: p(x) > 0} q(x) f\left(\frac{p(x)}{q(x)}\right) \geq f\left(\sum_{x \in S: p(x) > 0} q(x) \frac{p(x)}{q(x)}\right)$$

$$= f(1) = 0.$$

□

KL-divergence between $\text{Ber}(x)$ and $\text{Ber}(y)$ distributions is denoted by

$$\text{kl}(x, y) := x \log \frac{x}{y} + (1 - x) \log \frac{1 - x}{1 - y}, \quad (x, y) \in [0, 1] \times (0, 1),$$

where

$$\text{kl}(x, y) = \begin{cases} 0, & x = y = 0 \text{ or } x = y = 1 \\ \infty, & \text{otherwise} \end{cases}.$$

LEMMA 3.8.

$KL(p, q) \geq \text{kl}(p(A), q(A))$ for all $A \subset S$ and $p, q \in \mathcal{P}(S)$.

Proof. Let $p, q \in \mathcal{P}(S)$ and $A \subset S$. We prove the case $0 < p(A), q(A) < 1$.

For this calculation, write

$$\begin{aligned} Q_1 &:= p(A) \underbrace{\sum_{x \in A} \frac{p(x)}{p(A)} \log \frac{p(x)/p(A)}{q(x)/q(A)}}_{\geq 0} + \sum_{x \in A} p(x) \log \frac{p(A)}{q(A)} \\ &\quad + p(A^c) \underbrace{\sum_{x \in A^c} \frac{p(x)}{p(A^c)} \log \frac{p(x)/p(A^c)}{q(x)/q(A^c)}}_{\geq 0} + \sum_{x \in A^c} p(x) \log \frac{p(A^c)}{q(A^c)} \end{aligned}$$

Then

$$\begin{aligned} KL(p, q) \\ = Q_1 \geq \text{kl}(p(A), q(A)). \end{aligned}$$

□

LEMMA 3.9 (Pinsker's Inequality).

$KL(p, q) \geq 2(p(A) - q(A))^2$ for all $A \subset S$ and $p, q \in \mathcal{P}(S)$.

Proof. Sufficient to prove $f(x, y) := \text{kl}(x, y) - 2(x - y)^2 \geq 0$ for all $x, y \in [0, 1]$. The case $y \in \{0, 1\}$ is trivial. Assume $x \in [0, 1], y \in (0, 1)$. Since

$$\frac{\partial f(x, y)}{\partial y} = \frac{1 - 4y(1 - y)}{y(1 - y)}(y - x),$$

the *mean value theorem* implies there is a $\tilde{y}$ between x and y such that

$$f(x, y) = f(x, x) + \frac{\partial f(x, \tilde{y})}{\partial y}(y - x) = \frac{1 - 4\tilde{y}(1 - \tilde{y})}{\tilde{y}(1 - \tilde{y})}(\tilde{y} - x)(y - x) \geq 0.$$

Let $p_n, q_n \in \mathcal{P}(S^n)$, for all $n = 1, \dots, N$ and some $N \in \mathbb{N}$, with

$$\sum_{x_n \in S} p_n(\mathbf{x}_{n-1}, x_n) = p_{n-1}(\mathbf{x}_{n-1})$$

for all $\mathbf{x}_{n-1} = (x_1, \dots, x_{n-1}) \in S^{n-1}$ and all $x_n \in S$. For all $n = 2, \dots, N$ and $\mathbf{x}_{n-1} \in S^{n-1}$ such that $p_{n-1}(\mathbf{x}_{n-1}) > 0$, define the conditional probability distribution $p_n(\cdot \mid \mathbf{x}_{n-1}) \in \mathcal{P}(S)$ by

$$p_n(A \mid \mathbf{x}_{n-1}) := \sum_{x \in A} \frac{p_n(\mathbf{x}_{n-1}, x)}{p_{n-1}(\mathbf{x}_{n-1})}, \quad \forall A \subset S.$$

Define $q_n(\cdot \mid \mathbf{x}_{n-1})$ in a similar manner, for all $n \in [N]$ and $\mathbf{x}_{n-1} \in S^{n-1}$.

□

LEMMA 3.10 (Chain Rule for KL-Divergence).

Sum of the expected KL divergences of the conditional distributions given random histories:

$$KL(p_N, q_N) = KL(p_1, q_1) + \sum_{n=2}^N \mathbb{E}_{\mathbf{x}_{n-1} \sim p_{n-1}} [KL(p_n(\cdot | \mathbf{x}_{n-1}), q_n(\cdot | \mathbf{x}_{n-1}))].$$

Proof. For all $n \in \{2, \dots, N\}$,

For this calculation, write

$$\begin{aligned} Q_1 &:= \sum_{\mathbf{x}_{n-1} \in S^{n-1}} p_{n-1}(\mathbf{x}_{n-1}) \sum_{x_n \in S} p_n(x_n | \mathbf{x}_{n-1}) \log \frac{p_n(x_n | \mathbf{x}_{n-1})}{q_n(x_n | \mathbf{x}_{n-1})} \\ &\quad + \sum_{\mathbf{x}_{n-1} \in S^{n-1}} p_{n-1}(\mathbf{x}_{n-1}) \sum_{x_n \in S} p_n(x_n | \mathbf{x}_{n-1}) \log \frac{p_{n-1}(\mathbf{x}_{n-1})}{q_{n-1}(\mathbf{x}_{n-1})} \end{aligned}$$

Then

$$\begin{aligned} KL(p_n, q_n) &= \sum_{(\mathbf{x}_{n-1}, x_n) \in S^n} p_n(\mathbf{x}_{n-1}, x_n) \log \frac{p_n(\mathbf{x}_{n-1}, x_n)}{q_n(\mathbf{x}_{n-1}, x_n)} \\ &= \sum_{(\mathbf{x}_{n-1}, x_n) \in S^n: p_{n-1}(\mathbf{x}_{n-1}) > 0} p_n(\mathbf{x}_{n-1}, x_n) \log \frac{p_n(x_n | \mathbf{x}_{n-1}) p_{n-1}(\mathbf{x}_{n-1})}{q_n(x_n | \mathbf{x}_{n-1}) q_{n-1}(\mathbf{x}_{n-1})} \\ &= Q_1 = \mathbb{E}_{\mathbf{x}_{n-1} \sim p_{n-1}} [KL(p_n(\cdot | \mathbf{x}_{n-1}), q_n(\cdot | \mathbf{x}_{n-1}))] + KL(p_{n-1}, q_{n-1}). \end{aligned}$$

The lemma follows by iteratively applying this equality. $\square$

COROLLARY 3.11.

Suppose that, for all $\mathbf{x}_n \in S^n$ and all $n = 2, \dots, N$,

$$p_n(\mathbf{x}_n) = \prod_{i=1}^n p_1(x_i) \quad \text{and} \quad q_n(\mathbf{x}_n) = \prod_{i=1}^n q_1(x_i) \quad \text{are independent.}$$

Then

$$p_n(\cdot | \mathbf{x}_{n-1}) \equiv p_1(\cdot) \quad \text{and} \quad q_n(\cdot | \mathbf{x}_{n-1}) \equiv q_1(\cdot),$$

for all $\mathbf{x}_{n-1}$ and all $n = 2, \dots, N$, and

$$KL(p_N, q_N) = KL(p_1, q_1) + \sum_{n=2}^N \mathbb{E}_{\mathbf{x}_{n-1} \sim p_{n-1}} [KL(p_n(\cdot | \mathbf{x}_{n-1}), q_n(\cdot | \mathbf{x}_{n-1}))] = N \cdot KL(p_1, q_1).$$

The KL-divergence of the distributions of two N -sized independent samples is simply N times the KL-divergence between the two sampling distributions.

EXAMPLE 3.12 (Application to a Hypothesis Test). Suppose $\mathbf{Y} = (Y_1, \dots, Y_n)$ i.i.d. sample from $\text{Ber}(\lambda)$, unknown $\lambda \in [0, 1]$. We want to know whether $\lambda \geq \mu = 0.50$. Let $r : \{0, 1\}^n \rightarrow \{\text{yes}, \text{no}\}$ be our decision rule. Let p and q be the probability distribution of $\mathbf{Y}$ under λ and μ , respectively. If $\lambda \geq 0.51$, we want $p(\{r(\mathbf{Y}) = \text{yes}\}) \geq 0.95$ and $q(\{r(\mathbf{Y}) = \text{yes}\}) \leq 0.05$. We currently have $n = 8000$ samples. Is that enough?

Proof. Let $A = \{r(\mathbf{Y}) = \text{yes}\}$. By Pinsker's inequality and independence,

$$2(0.90)^2 \leq 2(p(A) - q(A))^2 \leq n \text{kl}(\lambda, 0.5), \quad \forall \lambda \in [0.51, 1].$$

Since $\text{kl}(\lambda, 0.5)$ is increasing in λ , on $\lambda \geq 0.5$, we obtain

$$n \geq \frac{2(0.90)^2}{\text{kl}(0.51, 0.50)} \approx 8100.$$

Hence, our error requirements cannot be satisfied with 8000 samples.

□

SECTION 3.5

| Regret Lower Bounds

LEMMA 3.13 (Bretagnolle-Huber-Carol Inequality).

For all $A \subset S$ and all $p, q \in \mathcal{P}(S)$,

$$p(A) + q(A^c) \geq \frac{1}{2} \exp(-KL(p, q)),$$

where $\exp(-\infty) := 0$.

Proof. Since $KL(p, q) \geq \text{kl}(p(A), q(A))$ it suffices to prove

$$p + 1 - q \geq \frac{1}{2} \exp(-\text{kl}(p, q)) \quad \text{for all } p, q \in [0, 1].$$

We prove the case $p, q \in (0, 1)$. Sufficient to prove:

$$f(p, q) := -\text{kl}(p, q) - \log(2(p + 1 - q)) \leq 0 \quad \forall p, q \in (0, 1).$$

Since

$$\frac{\partial f(p, q)}{\partial q} = \frac{2p(\frac{p+1}{2} - q)}{q(1-q)(p+1-q)} > 0 \quad \text{if } q < \frac{p+1}{2},$$

it follows that, for all $p \in (0, 1)$,

$$f(p, q) \leq f\left(p, \frac{p+1}{2}\right) \leq -\log(p+1) < 0.$$

□

Consider the k -armed MAB with $v = (v_1, \dots, v_k)$ vector of reward distributions and fixed policy π . Suppose rewards have a pmf $p_a^v(x) = v_a(\{x\})$ for x in a countable set $\mathcal{X}$. Let $\mathbb{P}_v^\pi$ denote the distribution of $H_n = \{(A_s, X_s) : 1 \leq s \leq n\}$. Then

$$p_v^\pi(a_1, x_1, \dots, a_n, x_n) := \mathbb{P}_v^\pi(\{(A_1, X_1, \dots, A_n, X_n) = (a_1, x_1, \dots, a_n, x_n)\})$$

is the probability mass function of H_n , taking values in $(\mathcal{A} \times \mathcal{X})^n$.

For this calculation, write

$$Q_1 := \pi_1(a_1 \mid \emptyset) \cdot p_{a_1}^v(x_1) \cdot \pi_2(a_2 \mid a_1, x_1) \cdot p_{a_2}^v(x_2) \cdots \pi_n(a_n \mid a_1, x_1, \dots, a_{n-1}, x_{n-1}) \cdot p_{a_n}^v(x_n)$$

Then

$$\begin{aligned} p_v^\pi(a_1, x_1, \dots, a_n, x_n) \\ = Q_1 = \prod_{t=1}^n \pi_t(a_t \mid \{(a_s, x_s)\}_{1 \leq s \leq t-1}) p_{a_t}^v(x_t). \end{aligned}$$

Now consider two vectors v and v' of discrete reward distributions. Let $\mathbb{P}_v^\pi, \mathbb{P}_{v'}^\pi$ be the distribution of H_n under v resp. v' .

LEMMA 3.14 (Divergence Decomposition).

$$KL(\mathbb{P}_v^\pi, \mathbb{P}_{v'}^\pi) = \sum_{i=1}^K KL(v_i, v'_i) \cdot \mathbb{E}_v^\pi[T_i(n)].$$

Proof.

$$\begin{aligned}
\text{KL}(\mathbb{P}_v^\pi, \mathbb{P}_{v'}^\pi) &= \sum_{(a_s, x_s)_{1 \leq s \leq n}} p_v^\pi(a_1, x_1, \dots, a_n, x_n) \log \frac{p_v^\pi(a_1, x_1, \dots, a_n, x_n)}{p_{v'}^\pi(a_1, x_1, \dots, a_n, x_n)} \\
&= \mathbb{E}_v^\pi \left[\log \frac{\prod_{t=1}^n \pi_t(A_t \mid (A_s, X_s)_{1 \leq s \leq t-1}) p_{A_t}^v(X_t)}{\prod_{t=1}^n \pi_t(A_t \mid (A_s, X_s)_{1 \leq s \leq t-1}) p_{A_t}^{v'}(X_t)} \right] = \mathbb{E}_v^\pi \left[\sum_{t=1}^n \log \frac{p_{A_t}^v(X_t)}{p_{A_t}^{v'}(X_t)} \right] \\
&= \sum_{t=1}^n \sum_{i \in A} \mathbb{E}_v^\pi \left[\log \frac{p_{A_t}^v(X_t)}{p_{A_t}^{v'}(X_t)} \mid A_t = i \right] \mathbb{P}_v^\pi(A_t = i) = \sum_{i \in A} \text{KL}(v_i, v'_i) \mathbb{E}_v^\pi \left[\sum_{t=1}^n \mathbb{1}_{\{A_t=i\}} \right].
\end{aligned}$$

□

THEOREM 3.15.

Let $k \geq 2$ and $n \geq k - 1$. For all policies π there are 1-subgaussian $(v_1, \dots, v_k) = v$ such that the regret

$$R_v^\pi(n) \geq c \cdot \sqrt{nk}$$

for some $c > 0$ independent of π, v, n, k .

Proof. Fix π , let $\Delta \in (0, \frac{1}{2})$. Consider Bernoulli rewards with mean

$$\mu = \left(\frac{1}{2}, \frac{1}{2} - \Delta, \dots, \frac{1}{2} - \Delta \right)$$

where arm 1 is optimal. Let $i^* \in \arg \min_{j=2, \dots, k} \mathbb{E}_\mu^\pi[T_j(n)]$ be a least-chosen arm (in expectation),

$$\mu' = \left(\frac{1}{2}, \frac{1}{2} - \Delta, \dots, \frac{1}{2} - \Delta, \underbrace{\frac{1}{2} + \Delta}_{\text{position } i^*}, \frac{1}{2} - \Delta, \dots, \frac{1}{2} - \Delta \right)$$

where i^* is optimal. Using the regret decomposition and conditioning on $T_1(n) < \frac{n}{2}$,

$$R_\mu^\pi(n) \geq \mathbb{P}_\mu^\pi \left(T_1(n) < \frac{n}{2} \right) \frac{n\Delta}{2} \quad \text{and} \quad R_{\mu'}^\pi(n) \geq \mathbb{P}_{\mu'}^\pi \left(T_1(n) \geq \frac{n}{2} \right) \frac{n\Delta}{2}.$$

Define the event $A = \{T_1(n) < \frac{n}{2}\}$.

$$R_\mu^\pi(n) + R_{\mu'}^\pi(n) \geq (\mathbb{P}_\mu^\pi(A) + \mathbb{P}_{\mu'}^\pi(A^c)) \cdot \frac{n\Delta}{2} \geq \frac{1}{2} \exp(-\text{KL}(\mathbb{P}_\mu^\pi, \mathbb{P}_{\mu'}^\pi)) \cdot \frac{n\Delta}{2}.$$

BHC inequality.

Now, by the divergence decomposition lemma:

$$\text{KL}(\mathbb{P}_\mu^\pi, \mathbb{P}_{\mu'}^\pi) = \sum_{i=1}^k \text{kl}(\mu_i, \mu'_i) \cdot \mathbb{E}_\mu^\pi[T_i(n)] = \text{kl}\left(\frac{1}{2} - \Delta, \frac{1}{2} + \Delta\right) \cdot \mathbb{E}_\mu^\pi[T_{i^*}(n)] \leq \frac{4\Delta^2}{(\frac{1}{2} - \Delta)^2} \cdot \frac{n}{k-1},$$

using

$$\text{kl}\left(\frac{1}{2} - \Delta, \frac{1}{2} + \Delta\right) \leq \frac{4\Delta^2}{(\frac{1}{2} - \Delta)^2}$$

and

$$n = \sum_{i=1}^k \mathbb{E}_\mu^\pi[T_i(n)] \geq (k-1) \mathbb{E}_\mu^\pi[T_{i^*}(n)],$$

we have

$$\max_{\mu \in [0,1]^k} R_\mu^\pi(n) \geq \frac{R_\mu^\pi(n) + R_{\mu'}^\pi(n)}{2} \geq \frac{1}{4} \exp\left(-\frac{4\Delta^2}{(\frac{1}{2} - \Delta)^2} \cdot \frac{n}{k-1}\right) \cdot \frac{n\Delta}{2}.$$

Choose

$$\Delta = \frac{1}{4} \sqrt{\frac{(k-1)}{n}} \in \left(0, \frac{1}{4}\right].$$

Then

$$\max_{\mu \in [0,1]^k} R_{\mu}^{\pi}(n) \geq \frac{1}{4} \exp(-4) \cdot \frac{n^{\frac{1}{4}} \sqrt{\frac{(k-1)}{n}}}{2} \geq \frac{\exp(-4)}{32\sqrt{2}} \sqrt{kn}$$

using

$$\sqrt{k-1} \geq \frac{1}{\sqrt{2}} \sqrt{k}.$$

□

Remark. (i) We create two difficult instances μ, μ' means are close, but optimal action changes from 1 to $i^* \neq 1$. The idea of creating a difficult instance is standard in regret-lower-bounds-proof.

(ii) $R_{\mu}^{\pi}(n)$ is bounded from below in terms of $\mathbb{P}_{\mu}^{\pi}(T_1(n) < \frac{n}{2})$; $R_{\mu'}^{\pi}(n)$ is bounded from below in terms of $\mathbb{P}_{\mu'}^{\pi}(T_1(n) \geq \frac{n}{2})$.

(iii) Bretagnolle-Huber-Carol inequality (BHC) and divergence decomposition lemma translates this into KL-divergence between μ and μ' . Observe that a strong statement (about all policies) follows relatively quickly from KL machinery.

(iv) Optimise how close μ and μ' should be.

(v) We have proven a bound on 1-subgaussian reward vectors.

(vi) The optimal constant $c \cdot \sqrt{kn}$ is still unknown.

Instance-Dependent Lower Bound

For all v with corresponding optimal action a^* , the policy with

$$\pi_t(a^* | h) = 1, \quad \forall t, h,$$

has $R_v^{\pi}(n) = 0$ for all n . However, for all v such that a^* is suboptimal, this policy has linear regret $R_v^{\pi}(n)$. For informative v -dependent lower-bounds we restrict the policy space.

DEFINITION 3.16.

A policy π is called *consistent* over a class $\mathcal{V}$ if

$$\lim_{n \rightarrow \infty} \frac{R_v^{\pi}(n)}{n^p} = 0$$

for all $p > 0$ and all $v \in \mathcal{V}$.

EXAMPLE 3.17. For all v , there is a C_v with $R_v^{\pi}(n) \leq C_v \log n$ for all $n \geq 2$.

In the worst-case bound, we *slightly changed* a reward distribution v_a to v'_a , and needed to know $\text{KL}(v_a, v'_a)$. Let $\mathcal{M}$ be a class of distributions with support $S \subseteq \mathbb{R}$ and finite mean. Let

$$\mu(P) := \int_S x dP(x)$$

denote the mean of P . For all $P \in \mathcal{M}$ and $\mu^* > \mu(P)$, define

$$d_{\inf}(P, \mu^*, \mathcal{M}) := \inf_{P' \in \mathcal{M}} \{\text{KL}(P, P') : \mu(P') > \mu^*\}.$$

Smallest change in KL-divergence needed to increase the mean to μ^* .

EXAMPLE 3.18. $\mathcal{M} = \mathcal{B}$, P has mean $\mu(P) = x \in [0, 1]$. For all $\mu^* \in (x, 1]$,

$$d_{\inf}(P, \mu^*, \mathcal{B}) = \inf_{P' \in \mathcal{M}} \{kl(\mu(P), \mu(P')) : \mu(P') > \mu^*\} = \inf_{y \in [0, 1]} \{kl(x, y) : y > \mu^*\} = kl(x, \mu^*).$$

Let $\mathcal{V} = \mathcal{M}_1 \times \dots \times \mathcal{M}_k$, and let π be a consistent policy over $\mathcal{V}$.

THEOREM 3.19.

For all $v \in \mathcal{V}$, with corresponding μ^* and $(\Delta_i : i \in \mathcal{A})$,

$$\liminf_{n \rightarrow \infty} \frac{R_v^\pi(n)}{\log(n)} \geq \sum_{i \in \mathcal{A} : \Delta_i > 0} \frac{\Delta_i}{d_{\inf}(v_i, \mu^*, \mathcal{M}_i)}.$$

Bernoulli rewards: $\mathcal{M}_1 = \dots = \mathcal{M}_k = \mathcal{B}$:

$$\liminf_{n \rightarrow \infty} \frac{R_v^\pi(n)}{\log(n)} \geq \sum_{i \in \mathcal{A} : \Delta_i > 0} \frac{\Delta_i}{kl(\mu_i, \mu^*)} \geq \sum_{i \in \mathcal{A} : \Delta_i > 0} \frac{\mu^*(1 - \mu^*)}{\Delta_i}$$

using

$$kl(x, y) \leq \frac{(x - y)^2}{y(1 - y)}.$$

Thus indeed: small Δ_i gives larger regret. The KL-UCB policy for Bernoulli rewards has asymptotically optimal:

$$\limsup_{n \rightarrow \infty} \frac{R_v^\pi(n)}{\log(n)} \leq \sum_{i \in \mathcal{A} : \Delta_i > 0} \frac{\Delta_i}{kl(\mu_i, \mu^*)}.$$

Proof. Fix v and consistent policy π . Let $d_i := d_{\inf}(v_i, \mu^*, \mathcal{M}_i)$ for all suboptimal arms i . Observe:

$$\liminf_{n \rightarrow \infty} \frac{R_v^\pi(n)}{\log(n)} = \sum_{i : \Delta_i > 0} \Delta_i \cdot \underbrace{\liminf_{n \rightarrow \infty} \frac{\mathbb{E}_v^\pi[T_i(n)]}{\log(n)}}_{\text{To show: } \geq \frac{1}{d_i}}.$$

Let i be a suboptimal arm, let $\epsilon > 0$. Let v'_i be such that $KL(v_i, v'_i) \leq d_i + \epsilon$ and $\mu'_i := \mu(v'_i) > \mu^*$. Let $v'_j = v_j$ for all $j \neq i$, write $v' = (v'_1, \dots, v'_k)$. We have made a suboptimal arm optimal, at 'costs' approximately $d_i + \epsilon$.

$$\begin{aligned} & R_v^\pi(n) + R_{v'}^\pi(n) \\ & \geq \Delta_i \frac{n}{2} \mathbb{P}_v^\pi\left(T_i(n) \geq \frac{n}{2}\right) + (\mu'_i - \mu^*) \frac{n}{2} \mathbb{P}_{v'}^\pi\left(T_i(n) \leq \frac{n}{2}\right) \\ & \geq \frac{n}{2} \min\{\Delta_i, \mu'_i - \mu^*\} \frac{1}{2} \exp(-KL(\mathbb{P}_v^\pi, \mathbb{P}_{v'}^\pi)) \\ & = \frac{n}{4} \min\{\Delta_i, \mu'_i - \mu^*\} \exp(-\mathbb{E}_v^\pi[T_i(n)] KL(v_i, v'_i)) \geq \frac{n}{4} \min\{\Delta_i, \mu'_i - \mu^*\} \exp(-\mathbb{E}_v^\pi[T_i(n)](d_i + \epsilon)). \end{aligned}$$

Bretagnolle-Huber-Carol inequality; Divergence decomposition; by construction.

$$\mathbb{E}_v^\pi[T_i(n)] \geq \frac{1}{(d_i + \epsilon)} \log\left(\frac{n \min\{\Delta_i, \mu'_i - \mu^*\}}{4(R_v^\pi(n) + R_{v'}^\pi(n))}\right)$$

Then

$$\liminf_{n \rightarrow \infty} \frac{\mathbb{E}_v^\pi[T_i(n)]}{\log(n)} \geq \frac{1}{(d_i + \epsilon)} \liminf_{n \rightarrow \infty} \left(1 - \frac{\log(R_v^\pi(n) + R_{v'}^\pi(n))}{\log(n)} \right) = \frac{1}{(d_i + \epsilon)}$$

since for all $p > 0$ there is a $C_p > 0$ s.t. $R_v^\pi(n) \leq C_p n^p$ for all n . Because $\epsilon > 0$ is arbitrary, the result follows. $\square$

Bayesian Bandits, Gittins Index, and Thompson Sampling

SECTION 4.1

| Bayesian Bandits

Frequentist Viewpoint. There is an unknown, non-random parameter μ out there. Problem-dependent regret $R_\pi(T; \mu)$. Worst-case regret $\sup_{\mu \in [0,1]^K} R_\pi(T; \mu)$.

Bayesian Viewpoint. There is an unknown, random parameter μ out there. Bayesian regret $\mathbb{E}_\mu[R_\pi(T; \mu)]$. Two common flavours: finite T , or infinite T with discounting.

Beta-Bernoulli Bandit with $T < \infty$

K -armed Bernoulli-bandit, means $\mu_1, \dots, \mu_K \in (0, 1)$. Each μ_a is a random variable, with prior distribution $\pi_a \sim \text{Beta}(n_a, m_a)$, for $n_a, m_a \in \mathbb{N}$.

$$\pi_a(x) = \frac{x^{n_a-1}(1-x)^{m_a-1}}{B(n_a, m_a)}, \quad x \in (0, 1).$$

Bayes' rule: if $y_{a,1}, \dots, y_{a,t} \sim Y_a$, then, for $x \in (0, 1)$,

$$\underbrace{\pi_a(x \mid y_{a,1}, \dots, y_{a,t})}_{\text{posterior}} \sim \underbrace{\mathbb{P}((Y_{a,1}, \dots, Y_{a,t}) = (y_{a,1}, \dots, y_{a,t}) \mid \mu_a = x)}_{\text{likelihood}} \cdot \underbrace{\pi_a(x)}_{\text{prior}}$$

$$\sim x^{\sum_{i=1}^t y_{a,i}} (1-x)^{\sum_{i=1}^t (1-y_{a,i})} \cdot x^{n_a-1} (1-x)^{m_a-1} \Rightarrow \text{Beta}\left(n_a + \sum_{i=1}^t y_{a,i}, m_a + \sum_{i=1}^t (1-y_{a,i})\right)$$

$$\mathbb{P}(Y_{a,t+1} = 1 \mid (Y_{a,1}, \dots, Y_{a,t}) = (y_{a,1}, \dots, y_{a,t})) = \frac{n_a + \sum_{i=1}^t y_{a,i}}{n_a + \sum_{i=1}^t y_{a,i} + m_a + \sum_{i=1}^t (1-y_{a,i})}.$$

We have a Markov decision process with *information states*. State space $\mathcal{S} := \mathbb{N}^K \times \mathbb{N}^K$, let $(X_t)_{t \in \mathbb{N}_0}$ be a Markov chain on $\mathcal{S}$. Reward: in state $\mathbf{x} = (\mathbf{s}, \mathbf{f})$,

$$R_a(\mathbf{x}) \sim \text{Bernoulli}\left(\frac{s_a}{s_a + f_a}\right),$$

for all $a \in \mathcal{A}$. Transitions: in state $\mathbf{x} = (\mathbf{s}, \mathbf{f})$, if you take action a , then

$$\mathbb{P}(X_{t+1} = \mathbf{x} + (\mathbf{e}_a, 0) \mid X_t = \mathbf{x}) = \frac{s_a}{s_a + f_a}, \quad \mathbb{P}(X_{t+1} = \mathbf{x} + (0, \mathbf{e}_a) \mid X_t = \mathbf{x}) = \frac{f_a}{s_a + f_a}.$$

Policy: map $\pi : \mathcal{S} \rightarrow \mathcal{A}$ (stationary deterministic policy is optimal (Puterman, 1994)).

Optimisation Problem

$$\max_{\pi: \mathcal{S} \rightarrow \mathcal{A}} \sum_{t=0}^{T-1} \mathbb{E}_\pi[R_{A_t}(X_t) \mid X_0 = x_0], \quad x_0 \in \mathcal{S},$$

where $A_t = \pi(X_t)$ for all $t = 0, \dots, T-1$. Optimal value-to-go function $V_n(\mathbf{x})$ for $\mathbf{x} \in \mathcal{S}$ and $n = 0, \dots, T-1$:

$$V_n(\mathbf{x}) = \max_{\pi: \mathcal{S} \rightarrow \mathcal{A}} \sum_{t=n}^{T-1} \mathbb{E}_\pi[R_{A_t}(X_t) \mid X_n = \mathbf{x}].$$

Dynamic Program

If $\mathbf{x} = (\mathbf{s}, \mathbf{f}) \in \mathcal{S}$ and $n = 0, \dots, T-1$:

For this calculation, write

$$\begin{aligned} Q_1 &:= \max_{\pi: \mathcal{S} \rightarrow \mathcal{A}} \frac{s_{\pi(\mathbf{x})}}{s_{\pi(\mathbf{x})} + f_{\pi(\mathbf{x})}} \cdot 1 \\ &\quad + \sum_{t=n+1}^{T-1} \mathbb{E}_\pi[R_{A_t}(X_t) \mid X_n = \mathbf{x}, X_{n+1} = \mathbf{x} + (\mathbf{e}_{\pi(\mathbf{x})}, 0)] \frac{s_{\pi(\mathbf{x})}}{s_{\pi(\mathbf{x})} + f_{\pi(\mathbf{x})}} \\ &\quad + \sum_{t=n+1}^{T-1} \mathbb{E}_\pi[R_{A_t}(X_t) \mid X_n = \mathbf{x}, X_{n+1} = \mathbf{x} + (0, \mathbf{e}_{\pi(\mathbf{x})})] \frac{f_{\pi(\mathbf{x})}}{s_{\pi(\mathbf{x})} + f_{\pi(\mathbf{x})}} \end{aligned}$$

Then

$$V_n(\mathbf{x})$$

$$= Q_1 = \max_{a \in \mathcal{A}} \frac{s_a}{s_a + f_a} (1 + V_{n+1}(\mathbf{x} + (\mathbf{e}_a, 0))) + \frac{f_a}{s_a + f_a} V_{n+1}(\mathbf{x} + (0, \mathbf{e}_a))$$

with $V_T(\mathbf{x}) := 0$ for all $\mathbf{x} \in \mathcal{S}$.

Bellman Equation

$$V_n(\mathbf{x}) = \max_{a \in \mathcal{A}} \underbrace{\frac{s_a}{s_a + f_a}}_{\text{immediate reward}} + \underbrace{\frac{s_a}{s_a + f_a} V_{n+1}(\mathbf{x} + (\mathbf{e}_a, 0)) + \frac{f_a}{s_a + f_a} V_{n+1}(\mathbf{x} + (0, \mathbf{e}_a))}_{\text{future reward}}$$

for $n = 0, 1, \dots, T-1$, and $V_T(\mathbf{x}) := 0$ for all $\mathbf{x} \in \mathcal{S}$. This equation can be computed backwards to get optimal solution, and other Markov decision process techniques available (policy iteration, value iteration).

Limitations

- (i) State space grows exponentially in K .
- (ii) Works well for conjugate priors, otherwise computationally more challenging.
- (iii) More fundamentally: you assume that your prior is correct.
- (iv) Conjecture (Berry, 1972): If, for the 2-armed Beta-Bernoulli bandit, the priors satisfy

$$\frac{s_1}{s_1 + f_1} = \frac{s_2}{s_2 + f_2}, \quad \text{and} \quad s_1 + f_1 < s_2 + f_2,$$

then it is optimal to start play with arm 1.

SECTION 4.2

| Gittins Index

Consider K -armed bandit Markov decision process, with states $\mathbf{x} = (x_1, \dots, x_K)$ on discrete state-space $\mathcal{S}$, actions determined by a policy $\pi : \mathcal{S} \rightarrow \mathcal{A}$, random rewards $R_a(\mathbf{x})$ (e.g. Bernoulli).

$$\max_{\pi} \sum_{t=0}^{\infty} \beta^t \mathbb{E}[R_{A_t}(X_t) \mid X_0 = x_0], \quad x_0 \in \mathcal{S},$$

where $X_t = (X_{t,1}, \dots, X_{t,K}) \in \mathcal{S}$ and $A_t = \pi(X_t)$ for all $t = 0, 1, \dots$, and discount factor $\beta \in (0, 1)$.

THEOREM 4.1 (Gittins).

There are functions $G_a(\cdot)$, $a \in \mathcal{A}$, such that, for all t , it is optimal to play an arm¹ that maximises $G_a(X_{t,a})$ over all arms $a \in \mathcal{A}$. The functions $G_a(\cdot)$ are called Gittins indices.

Proof. First, consider a single arm j and an outside option. Action space = {play, don't play} or, equivalently, $\{j, 0\}$. State $x_j \in \mathcal{S}_j$ encodes information about arm j . State process $X_{t,j}$, initial state $X_{0,j}$, actions $A_t \in \{j, 0\}$. If gambler does not play: stays in the same state (no new information). Playing costs $\gamma_j \in \mathbb{R}$ per time period and *prevailing charge*. Arm 0 has reward 0 with probability one; arm j has (random) reward $R_j(X_{t,j}) - \gamma_j$. Policy $\pi : \mathcal{S}_j \rightarrow \{j, 0\}$. Earnings from policy π :

$$\sum_{t=0}^{\infty} \beta^t \mathbb{E}[R_j(X_{t,j}) - \gamma_j \mid X_{0,j} = x_j].$$

Hence: if don't play is optimal, it stays optimal:

$$\sup_{\pi} \sum_{t=0}^{\tau(\pi)-1} \beta^t \mathbb{E}[R_j(X_{t,j}) - \gamma_j \mid X_{0,j} = x_j]$$

where $\tau(\pi) = \min\{t \geq 0 : \pi(X_{t,j}) = 0\}$ (state-dependent). Now you are the casino owner. Start in state x_j . How to set a *fair charge* γ_j such that the gambler does not walk away immediately?

For this calculation, write

$$Q_1 := \sup \left\{ \gamma : \sup_{\pi} \sum_{t=0}^{\tau(\pi)-1} \beta^t \mathbb{E}[R_j(X_{t,j}) \mid X_{0,j} = x_j] - \sum_{t=0}^{\tau(\pi)-1} \beta^t \gamma \geq 0 \right\}$$

Then

$$\begin{aligned} G_j(x_j) &= \sup \left\{ \gamma \in \mathbb{R} : \underbrace{\sup_{\pi} \sum_{t=0}^{\tau(\pi)-1} \beta^t \mathbb{E}[R_j(X_{t,j}) - \gamma \mid X_{0,j} = x_j]}_{\text{reward under optimal play}} \geq 0 \right\} \\ &= Q_1 = \sup_{\pi} \frac{\sum_{t=0}^{\tau(\pi)-1} \beta^t \mathbb{E}[R_j(X_{t,j}) \mid X_{0,j} = x_j]}{\sum_{t=0}^{\tau(\pi)-1} \beta^t}. \end{aligned}$$

This function is called the *Gittins index* for arm j . If you play the *optimal policy*, and *prevailing charge* $\gamma_j = G_j(x_j)$ and *initial state* $X_{0,j} = x_j$, then expected total discounted *reward equals zero*. Because the system is Markovian: optimal $\pi(x_j) = j$ if and only if $\gamma_j \leq G_j(x_j)$ (Lemma 35.7(c)).

¹ Dimension reduction: only depends on information of the single arm.

With a fixed prevailing charge γ_j , play may end after finitely many periods. But, suppose we lower the prevailing charge to the fair charge at each time the gambler would otherwise stop, e.g., each time the fair charge drops below the prevailing charge $\gamma_j(t) = \min\{\gamma_j(t-1), G_j(X_{t,j})\}$. You *restart*, as it were, with the current state $X_{t,j}$ as initial state. In each epoch between restarts, the prevailing charge equals the fair charge. By definition of G_j , optimal play gives zero total reward in each epoch. By lowering the prevailing charge in this way, the gambler continues forever, with expected total discounted reward still zero. $\square$

K Bandits (No Outside Option)

State space $\mathcal{S} = \prod_{k=1}^K \mathcal{S}_k$. State process $(X_t)_{t \in \mathbb{N}_0} = (X_{t,1}, \dots, X_{t,K})_{t \in \mathbb{N}_0}$ (e.g. vector of beliefs). Initial state $\mathbf{x}_0 \in \mathcal{S}$. Objective:

$$\sup_{\pi: \mathcal{S} \rightarrow \mathcal{A}} \sum_{t=0}^{\infty} \beta^t \mathbb{E}[R_{A_t}(X_t) \mid X_0 = \mathbf{x}_0],$$

where $A_t = \pi(X_t)$. Set prevailing charges to fair charges, reduce if fair charge below prevailing charge, for all arms. In fact, always playing arm with largest fair charge is optimal:

$$G_j(x_j) = \sup_{\pi} \frac{\sum_{t=0}^{\tau(\pi)-1} \beta^t \mathbb{E}_{\pi}[R_j(X_j(t)) \mid X_j(0) = x_j]}{\sum_{t=0}^{\tau(\pi)-1} \beta^t}.$$

Remark. (i) Exact solution in infinite-horizon discounted-reward case.

(ii) Nice structure: index G_a only depends on information about arm a .

(iii) Bad news: index functions often hard to compute.

(iv) Misspecification risk.

(v) Hard to generalise: e.g., no index policy if $T < \infty$, or for restless MAB's (but good heuristic).

(vi) With positive probability, the optimal arm is played only finitely many times.

SECTION 4.3

Thompson Sampling

Consider K -armed bandit. Reward R_a Bernoulli with mean μ_a , for all $a \in \mathcal{A}$. Each μ_a is random, abs. cont. prior distribution $\pi_a(z)$, $z \in (0, 1)$. $h_t := (a_1, y_1, \dots, a_t, y_t)$ history of actions and observed rewards, up to time t . $\pi_a(z | h_t)$ posterior distribution of μ_a given h_t , for $z \in (0, 1)$.

Sample from posteriors. For each $t \in \mathbb{N}_0$: draw $\theta_a \sim \pi_a(\cdot | h_t)$, for each $a \in \mathcal{A}$, select $a_{t+1} \in \arg \max_{a \in \mathcal{A}} \theta_a$. Let

$$\mu^* = \max_{a \in \mathcal{A}} \mu_a \quad a^*(\mu) = \arg \max_{a \in \mathcal{A}} \mu_a$$

random variable; random variable, a.s. unique.

For all $a \in \mathcal{A}$,

$$\mathbb{P}(a^*(\mu) = a | h_t) = \mathbb{P}(\mu_a \geq \mu_{a'} \forall a' \neq a | h_t) = \int_{z \in (0,1)^K} \mathbb{1}_{\{z_a \geq z_{a'} \forall a' \neq a\}} \cdot \prod_{i=1}^K \pi_i(z_i | h_t) dz,$$

and

$$\mathbb{P}(a_{t+1} = a | h_t) = \mathbb{P}(\theta_a \geq \theta_{a'} \forall a' \neq a) = \int_{z \in (0,1)^K} \mathbb{1}_{\{z_a \geq z_{a'} \forall a' \neq a\}} \cdot \prod_{i=1}^K \pi_i(z_i | h_t) dz.$$

Thus, a_{t+1} is distributed as the optimal arm $a^*(\mu)$, given history h_t . Bayesian regret:

$$BR_\pi(T) = \mathbb{E}_\mu \left[\sum_{t=1}^T (\mu^* - \mu_{a_t}) \right]$$

(Expectation both over μ and over distribution of a_t , $t = 1, \dots, T$).

THEOREM 4.2.

For all $T \geq K$, $BR_{TS}(T) \leq 1 + 8\sqrt{KT \log T}$.

Proof. ²Let $N_t(a) := \sum_{i=1}^t \mathbb{1}\{a_i = a\}$ “How often arm a used up to period t ”. Let $\hat{\mu}_t(a) := N_t(a)^{-1} \sum_{i=1}^t y_i \mathbb{1}\{a_i = a\}$ “average reward arm a ”. Let $r_t(a) := \sqrt{\frac{4 \log T}{1 + N_t(a)}}$ “a confidence radius”. Define the good event

$$E = \{|\hat{\mu}_t(a) - \mu(a)| \leq r_t(a), \forall a \in \mathcal{A}, \forall t \leq T \text{ s.t. } N_t(a) \geq 1\}$$

“all estimates reasonably good, for all t s.t. $\hat{\mu}_t(a)$ defined”. Just as in UCB1’s analysis, $\mathbb{P}(E^c) \leq T^{-1}$ if $T \geq K$ using Hoeffding’s Inequality (even though μ random). Let $U_t(a) = \hat{\mu}_t(a) + r_t(a)$, for $t = 1, \dots, T$ and $a \in \mathcal{A}$. On E , $\mu(a) - U_t(a) \leq 0$ a.s. and $U_t(a) - \mu(a) \leq 2r_t(a)$ a.s., for all t and $a \in \mathcal{A}$. The Bayesian regret in period t :

$$\begin{aligned} \mathbb{E}_\mu[(\mu^* - \mu(a_t)) \mathbb{1}_E] &= \mathbb{E}_\mu[\mathbb{E}[(\mu^* - \mu(a_t)) \mathbb{1}_E | h_{t-1}]] \\ &\leq \mathbb{E}_\mu[\mathbb{E}[(\mu^* - U_t(a^*)) \mathbb{1}_E + (U_t(a_t) - \mu(a_t)) \mathbb{1}_E | h_{t-1}]] \leq \mathbb{E}_\mu[2r_t(a_t)] \end{aligned}$$

and

$$\mathbb{E}_\mu[(\mu^* - \mu(a_t)) \mathbb{1}_{E^c}] \leq \mathbb{P}(E^c) \leq T^{-1},$$

² This proof is adapted from Slivkins (2022) section 3.3.

so

$$\sum_{t=1}^T \mathbb{E}_\mu[(\mu^* - \mu(a_t))] \leq 1 + 2 \sum_{t=1}^T \mathbb{E}_\mu \left[\sqrt{\frac{4 \log T}{1 + N_t(a_t)}} \right].$$

The Bayesian regret satisfies:

$$\begin{aligned} \sum_{t=1}^T \mathbb{E}_\mu[(\mu^* - \mu(a_t))] &\leq 1 + 2 \sum_{t=1}^T \mathbb{E}_\mu \left[\sqrt{\frac{4 \log T}{1 + N_t(a_t)}} \right] = 1 + 4\sqrt{\log T} \mathbb{E}_\mu \left[\sum_{a \in \mathcal{A}} \sum_{t: a_t = a} \sqrt{\frac{1}{1 + N_t(a)}} \right] \\ &= 1 + 4\sqrt{\log T} \mathbb{E}_\mu \left[\sum_{a \in \mathcal{A}} \sum_{j=1}^{n_T(a)} \sqrt{\frac{1}{1+j}} \right] \leq 1 + 8\sqrt{\log T} \cdot K \cdot \mathbb{E}_\mu \left[\sum_{a \in \mathcal{A}} \frac{1}{K} \sqrt{N_T(a)} \right] \\ &\leq 1 + 8\sqrt{\log T} \cdot K \cdot \mathbb{E}_\mu \left[\sqrt{\sum_{a \in \mathcal{A}} \frac{1}{K} N_T(a)} \right] \leq 1 + 8\sqrt{\log T} \cdot \sqrt{TK}. \end{aligned}$$

$\sum_{j=1}^n \frac{1}{\sqrt{1+j}} \leq 2\sqrt{n}$; Jensen's inequality.

□

Remark. (i) Frequentist-analysis of Beta-Bernoulli TS: problem-independent upper bound $O(\sqrt{KT \log T})$; problem-dependent bound

$$R_{TS}(T) \leq (1 + \epsilon) \sum_{a: \Delta_a > 0} \frac{\Delta_a}{kl(\mu_a, \mu^*)} \log T + O\left(\frac{K}{\epsilon^2}\right).$$

(ii) Thompson sampling is a hot research topic (e.g. relation with UCB1, effect of misspecified priors, nonparametric TS, generalisability). Very good numerical performance.

Dynamic Pricing and Learning

SECTION 5.1

| Optimal Selling Price

A firm sells a single product in discrete time periods $t \in \mathbb{N}$. For now: no competition, infinite supply, marginal costs zero. Each period t :

- (i) choose *selling price* $p_t \in [p_{\min}, p_{\max}]$, for some $0 \leq p_{\min} < p_{\max} < \infty$;
- (ii) observe *demand* from *linear demand model*:

$$d_t = \theta_1 + \theta_2 p_t + \epsilon_t,$$

where $\theta = (\theta_1, \theta_2)$ are *unknown* parameters in compact known set $\Theta \subset (0, \infty) \times (-\infty, 0)$, and ϵ_t is unobservable random disturbance term with

$$\mathbb{E}[\epsilon_t \mid p_t] = 0, \quad \mathbb{E}[\epsilon_t^2 \mid p_t] < \sigma^2 \text{ a.s.}$$

for some $\sigma^2 > 0$ and all $t \in \mathbb{N}$;

- (iii) collect *revenue* $p_t d_t$.

A *policy* π is a collection of probability distributions $\pi(\cdot \mid h)$ on $[p_{\min}, p_{\max}]$, such that

$$p_t \mid (p_1, d_1, \dots, p_{t-1}, d_{t-1}) = h$$

has distribution $\pi(\cdot \mid h)$, for all possible *histories* $h \in H$, where

$$H := \bigcup_{t \in \mathbb{N}} ([p_{\min}, p_{\max}] \times \mathbb{R})^{t-1}.$$

Regret of a policy: for $T \in \mathbb{N}$ and $\theta \in \Theta$,

$$\Delta_{\theta}^{\pi}(T) := T \cdot \max_{p \in [p_{\min}, p_{\max}]} p(\theta_1 + \theta_2 p) - \mathbb{E}_{\theta}^{\pi} \left[\sum_{t=1}^T p_t d_t \right].$$

Worst-case regret

$$\Delta^{\pi}(T) := \sup_{\theta \in \Theta} \Delta_{\theta}^{\pi}(T).$$

Since $\min_{\pi} \Delta^{\pi}(T)$ is intractable, focus on asymptotically low regret.

Certainty Equivalent Pricing

¹ Choose arbitrary initial prices $p_1 \neq p_2$ in $[p_{\min}, p_{\max}]$. For each $t \geq 2$:

(i) determine *least-squares estimate* based on available data:

$$\hat{\theta}(t) = \mathcal{P}_{\Theta} \left\{ \arg \min_{\vartheta \in \mathbb{R}^2} \sum_{i=1}^t (d_i - \vartheta_1 - \vartheta_2 p_i)^2 \right\},$$

where $\mathcal{P}_{\Theta}$ is the projection operator on Θ ;

(ii) set

$$p_{t+1} \in \arg \max_{p \in [p_{\min}, p_{\max}]} p \left(\hat{\theta}_1(t) + \hat{\theta}_2(t)p \right)$$

and always choose the *perceived optimal action*.

Remark. As $t \rightarrow \infty$, $\hat{\theta}(t)$ always converges, but with probability zero to the true θ ; p_t always converges, but with probability zero to the true optimal price.

The revenue $p(\theta_1 + \theta_2 p)$ is maximised by $p = -\frac{\theta_1}{2\theta_2}$ (if $\theta_2 < 0$). Assume for convenience that $p \in (p_{\min}, p_{\max})$ for all $\theta \in \Theta$. Suppose we charge an estimated optimal price:

$$\theta_1 + \theta_2 \left(-\frac{\hat{\theta}_1}{2\hat{\theta}_2} \right) \quad \hat{\theta}_1 + \hat{\theta}_2 \left(-\frac{\hat{\theta}_1}{2\hat{\theta}_2} \right)$$

true expected demand; predicted expected demand.

If they equal, then $\hat{\theta}$ is an *indeterminate equilibrium*, and $-\frac{\hat{\theta}_1}{2\hat{\theta}_2}$ is an *uninformative action/price*. Model output ‘confirms’ correctness of the (incorrect) estimates.

Variation in controls gives better estimates. Certainty-equivalent pricing induces insufficient variation in the prices. To ensure convergence of $\hat{\theta}_t$, some amount of experimentation is necessary, but, not too much.

Tools from Linear Algebra

Let A be a symmetric $d \times d$ real-valued positive definite matrix ($x^\top A x > 0, \forall x$).

- (i) All eigenvalues of A are real-valued and positive.
- (ii) The eigenvalues of A^{-1} are the reciprocals of the eigenvalues of A .
- (iii) A^{-1} is symmetric and positive definite.
- (iv) There is a unique positive definite *square root* B such that $B^2 = A$.
- (v) For any U, V , $\text{tr}(UV) = \text{tr}(VU)$.
- (vi) $\det(AB) = \det(BA)$.
- (vii) Let $0 < y_1 < y_2 < \dots$ be an increasing and strictly positive sequence. Then

$$\sum_{s=2}^t \frac{y_s - y_{s-1}}{y_s} = \sum_{s=2}^t \int_{y_{s-1}}^{y_s} \frac{1}{y_s} dx \leq \sum_{s=2}^t \int_{y_{s-1}}^{y_s} \frac{1}{x} dx = \sum_{s=2}^t \log \frac{y_s}{y_{s-1}} = \log \frac{y_t}{y_1}.$$

¹ Also known as myopic pricing, greedy pricing, and passive learning.

THEOREM 5.1 (Sylvester's Determinant Theorem).

$$\det(I + AB) = \det(I + BA).$$

THEOREM 5.2 (Sherman-Morrison Formula).

If A square invertible and $1 + v^\top A^{-1}u \neq 0$,

$$(A + uv^\top)^{-1} = A^{-1} - \frac{A^{-1}uv^\top A^{-1}}{1 + v^\top A^{-1}u}, \quad \forall u, v.$$

Convergence Rates of Ordinary Least Squares

Let $y_t = \theta^\top x_t + \epsilon_t$ for all $t \in \mathbb{N}$, where $\theta \in \mathbb{R}^d$ unknown parameter, $x_t \in \mathbb{R}^d$ covariates, $\mathbb{E}[\epsilon_t] = 0$, $\mathbb{E}[\epsilon_i \epsilon_j] = 0, \forall i \neq j$, $\mathbb{E}[\epsilon_t^2] \leq \sigma^2$. First consider the *fixed design* case where $\{x_t : t \in \mathbb{N}\}$ are non-random. Least-squares estimate defined by

$$\min_{\vartheta \in \mathbb{R}^d} \sum_{i=1}^t (y_i - \vartheta^\top x_i)^2.$$

Unique minimiser

$$\hat{\theta}(t) := \left(\sum_{i=1}^t x_i x_i^\top \right)^{-1} \sum_{i=1}^t x_i y_i$$

provided $\sum_{i=1}^t x_i x_i^\top$ is invertible. In vector notation:

$$X = \underbrace{\begin{pmatrix} x_1^\top \\ \vdots \\ x_t^\top \end{pmatrix}}_{t \times d \text{ matrix}}, \quad E = \underbrace{\begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_t \end{pmatrix}}_{t \times 1 \text{ vector}}, \quad Y = \underbrace{X\theta + E}_{t \times 1 \text{ vector}}.$$

Ordinary least squares: $\min_{\vartheta} \|X\vartheta - Y\|^2$. If $X^\top X$ invertible,

$$\hat{\theta} = (X^\top X)^{-1} X^\top Y = \theta + \underbrace{(X^\top X)^{-1}}_{d \times d} \underbrace{X^\top E}_{d \times 1}.$$

THEOREM 5.3 (Fixed Design).

If $X^\top X$ invertible, then

$$\mathbb{E}[\|\hat{\theta} - \theta\|^2] \leq \frac{d\sigma^2}{\lambda_{\min}(X^\top X)}.$$

Proof.

$$\begin{aligned} \mathbb{E}[\|\hat{\theta} - \theta\|^2] &= \mathbb{E}[\|(X^\top X)^{-1} X^\top E\|^2] = \mathbb{E}[E^\top X (X^\top X)^{-2} X^\top E] \leq \sigma^2 \operatorname{tr}((X^\top X)^{-2} X^\top X) \\ &= \sigma^2 \operatorname{tr}((X^\top X)^{-1}) \\ &\leq \sigma^2 d \lambda_{\max}((X^\top X)^{-1}) = \sigma^2 d \lambda_{\min}(X^\top X)^{-1}. \end{aligned}$$

□

Remark. If $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$, then $\hat{\theta} \sim \mathcal{N}(\theta, \sigma^2(X^\top X)^{-1})$. In all directions, the covariates should have enough variation.

In data-driven decision problems, covariates are typically random: x_{t+1} depends on $\hat{\theta}_t$. Need analysis for linear regression with *adaptive design*.

THEOREM 5.4 (Adaptive Design).

If $\{x_t : t \in \mathbb{N}\}$ are bounded and $\lambda_{\min}(\mathcal{I}_t) \geq \lambda > 0$ a.s., then there is a $C > 0$ independent of t and θ , such that

$$\mathbb{E}[\|\hat{\theta} - \theta\|^2] \leq \frac{C \log t}{\lambda}.$$

Proof. Suppose that $\sum_{i=1}^t x_i x_i^\top$ is invertible. Since $y_i = x_i^\top \theta + \epsilon_i$, for all $t \geq t_0$, the unique minimiser

$$\hat{\theta}(t) = \left(\sum_{i=1}^t x_i x_i^\top \right)^{-1} \sum_{i=1}^t x_i y_i = \theta + \left(\sum_{i=1}^t x_i x_i^\top \right)^{-1} \sum_{i=1}^t x_i \epsilon_i,$$

Suppose

$$\lambda_{\min} \left(\sum_{i=1}^t x_i x_i^\top \right) \geq \lambda > 0, \quad \text{a.s.},$$

$$\begin{aligned} & \mathbb{E}[\|\hat{\theta}(t) - \theta\|^2] \\ &= \mathbb{E} \left[\left\| \left(\sum_{i=1}^t x_i x_i^\top \right)^{-1} \sum_{i=1}^t x_i \epsilon_i \right\|^2 \right] \\ &\leq \mathbb{E} \left[\left\| \left(\sum_{i=1}^t x_i x_i^\top \right)^{-\frac{1}{2}} \right\|^2 \cdot \left\| \left(\sum_{i=1}^t x_i x_i^\top \right)^{-\frac{1}{2}} \sum_{i=1}^t x_i \epsilon_i \right\|^2 \right] \leq \frac{1}{\lambda_{\min}} \mathbb{E} \left[\underbrace{\sum_{i=1}^t x_i^\top \epsilon_i \left(\sum_{i=1}^t x_i x_i^\top \right)^{-1} \sum_{i=1}^t x_i \epsilon_i}_{\text{need to bound this}} \right]. \end{aligned}$$

LEMMA 5.5.

For all $t \geq t_0$,

$$\mathbb{E} \left[\sum_{i=1}^t x_i^\top \epsilon_i \left(\sum_{i=1}^t x_i x_i^\top \right)^{-1} \sum_{i=1}^t x_i \epsilon_i \right] \leq \mathbb{E} \left[M_{t_0}^\top \mathcal{I}_{t_0}^{-1} M_{t_0} + o^2 \sum_{s=t_0+1}^t x_s^\top \mathcal{I}_s^{-1} x_s \right].$$

Proof.

For this calculation, write

$$\mathcal{Q}_1 := (M_t + x_{t+1} \epsilon_{t+1})^\top \underbrace{\left(\mathcal{I}_t^{-1} - \frac{\mathcal{I}_t^{-1} x_{t+1} x_{t+1}^\top \mathcal{I}_t^{-1}}{1 + x_{t+1}^\top \mathcal{I}_t^{-1} x_{t+1}} \right)}_{\text{Sherman-Morrison formula}} (M_t + x_{t+1} \epsilon_{t+1})$$

Then

$$\begin{aligned} M_{t+1}^\top \mathcal{I}_{t+1}^{-1} M_{t+1} &= (M_t + x_{t+1} \epsilon_{t+1})^\top (\mathcal{I}_t + x_{t+1} x_{t+1}^\top)^{-1} (M_t + x_{t+1} \epsilon_{t+1}) \\ &= \mathcal{Q}_1 = M_t^\top \mathcal{I}_t^{-1} M_t - \underbrace{(\cdots)}_{>0} + \epsilon_{t+1} \cdot \underbrace{(\cdots)}_{>0} + \epsilon_{t+1}^2 \cdot x_{t+1}^\top \mathcal{I}_{t+1}^{-1} x_{t+1}. \end{aligned}$$

Take expectations, telescope the sum. □

LEMMA 5.6.

Suppose $\sup_{t \in \mathbb{N}} \|x_t\| < M < \infty$ for some non-random M . Then

$$\sum_{s=t_0+1}^t x_s^\top I_s^{-1} x_s = O(\log t).$$

Proof. For all $s > t_0$,

$$\det(I_{s-1}) = \det(I_s - x_s x_s^\top) = \det(I_s) \det(I - I_s^{-1} x_s x_s^\top) = \det(I_s) (1 - x_s^\top I_s^{-1} x_s)$$

Sylvester's determinant theorem.

Then

$$\begin{aligned} \sum_{s=t_0+1}^t x_s^\top I_s^{-1} x_s &= \sum_{s=t_0+1}^t \frac{\det(I_s) - \det(I_{s-1})}{\det(I_s)} = \sum_{s=t_0+1}^t \frac{\det(I_s) - \det(I_{s-1})}{\det(I_{t_0})} \leq \log \left(\frac{\det(I_t)}{\det(I_{t_0})} \right) \\ &= O(\log(\lambda_{\max}(I_t))) = O(\log(t)). \end{aligned}$$

□

□

Remark. (i) In expectation, therefore a non-random lower bound.

(ii) Additional $\log t$ term compared to fixed design. But unclear if/when this term 'is really there': at least sometimes it is.

(iii) Exploit this relation between 'quality of estimates' and 'amount of exploration' to construct consistent pricing policy.

SECTION 5.2

| Construct Pricing Policy

In the pricing problem, we have

$$d_t = (\theta_1, \theta_2) \begin{pmatrix} 1 \\ p_t \end{pmatrix} + \epsilon_t = \theta_1 + \theta_2 p_t + \epsilon_t,$$

and thus

$$\mathcal{I}_t = \sum_{i=1}^t \begin{pmatrix} 1 \\ p_i \end{pmatrix} \begin{pmatrix} 1 & p_i \end{pmatrix} = \sum_{i=1}^t \begin{pmatrix} 1 & p_i \\ p_i & p_i^2 \end{pmatrix}.$$

For all $t \geq 2$,

$$\begin{aligned} \det(\mathcal{I}_t) &= \sum_{i=1}^t 1 \sum_{i=1}^t p_i^2 - \left(\sum_{i=1}^t p_i \right)^2 \\ &= t^2 \text{Var}(p_1, \dots, p_t), \quad \text{tr}(\mathcal{I}_t) = \sum_{i=1}^t (1 + p_i^2) \in \left[(1 + p_{\min}^2)t, (1 + p_{\max}^2)t \right], \end{aligned}$$

and thus

$$\begin{aligned} \lambda_{\min}(\mathcal{I}_t) &= \frac{\det(\mathcal{I}_t)}{\lambda_{\max}(\mathcal{I}_t)} \geq \frac{\det(\mathcal{I}_t)}{\text{tr}(\mathcal{I}_t)} \geq \frac{t \text{Var}(p_1, \dots, p_t)}{1 + p_{\max}^2}, \quad \lambda_{\min}(\mathcal{I}_t) = \frac{\det(\mathcal{I}_t)}{\lambda_{\max}(\mathcal{I}_t)} \\ &\leq \frac{\det(\mathcal{I}_t)}{\frac{1}{2} \text{tr}(\mathcal{I}_t)} \leq \frac{2t \text{Var}(p_1, \dots, p_t)}{1 + p_{\min}^2}. \end{aligned}$$

Thus, $\lambda_{\min}(\mathcal{I}_t) \sim t \text{Var}(p_1, \dots, p_t)$.

Controlled Variance Pricing

Choose arbitrary initial prices $p_1 \neq p_2$ in $[p_{\min}, p_{\max}]$. For each $t \geq 2$:

$$\begin{aligned} \hat{\theta}(t) &:= \mathcal{P}_{\Theta} \left\{ \arg \min_{\vartheta \in \mathbb{R}^2} \sum_{i=1}^t (d_i - \vartheta_1 - \vartheta_2 p_i)^2 \right\} \\ p_{t+1} &\in \arg \max_{p \in [p_{\min}, p_{\max}]} p \left(\hat{\theta}_1(t) + \hat{\theta}_2(t) p \right), \end{aligned}$$

$$\text{such that } (t+1) \text{Var}(p_1, \dots, p_{t+1}) \geq f(t+1),$$

information constraint.

for an increasing function $f : \mathbb{N} \rightarrow [0, \infty)$ to determine. Always choose the perceived optimal action that induces sufficient experimentation. Feasibility:

$$f(2) \leq \frac{1}{2}(p_{\max} - p_{\min})^2, \quad \text{and} \quad f(t+1) - f(t) \leq \frac{1}{8}(p_{\max} - p_{\min})^2, \quad \forall t \geq 2.$$

THEOREM 5.7 (Regret of CVP).

Let π correspond to the controlled variance pricing (CVP). Then

$$\Delta^\pi(T) = O\left(\sqrt{T \log T}\right).$$

Proof. Let π correspond to CVP. Let $\theta \in \Theta$, and let $T \in \mathbb{N}$.

$$\Delta_{\theta}^{\pi}(T) = T \cdot \max_{p \in [p_{\min}, p_{\max}]} p(\theta_1 + \theta_2 p) - \mathbb{E}_{\theta}^{\pi} \left[\sum_{t=1}^T p_t(\theta_1 + \theta_2 p_t) \right].$$

By assumption,

$$p^*(\theta) := \arg \max_{p \in [p_{\min}, p_{\max}]} p(\theta_1 + \theta_2 p) = -\frac{\theta_1}{2\theta_2} \in (p_{\min}, p_{\max}).$$

For all p ,

$$\underbrace{p^*(\theta) \cdot (\theta_1 + \theta_2 p^*(\theta)) - p(\theta_1 + \theta_2 p)}_{\text{revenue loss}} = -\theta_2 \underbrace{(p^*(\theta) - p)^2}_{\text{(squared) decision error}}.$$

Remark. More generally, if x^* maximises a C^2 function r on a compact convex $\mathcal{X} \subset \mathbb{R}^d$ and satisfies first order conditions, then for all $x \in \mathcal{X}$,

$$r(x^*) - r(x) = r'(x^*)(x - x^*) + \frac{1}{2}(x - x^*)^{\top} r''(\tilde{x})(x - x^*) \leq \sup_{\tilde{x} \in \mathcal{X}} \frac{1}{2} \|r''(\tilde{x})\| \cdot \|x - x^*\|^2.$$

For all $p, \hat{\theta}$, since $(u + v)^2 \leq 2u^2 + 2v^2$,

$$(p^*(\theta) - p)^2 \leq 2(p^*(\theta) - p^*(\hat{\theta}))^2 + 2(p^*(\hat{\theta}) - p)^2,$$

and by the Taylor series expansion,

$$|p^*(\theta) - p^*(\hat{\theta})| \leq \|\theta - \hat{\theta}\| \cdot \sup_{\theta \in \Theta} \left| \frac{d}{d\theta} \left(-\frac{\theta_1}{2\theta_2} \right) \right|,$$

hence, there is a K_1 such that, for all $t \geq 2$,

$$(p^*(\theta) - p_{t+1})^2 \leq 2K_1 \underbrace{\|\theta - \hat{\theta}_t\|^2}_{\text{(squared) estimation error}} + 2 \underbrace{(p^*(\hat{\theta}_t) - p_{t+1})^2}_{\text{(squared) exploration error}}.$$

Remark. More generally, if $x^*(\theta)$ satisfies $r'(x; \theta) = 0$ and $r''(x^*(\theta); \theta)$ is invertible for all θ in an open set containing Θ , then (implicit function theorem) $x^*(\theta)$ is continuously differentiable in Θ , hence $\sup_{\theta \in \Theta} \left\| \frac{d}{d\theta} x^*(\theta) \right\| < \infty$.

Squared Estimation Error. Let $\hat{\vartheta}_t$ be the *unprojected* ordinary least squares estimate. For all $t \geq 2$, since

$$\lambda_{\min}(\mathcal{I}_t) \geq \frac{t \text{Var}(p_1, \dots, p_t)}{(1 + p_{\max})^2} \geq \frac{f(t)}{(1 + p_{\max})^2} \quad a.s.,$$

there is a $C > 0$ independent of t and θ , such that

$$\mathbb{E}[\|\theta - \hat{\theta}_t\|^2] = \mathbb{E}[\|\mathcal{P}_{\Theta}\{\theta\} - \mathcal{P}_{\Theta}\{\hat{\theta}_t\}\|^2] \leq \mathbb{E}[\|\theta - \hat{\theta}_t\|^2] \leq \frac{C \log(t)}{f(t)}. \quad (5.1)$$

projection is non-expansive.

Squared Exploration Error. Let $V_t := t \cdot \text{Var}(p_1, \dots, p_t)$, for all $t \in \mathbb{N}$.

$$V_{t+1} = V_t + \frac{t}{t+1} (p_{t+1} - \bar{p}_t)^2$$

where $\bar{p}_t = t^{-1} \sum_{i=1}^t p_i$. If

$$V_t + \frac{t}{t+1} (p^*(\hat{\theta}_t) - \bar{p}_t)^2 \geq f(t+1),$$

then $p_{t+1} = p^*(\hat{\theta}_t)$, no exploration. If not, let δ such that

$$V_t + \frac{t}{t+1}\delta^2 = f(t+1).$$

Then $p^*(\hat{\theta}_t)$ lies in the infeasible price range $(\bar{p}_t - \delta, \bar{p}_t + \delta)$. Since $p \cdot (\hat{\theta}_1 + \hat{\theta}_2 p)$ is concave on $[p_{\min}, p_{\max}]$, the revenue-maximal price p_{t+1} is the feasible price closest to $p^*(\hat{\theta}_t)$. And

$$(p_{t+1} - p^*(\hat{\theta}_t))^2 \leq (2\delta)^2 = 4 \cdot \frac{t+1}{t}(f(t+1) - V_t) \leq 6(f(t+1) - f(t)),$$

by construction of the policy.

Putting everything together,

$$\begin{aligned} \Delta_{\theta}^{\pi}(T) &= \sum_{t=1}^T \mathbb{E}_{\theta}^{\pi} \left[\max_{p \in [p_{\min}, p_{\max}]} p(\theta_1 + \theta_2 p) - p_t(\theta_1 + \theta_2 p_t) \right] \\ &\leq \underbrace{2 \max_{p \in [p_{\min}, p_{\max}]} p(\theta_1 + \theta_2 p)}_{=: K_2} + \underbrace{2(-\theta_2)}_{\leq K_3 := \sup_{\theta} (-2\theta_2)} \sum_{t=2}^{T-1} \mathbb{E}_{\theta}^{\pi} \left[(p^*(\theta) - p_{t+1})^2 \right] \\ &\leq K_2 + K_3 \sum_{t=2}^{T-1} \mathbb{E}_{\theta}^{\pi} \left[2K_1 \|\theta - \hat{\theta}_t\|^2 + 2(p^*(\theta) - p_{t+1})^2 \right] \\ &\leq K_2 + K_3 \sum_{t=2}^{T-1} \left\{ \frac{2K_1 C \log(t)}{f(t)} + 12(f(t+1) - f(t)) \right\} \leq K_2 + K_3 \sum_{t=2}^{T-1} \left\{ \frac{2K_1 C \log(t)}{f(t)} \right\} + 12K_3 f(T). \end{aligned}$$

The function f captures the exploration-exploitation trade-off. Optimal choice $f(t) = c\sqrt{t \log(t)}$, for sufficiently small $c > 0$.

$$\Delta_{\theta}^{\pi}(T) = O\left(\sum_{t=2}^{T-1} \frac{\sqrt{\log(t)}}{\sqrt{t}}\right) + \sqrt{T \log T} = O\left(\sqrt{T \log T}\right).$$

In fact, since all constants are independent of θ ,

$$\Delta^{\pi}(T) = O\left(\sqrt{T \log T}\right).$$

□

Remark. (i) Optimal regret (of this policy) close to $\sqrt{T}$.

(ii) Price variance instead of $\lambda_{\min}$ easier to communicate in practice.

(iii) Results can be extended to generalised linear models: $d_t = h(\theta_1 + \theta_2 p_t) + \epsilon_t$ and multiple products.

SECTION 5.3

| Regret Lower Bounds

Cramér-Rao Lower Bound

Let X be a random variable with support $\mathcal{X} \subset \mathbb{R}^n$, and C^1 PDF $f(\cdot | \theta)$, for some parameter $\theta \in \mathbb{R}$. Let $\hat{\theta} : \mathcal{X} \rightarrow \mathbb{R}$ be an estimator. Let $b(\theta) := \mathbb{E}[\hat{\theta}(X) - \theta]$ be the bias. Assume f is ‘nice’: $f(x | \theta) > 0$ for all $x \in \mathcal{X}$, we can interchange integrals and derivatives, and

$$I(\theta) := \mathbb{E} \left[\left(\frac{d}{d\theta} \log f(X | \theta) \right)^2 \right]$$

exists. Then

$$\mathbb{E} \left[\left(\hat{\theta}(X) - \theta \right)^2 \right] \geq \frac{(1 + b'(\theta))^2}{I(\theta)}.$$

If $X = (X_1, \dots, X_n)$ i.i.d. and $b(\theta) \equiv 0$, then

$$\mathbb{E} \left[\left(\hat{\theta}(X) - \theta \right)^2 \right] \geq \frac{1}{n \cdot I_{X_1}(\theta)}.$$

Proof.

$$\begin{aligned} & \mathbb{E} \left[\left(\hat{\theta}(X) - \theta \right)^2 \right] \cdot I(\theta) \\ &= \int_{\mathcal{X}} \left(\hat{\theta}(x) - \theta \right)^2 f(x | \theta) dx \cdot \int_{\mathcal{X}} \left(\frac{\frac{d}{d\theta} f(x | \theta)}{f(x | \theta)} \right)^2 f(x | \theta) dx \\ &\geq \left(\int_{\mathcal{X}} \left(\hat{\theta}(x) - \theta \right) \sqrt{f(x | \theta)} \cdot \frac{\frac{d}{d\theta} f(x | \theta)}{\sqrt{f(x | \theta)}} dx \right)^2 = \left(\frac{d}{d\theta} \int_{\mathcal{X}} \hat{\theta}(x) f(x | \theta) dx - \theta \frac{d}{d\theta} \int_{\mathcal{X}} f(x | \theta) dx \right)^2 \\ &= \left(\frac{d}{d\theta} (\theta + b(\theta)) - \theta \frac{d}{d\theta} 1 \right)^2 = (1 + b'(\theta))^2. \end{aligned}$$

□

Maximum Likelihood Estimation (MLE) maximises $\theta \mapsto f(X | \theta)$. Equivalently, maximise $\theta \mapsto \log f(X | \theta)$. An interior solution satisfies

$$s(\theta) := \frac{d}{d\theta} \log f(X | \theta) = 0.$$

The ‘true θ ’, say θ^* , satisfies (under mild conditions)

$$\mathbb{E}[s(\theta^*)] = \int_{\mathcal{X}} f(x | \theta^*) \frac{f'(x | \theta^*)}{f(x | \theta^*)} dx = \frac{d}{d\theta} \int_{\mathcal{X}} f(x | \theta) dx \Big|_{\theta=\theta^*} = 0.$$

(Minus) the second derivative with respect to θ measures how ‘flat’ the expected score function is around θ^* :

$$\mathbb{E} \left[\frac{d}{d\theta} s(\theta) \right] \Big|_{\theta=\theta^*} = \int_{\mathcal{X}} f(x | \theta^*) \frac{f'(x | \theta^*) f''(x | \theta^*) - f'(x | \theta^*)^2}{f(x | \theta^*)^2} dx$$

$$= \frac{d^2}{d\theta^2} \int_{\mathcal{X}} f(x, \theta) dx \Big|_{\theta=\theta^*} - \mathbb{E} \left[\left(\frac{d}{d\theta} \log f(X | \theta) \right)^2 \right] = -I(\theta^*).$$

Very flat, difficult to learn the maximiser of the MLE from noisy data. We extend the Cramér-Rao bound in three directions.

- (i) Bayesian version, where θ is random, to allow for this argument:

$$\Delta^\pi(T) = \sup_{\theta \in \Theta} \Delta_\theta^\pi(T) \geq \mathbb{E}[\Delta_\theta^\pi(T)].$$

- (ii) Estimate a *function* of θ (e.g. price) instead of θ .
 (iii) Multidimensional θ (in pricing model: 2 dimensional).

Van Trees' Inequality: Bayesian Cramér-Rao Lower Bound

View θ as a random variable θ on a closed interval $\Theta = [\theta_{\min}, \theta_{\max}] \subset \mathbb{R}$. θ has C^1 PDF λ , with $\lambda(\theta) = 0$ on $\partial\Theta$ and $\lambda(\theta) > 0$ on $\text{int}(\Theta)$. Let

$$I(\lambda) := \mathbb{E} \left[\left(\frac{d}{d\theta} \log \lambda(\theta) \right)^2 \right].$$

Then

$$\mathbb{E} \left[\left(\hat{\theta}(X) - \theta \right)^2 \right] \geq \frac{1}{\mathbb{E}[I(\theta)] + I(\lambda)}.$$

Proof. By partial integration and λ is zero on $\partial\Theta$,

$$\begin{aligned} & \int_{\mathcal{X}} \int_{\Theta} (\hat{\theta}(x) - \theta) \frac{d}{d\theta} \{f(x | \theta) \lambda(\theta)\} d\theta dx \\ &= \int_{\mathcal{X}} \left\{ \left[(\hat{\theta}(x) - \theta) f(x | \theta) \lambda(\theta) \right]_{\theta_{\min}}^{\theta_{\max}} - \int_{\Theta} \frac{d}{d\theta} \{(\hat{\theta}(x) - \theta)\} f(x | \theta) \lambda(\theta) d\theta \right\} dx \\ &= \int_{\mathcal{X}} \int_{\Theta} f(x | \theta) \lambda(\theta) d\theta dx = 1. \end{aligned}$$

By *Cauchy-Schwarz inequality*,

$$1^2 \leq \underbrace{\left(\int_{\mathcal{X}} \int_{\Theta} (\hat{\theta}(x) - \theta)^2 f(x | \theta) \lambda(\theta) d\theta dx \right)}_{\mathbb{E}[(\hat{\theta}(X) - \theta)^2]} \cdot \left(\int_{\mathcal{X}} \int_{\Theta} \left(\frac{\frac{d}{d\theta} \{f(x | \theta) \lambda(\theta)\}}{f(x | \theta) \lambda(\theta)} \right)^2 f(x | \theta) \lambda(\theta) d\theta dx \right).$$

The latter part can be written as:

For this calculation, write

$$\begin{aligned} Q_1 &:= \mathbb{E}[I(\theta)] + 2 \underbrace{\int_{\Theta} \int_{\mathcal{X}} \frac{d}{d\theta} \{\log f(x | \theta)\} f(x | \theta) dx \frac{d}{d\theta} \{\log \lambda(\theta)\} \lambda(\theta) d\theta}_{\text{expected score is zero}} \\ &\quad + \int_{\Theta} \left(\frac{d}{d\theta} \log \lambda(\theta) \right)^2 \lambda(\theta) d\theta \end{aligned}$$

Then

$$\int_{\mathcal{X}} \int_{\Theta} \left(\frac{d}{d\theta} \log(f(x | \theta) \lambda(\theta)) \right)^2 f(x | \theta) \lambda(\theta) d\theta dx$$

$$\begin{aligned}
&= \int_X \int_{\Theta} \left(\frac{d}{d\theta} \log f(x | \theta) \right)^2 f(x | \theta) \lambda(\theta) d\theta dx \\
&\quad + \int_X \int_{\Theta} 2 \frac{d}{d\theta} \{\log f(x | \theta)\} \cdot \frac{d}{d\theta} \{\log \lambda(\theta)\} f(x | \theta) \lambda(\theta) d\theta dx \\
&\quad + \int_X \int_{\Theta} \left(\frac{d}{d\theta} \log \lambda(\theta) \right)^2 f(x | \theta) \lambda(\theta) d\theta dx \\
&= Q_1 = \mathbb{E}[I(\theta)] + I(\lambda).
\end{aligned}$$

□

Extension: if we estimate a C^1 function $\psi(\theta)$ by $\hat{\psi}(X)$, then

$$\mathbb{E} \left[\left(\hat{\psi}(X) - \psi(\theta) \right)^2 \right] \geq \frac{(\mathbb{E}[\psi'(\theta)])^2}{\mathbb{E}[I(\theta)] + I(\lambda)}.$$

Proof. By partial integration and λ is zero on $\partial\Theta$,

$$\begin{aligned}
&\int_X \int_{\Theta} (\hat{\psi}(x) - \psi(\theta)) \frac{d}{d\theta} \{f(x | \theta) \lambda(\theta)\} d\theta dx \\
&= \int_X \left\{ \left[(\hat{\psi}(x) - \psi(\theta)) f(x | \theta) \lambda(\theta) \right]_{\theta_{\min}}^{\theta_{\max}} - \int_{\Theta} \frac{d}{d\theta} \{(\hat{\psi}(x) - \psi(\theta))\} f(x | \theta) \lambda(\theta) d\theta \right\} dx \\
&= \int_X \int_{\Theta} \psi'(\theta) f(x | \theta) \lambda(\theta) d\theta dx = (\mathbb{E}[\psi'(\theta)])^2.
\end{aligned}$$

By *Cauchy-Schwarz inequality*,

$$(\mathbb{E}[\psi'(\theta)])^2 \leq \underbrace{\left(\int_X \int_{\Theta} (\hat{\psi}(x) - \psi(\theta))^2 f(x | \theta) \lambda(\theta) d\theta dx \right)}_{\mathbb{E}[(\hat{\psi}(X) - \psi(\theta))^2]} \cdot \left(\int_X \int_{\Theta} \frac{\left(\frac{d}{d\theta} \{f(x | \theta) \lambda(\theta)\} \right)^2}{f(x | \theta) \lambda(\theta)} d\theta dx \right).$$

As before the latter part $\mathbb{E}[I(\theta)] + I(\lambda)$.

□

Multivariate Van Trees Inequality

² Let X be a random variable with support $\mathcal{X} \subset \mathbb{R}^n$, with nice C^1 PDF $f(\cdot | \theta)$, for some parameter $\theta \in \Theta \subset \mathbb{R}^d$. Let Θ be compact with piecewise C^1 boundary. Let λ be a nice C^1 PDF with $\lambda(\theta) = 0$ on $\partial\Theta$ and $\lambda(\theta) > 0$ on $\text{int}(\Theta)$. Let $\psi : \Theta \rightarrow \mathbb{R}$ be a nice C^1 function with derivative ψ' (column vector). Let $\hat{\psi} : \mathcal{X} \rightarrow \Theta$ be an estimator of $\psi(\theta)$. Let $C : \Theta \rightarrow \mathbb{R}^d$ be a nice C^1 function (row vector). Let

$$I(\theta) := \mathbb{E} \left[\left(\frac{d}{d\theta} \log f(X | \theta) \right) \left(\frac{d}{d\theta} \log f(X | \theta) \right)^{\top} \right].$$

Then

$$\mathbb{E} \left[\left(\hat{\psi}(X) - \psi(\theta) \right)^2 \right] \geq \frac{\mathbb{E}[(C(\theta)^{\top} \psi'(\theta))^2]}{\mathbb{E}[C(\theta)I(\theta)C(\theta)^{\top}] + \tilde{I}_{\lambda}},$$

where

$$\tilde{I}_{\lambda} := \int_{\Theta} \sum_{k,l} \frac{\partial \{C_k(\theta) \lambda(\theta)\}}{\partial \theta_k} \frac{\partial \{C_l(\theta) \lambda(\theta)\}}{\partial \theta_l} d\theta.$$

² The actual statement (*Theorem 1* in Gill and Levit, 1995) is even more general, allowing for vector-valued functions ψ . See Keskin and Zeevi (2014) for the 1-dimensional version we use.

Proof. By bookkeeping and Cauchy-Schwarz. □

Remark. Useful result to prove lower bounds on mean squared errors.

EXAMPLE 5.8. Let π be a policy, $T \in \mathbb{N}$.

For this calculation, write

$$Q_1 := -\theta_{2,\max} \mathbb{E} \left[\sum_{t=1}^T (\psi(\theta) - \bar{p}_T)^2 + 2(\psi(\theta) - \bar{p}_T)(\bar{p}_T - p_t) + (\bar{p}_T - p_t)^2 \right]$$

Then

$$\begin{aligned} \Delta^\pi(T) &= \sup_{\theta \in \Theta} \mathbb{E} \left[\sum_{t=1}^T \max_{p \in [p_{\min}, p_{\max}]} p(\theta_1 + \theta_2 p) - p_t(\theta_1 + \theta_2 p_t) \right] \geq \mathbb{E} \left[-\theta_2 \sum_{t=1}^T (\psi(\theta) - p_t)^2 \right] \\ &\geq -\theta_{2,\max} \mathbb{E} \left[\sum_{t=1}^T (\psi(\theta) - p_t)^2 \right] \\ &= Q_1 \geq -\theta_{2,\max} T \cdot \mathbb{E}[(\psi(\theta) - \bar{p}_T)^2]. \end{aligned}$$

We now show

$$\mathbb{E}[(\psi(\theta) - \bar{p}_T)^2] \geq \frac{c_1^2}{\mathbb{E}[\sum_{t=1}^T (\psi(\theta) - p_t)^2] + \tilde{\mathcal{I}}_\lambda}.$$

Suppose that $d_t = \theta_1 + \theta_2 p_t + \epsilon_t$ where $\epsilon \sim N(0, 1)$ i.i.d. Suppose prices are non-random. Let

$$\begin{aligned} f(d_1, \dots, d_T \mid \theta) &= \prod_{t=1}^T \frac{1}{\sqrt{2\pi}} \exp \left(-\frac{1}{2} (d_t - \theta_1 - \theta_2 p_t)^2 \right), \\ \mathcal{I}(\theta) &= \sum_{t=1}^T \begin{pmatrix} 1 \\ p_t \end{pmatrix} \begin{pmatrix} 1 & p_t \end{pmatrix}. \end{aligned} \quad (\text{independent of } \theta)$$

Let $\hat{\psi}(X) = \bar{p}_T$. Choose appropriate λ . Then by *Van Trees' inequality*,

$$\mathbb{E} \left[\left(\hat{\psi}(X) - \psi(\theta) \right)^2 \right] \geq \frac{\mathbb{E}[(C(\theta)^\top \psi'(\theta))^2]}{\mathbb{E}[C(\theta) \mathcal{I}(\theta) C(\theta)^\top] + \tilde{\mathcal{I}}_\lambda},$$

where $\theta = (\theta_1, \theta_2)$, $\Theta = [\theta_{1,\min}, \theta_{1,\max}] \times [\theta_{2,\min}, \theta_{2,\max}]$, $\psi(\theta) = -\frac{\theta_1}{2\theta_2}$. High lower bound implies $C(\theta) \sim v_{\min}(\mathcal{I}(\theta))$. If $C(\theta) = [\psi(\theta), -1]$, then

$$C(\theta) \mathcal{I}(\theta) C(\theta)^\top = \sum_{t=1}^T (\psi(\theta) - p_t)^2 \sim \lambda_{\min}(\mathcal{I}(\theta)).$$

Then also $c_1 := \min_{\theta \in \Theta} C(\theta)^\top \psi'(\theta) > 0$.

$$\begin{aligned} \Delta^\pi(T) &\geq -\theta_{2,\max} \mathbb{E} \left[\sum_{t=1}^T (\psi(\theta) - p_t)^2 \right] \\ &\geq -\theta_{2,\max} \cdot T \cdot \mathbb{E}[(\psi(\theta) - \bar{p}_T)^2] \geq -\theta_{2,\max} \cdot T \cdot \frac{c_1^2}{\mathbb{E}[\sum_{t=1}^T (\psi(\theta) - p_t)^2] + \tilde{\mathcal{I}}_\lambda}. \end{aligned}$$

Implies there is a $c > 0$ such that, for all $T \in \mathbb{N}$, $\Delta^\pi(T) \geq c\sqrt{T}$.

SECTION 5.4

Convergence Rates of Kiefer-Wolfowitz Stochastic Approximation

Kiefer-Wolfowitz stochastic approximation scheme:

$$X_{n+1} = X_n + a_n \left(\frac{\tilde{f}(X_n + c_n) - \tilde{f}(X_n - c_n)}{c_n} \right), \quad n \in \mathbb{N},$$

where initial condition $X_1 \in \mathbb{R}$, can be random. $\{a_n\}$ and $\{c_n\}$ non-random strictly positive tuning sequences. $\tilde{f}(X_n + c_n)$ and $\tilde{f}(X_n - c_n)$ are drawn according to a conditional distribution with CDF $H(y \mid X_n + c_n)$ and $H(y \mid X_n - c_n)$.³

The *unknown* objective function that we wish to maximise:

$$f(x) := \int_{\mathbb{R}} y dH(y \mid x),$$

which is differentiable and has unique maximiser $x^* \in \mathbb{R}$.

There are positive constants K_0, K_1 , such that

$$K_0 |x - x^*| \leq |f'(x)| \leq K_1 |x - x^*|$$

for all $x \in \mathbb{R}$, preventing ‘flat regions’ away from x^* . Strongly unimodal: $f'(x)(x - x^*) < 0$ for all $x \in \mathbb{R} \setminus \{x^*\}$.

$$\sigma^2 := \sup_{x \in \mathbb{R}, n \in \mathbb{N}} \text{Var}[\tilde{f}(X_n + c_n) - \tilde{f}(X_n - c_n) \mid X_n = x] < \infty.$$

THEOREM 5.9 (Broadie, Cicek, Zeevi (2011)).

Let $a_n = \theta_a n^{-1}$, for some $\theta_a > \frac{1}{8K_0}$. Let $c_n = \theta_c n^{-\frac{1}{4}}$, for $\theta_c > 0$. There is a $C > 0$ such that, for all $n \in \mathbb{N}$,

$$\mathbb{E}[(X_{n+1} - x^*)^2] \leq C n^{-\frac{1}{2}}.$$

Proof.

□

Remark. (i) In the paper, much more general $\{a_n\}$ and $\{c_n\}$ are treated.

(ii) There are many other papers on Kiefer-Wolfowitz and variants; see Broadie, Cicek, and Zeevi (2012) for some references.

(iii) Truncation on $[l, u]$ does not change the convergence rate

$$X_{n+1} = \Pi_{I_{n+1}} \left\{ X_n + a_n \left(\frac{\tilde{f}(X_n + c_n) - \tilde{f}(X_n - c_n)}{c_n} \right) \right\}$$

where $I_{n+1} = [l + c_{n+1}, u - c_{n+1}]$, assuming $c_n < \frac{u-l}{2}$ for all $n \in \mathbb{N}$.

(iv) Bounds for the maximum:

$$f(x^*) - f(X_n) = |f'(\xi_n)| \cdot |x^* - X_n| \quad \text{for some } \xi_n \in (x^*, X_n)$$

³ Stochastic version of gradient ascent: $x_{n+1} = x_n + a_n f'(x_n)$.

$$\leq K_1 |\xi_n - x^*| \cdot |x^* - X_n| \leq K_1 (x^* - X_n)^2,$$

hence

$$\sum_{n=1}^T \mathbb{E}[f(x^*) - f(X_n)] \leq K_1 \sum_{n=1}^T \frac{C}{\sqrt{n}} \leq 2K_1 C \sqrt{T}.$$

(v) *If the truth is close to the parametric model, that usually performs better. Structural insights, generalisability. Numerical performance, more generally, data-driven performance.*

Application to pricing: x corresponds to price p , $f(p) = p d(p)$, where $d(p)$ is expected demand. All differentiable demand functions are satisfied, yield $O(\sqrt{T})$ regret.

SECTION 5.5

| Model Mis-Specification

Each period $t \in \mathbb{N}$, a seller:

- (i) chooses a selling price $p_t \in [p_{\min}, p_{\max}]$, for some $0 < p_{\min} < p_{\max} < \infty$;
- (ii) observes demand $d_t = \lambda(p_t) + \epsilon_t$, where $\lambda : [0, \infty) \rightarrow [0, \infty)$ is an *unknown function*, and ϵ_t unobservable random disturbance term with light-tailed

$$\mathbb{E}[\epsilon_t | p_t] = 0, \quad \mathbb{E}[\epsilon_t^2 | p_t] < \sigma^2 \quad a.s., \quad \mathbb{E}[\exp(s\epsilon_t)] < \infty, \quad \exists \rho > 0, \quad \forall s \in (-\rho, \rho);$$

- (iii) collect revenue $p_t d_t$.

Assumption on λ : positive, strictly decreasing and C^2 on $[p_{\min}, p_{\max}]$, $p \mapsto r(p) := p\lambda(p)$ has unique maximiser $p^* \in (p_{\min}, p_{\max})$. Technical assumption:

$$\sup_{p \in (p_{\min}, p_{\max})} \frac{1}{2} \frac{\lambda(p) |\lambda''(p)|}{\lambda'(p)^2} < 1. \quad (\text{compare log-concave: } \frac{\lambda(p) \lambda''(p)}{\lambda'(p)^2} < 1).$$

The seller *assumes* that demand is linear:

$$\lambda(p) = \alpha - \beta p, \quad \forall p \in [p_{\min}, p_{\max}],$$

and some $(\alpha, \beta) \in \Theta := [\alpha_{\min}, \alpha_{\max}] \times [\beta_{\min}, \beta_{\max}] \subset \mathbb{R}_{++}^2$. Assume

$$p_{\min} < \inf_{(a,b) \in \Theta} \frac{a}{2b} < \sup_{(a,b) \in \Theta} \frac{a}{2b} < p_{\max}.$$

A cycle-based policy $\pi = \pi(\hat{p}_1, \{n_i, \delta_i : i \in \mathbb{N}\})$: input: $\hat{p}_1 \in [p_{\min}, p_{\max}]$, sequences $(n_i)_{i \in \mathbb{N}}$ in $\mathbb{N}$, sequence $(\delta_i)_{i \in \mathbb{N}}$ in $[0, \infty)$ such that

$$\frac{a}{2b} + \delta_i \leq p_{\max}, \quad \forall (a, b) \in \Theta.$$

Set $t_i := \sum_{j=1}^{i-1} 2n_j$ for all $i \in \mathbb{N}$. For all $i \in \mathbb{N}$:

- (i) pricing:

$$p_t := \hat{p}_i \quad \text{for } t = t_i + 1, \dots, t_i + n_i \quad p_t := \hat{p}_i + \delta_i \quad \text{for } t = t_i + n_i + 1, \dots, t_i + 2n_i$$

- (ii) estimation:

$$(\hat{\alpha}_i, \hat{\beta}_i) := \arg \min_{(\alpha, \beta) \in \mathbb{R}^2} \sum_{t=t_i+1}^{t_i+2n_i} (\alpha - \beta p_t - d_t)^2 \quad (\hat{\alpha}_i, \hat{\beta}_i) := \mathbb{P}_{\Theta} \left\{ (\hat{\alpha}_i, \hat{\beta}_i) \right\} \quad \hat{p}_{i+1} := \frac{\hat{\alpha}_i}{2\hat{\beta}_i}$$

If true demand is linear, $\theta = (\alpha, \beta)$, $\hat{\theta}_i = (\hat{\alpha}_i, \hat{\beta}_i)$,

$$\mathbb{E}[|\hat{p}_{i+1} - p^*|^2] = O\left(\mathbb{E}[\|\hat{\theta}_i - \theta\|^2]\right) = O\left(\frac{\log(2n_i)}{2n_i \cdot (\frac{1}{2}\delta_i)^2}\right).$$

If $\delta_i \rightarrow 0$ and $\frac{\log(n_i)}{n_i \cdot \delta_i^2} \rightarrow 0$ as $i \rightarrow \infty$, then by *Chebyshev's inequality*,

$$\lim_{t \rightarrow \infty} \mathbb{P}(|p_t - p^*| > \epsilon) \rightarrow 0 \quad \text{for all } \epsilon > 0, \quad p_t \xrightarrow{P} p^*.$$

A (Surprising) Result

THEOREM 5.10.

Suppose that λ is not necessarily linear, but satisfies the given assumptions. If $\delta_i \rightarrow 0$ and

$$\frac{\delta_i \sqrt{n_i}}{\log(n_i)} \rightarrow \infty$$

as $i \rightarrow \infty$, then $p_t \xrightarrow{\mathcal{P}} p^*$.

Remark. Policy designed for linear demand functions, still ensures consistency for (class of) non-linear demand functions.

Without Noise

$$\hat{\beta}_i \approx -\frac{\lambda(\hat{p}_i + \delta_i) - \lambda(\hat{p}_i)}{\delta_i} \quad (\text{minus slope}), \quad \hat{\alpha}_i \approx \hat{\beta}_i \hat{p}_i + \lambda(\hat{p}_i) \quad (\text{intercept}).$$

Estimated optimal price (ignoring projection):

$$\hat{p}_{i+1} = \frac{\hat{\alpha}_i}{2\hat{\beta}_i} \approx \frac{1}{2} \left(\hat{p}_i + \frac{\lambda(\hat{p}_i)}{-\frac{\lambda(\hat{p}_i + \delta_i) - \lambda(\hat{p}_i)}{\delta_i}} \right) \approx \frac{1}{2} \left(\hat{p}_i - \frac{\lambda(\hat{p}_i)}{\frac{\lambda(\hat{p}_i + \delta_i) - \lambda(\hat{p}_i)}{\delta_i}} \right).$$

Suppose $\hat{p}_i \rightarrow p_\infty$ if $i \rightarrow \infty$. Then

$$p_\infty = \frac{1}{2} \left(p_\infty - \frac{\lambda(p_\infty)}{\lambda'(p_\infty)} \right) \text{ hence } p_\infty \lambda'(p_\infty) + \lambda(p_\infty) = 0.$$

But, $r'(p) = \frac{d}{dp}[p\lambda(p)] = p\lambda'(p) + \lambda(p)$ and thus

$$r'(p_\infty) = 0 \text{ which implies } p_\infty = p^*.$$

If there is a limit price, it must be the optimal price. Existence of p_∞ : for some $\gamma \in (0, 1)$,

$$|\hat{p}_{i+1} - p^*| \leq \gamma |\hat{p}_i - p^*| + O\left(\delta_i + \frac{\sqrt{\log n_i}}{\delta_i \sqrt{n_i}}\right) \text{ w.h.p.}$$

With Noise

For this calculation, write

$$Q_1 := -\frac{\lambda(\hat{p}_i + \delta_i) - \lambda(\hat{p}_i) + \frac{1}{n_i} \sum_{t=t_i+1}^{t_i+2n_i} \epsilon_t - \frac{1}{n_i} \sum_{t=t_i+1}^{t_i+n_i} \epsilon_t}{\delta_i}$$

Then

$$\begin{aligned} \hat{\beta}_i &= Q_1 = -\lambda'(\hat{p}_i) + \underbrace{\left(\lambda'(\hat{p}_i) - \frac{\lambda(\hat{p}_i + \delta_i) - \lambda(\hat{p}_i)}{\delta_i} \right)}_{O(\delta_i)} + \underbrace{\frac{1}{2\delta_i n_i} \sum_{t=t_i+1}^{t_i+2n_i} \tilde{\epsilon}_t}_{\xrightarrow{P} 0 \text{ if } \delta_i \sqrt{n_i} / \log(n_i) \rightarrow \infty} \quad (\tilde{\epsilon}_t := \pm \epsilon_t) \end{aligned}$$

$$\hat{\alpha}_i - \hat{\beta}_i \left(\hat{p}_i + \frac{1}{2} \delta_i \right) = \frac{\lambda(\hat{p}_i) + \lambda(\hat{p}_i + \delta_i)}{2} + \frac{1}{2n_i} \sum_{t=t_i+1}^{t_i+2n_i} \epsilon_t$$

$$\hat{\alpha}_i \xrightarrow{P} \alpha \text{ if } \hat{\beta}_i \xrightarrow{P} \beta \text{ and } n_i \rightarrow \infty.$$

THEOREM 5.11.

If $\delta_i = n_i^{-\frac{1}{4}}$, then

$$\Delta^\pi(T) = O\left(\sqrt{T}(\log T)^2\right).$$

Proof.

□

Remark. Not only consistency is preserved, but also regret (near) optimality. Of course, there are limits on the extent of robustness to modeling errors.

Exercises

EXERCISE 5.1 (Implementation of constrained ILS). Consider the canonical dynamic pricing and learning problem with linear demand function, as discussed in Keskin and Zeevi (2014), Section 1-3.3.

- (i) Implement a version of constrained iterated least squares with threshold parameter κ (Keskin and Zeevi, 2014, section 3.3).
- (ii) Show (in a picture) that $\kappa = 0$ leads to linear regret.
- (iii) For different values of T , show (in a picture) how the regret depends on κ and explain why.
- (iv) Modify your algorithm such that the ‘optimal’ κ is somehow adaptively learned over time.

SOLUTION. (i) Here we still consider the single-product setting as

$$d_t = \theta_1 + \theta_2 p_t + \epsilon_t,$$

where $p_t \in [p_{\min}, p_{\max}]$, $\theta = (\theta_1, \theta_2) \in \Theta = (0, \infty) \times (-\infty, 0)$, $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$. The constrained ILS algorithm is implemented with a parameter κ to adjust the greedy ILS price whenever its deviation from the historical average price is not large enough. Given that

$$\hat{\theta}(t) = \arg \min_{\vartheta \in \Theta} \sum_{i=1}^t (d_i - \vartheta_1 - \vartheta_2 p_i)^2, \quad \varphi(\theta) = \arg \max_{p \in [p_{\min}, p_{\max}]} p(\theta_1(t) + \theta_2(t)p),$$

let δ_s denote the difference between the greedy ILS price in period s and the average price in the first $s - 1$ periods, that is $\delta_s = \varphi(\hat{\theta}(s)) - \bar{p}_{s-1}$. In each period t , the CILS policy with threshold parameter κ charges the price

$$p_t = \begin{cases} \bar{p}_{t-1} + \text{sgn}(\delta_t) \kappa t^{-\frac{1}{4}} & \text{if } |\delta_t| < \kappa t^{-\frac{1}{4}} \\ \varphi(\hat{\theta}(t)) & \text{otherwise} \end{cases}.$$

Set $\theta = (1.1, -0.5)$, $\Theta = [1, 5] \times [-2, -0.1]$, $\sigma = 0.1$, $T = 100000$, and $[p_{\min}, p_{\max}] = [0.75, 2]$, then we implement the CILS algorithm with parameter $\kappa = 0.1$ and initial prices $p_1 = 1$, $p_2 = 1.5$. The relevant code is shown below:

```
import numpy as np
from scipy.optimize import minimize
from scipy.optimize import curve_fit

def generate_demand(theta, price, sigma):
    return theta[0] + theta[1] * price + np.random.normal(0, sigma, 1)

def sse(theta, demands, prices):
    return np.sum((demands - theta[0] - theta[1] * prices) ** 2)

def distance_squared(vartheta, theta_hat):
    return np.sum((vartheta - theta_hat) ** 2)

def cils_algorithm(kappa, theta, p_min, p_max, sigma, T, theta_1_min,
    ↪ theta_1_max, theta_2_min, theta_2_max):
    prices = np.zeros(T)
```

```

demands = np.zeros(T)

prices[0] = 1
demands[0] = generate_demand(theta, prices[0], sigma)

prices[1] = 1.5
demands[1] = generate_demand(theta, prices[1], sigma)

initial_params = np.array([0, 0])

bounds = [(theta_1_min, theta_1_max), (theta_2_min, theta_2_max)]

p_bar = np.mean(prices[:2])

for t in range(2, T):
    res_1 = minimize(sse, initial_params, args=(demands[:t],
        ↪ prices[:t]))
    theta_hat = res_1.x

    res_2 = minimize(distance_squared, theta_hat, args=(theta_hat,),
        ↪ bounds=bounds)
    vartheta = res_2.x

    phi_0 = -vartheta[0] / (2 * vartheta[1])
    phi = phi_0 if p_min <= phi_0 <= p_max else p_min if abs(p_min -
        ↪ phi_0) < abs(p_max - phi_0) else p_max

    delta = phi - p_bar

    if abs(delta) < kappa * t**(-1/4):
        prices[t] = p_bar + np.sign(delta) * kappa * t**(-1/4)
    else:
        prices[t] = phi

    demands[t] = generate_demand(theta, prices[t], sigma)

    p_bar += (prices[t] - p_bar) / (t + 1)

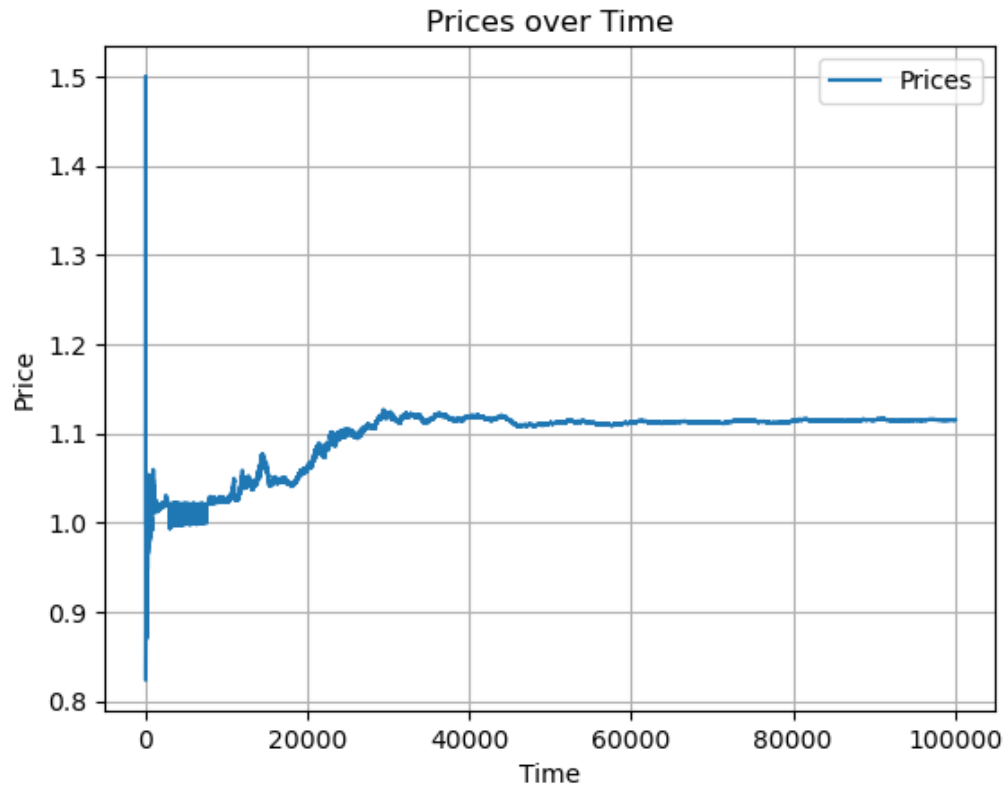
return prices, demands, vartheta

def calculate_regret(prices, theta):
    regrets = []

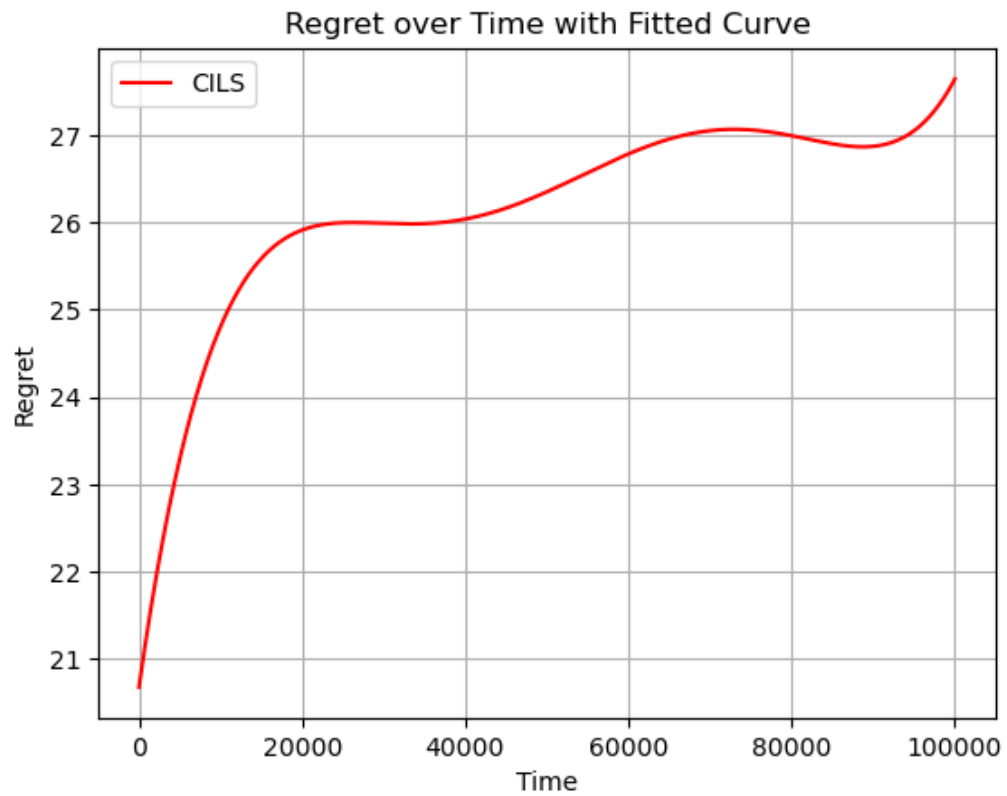
    for t in range(len(prices)):
        phi = -theta[0] / (2 * theta[1])
        revenue = prices[t] * (theta[0] + theta[1] * prices[t])
        regret = phi * (theta[0] + theta[1] * phi) - revenue
        regrets.append(regret)
    return np.cumsum(regrets)

```

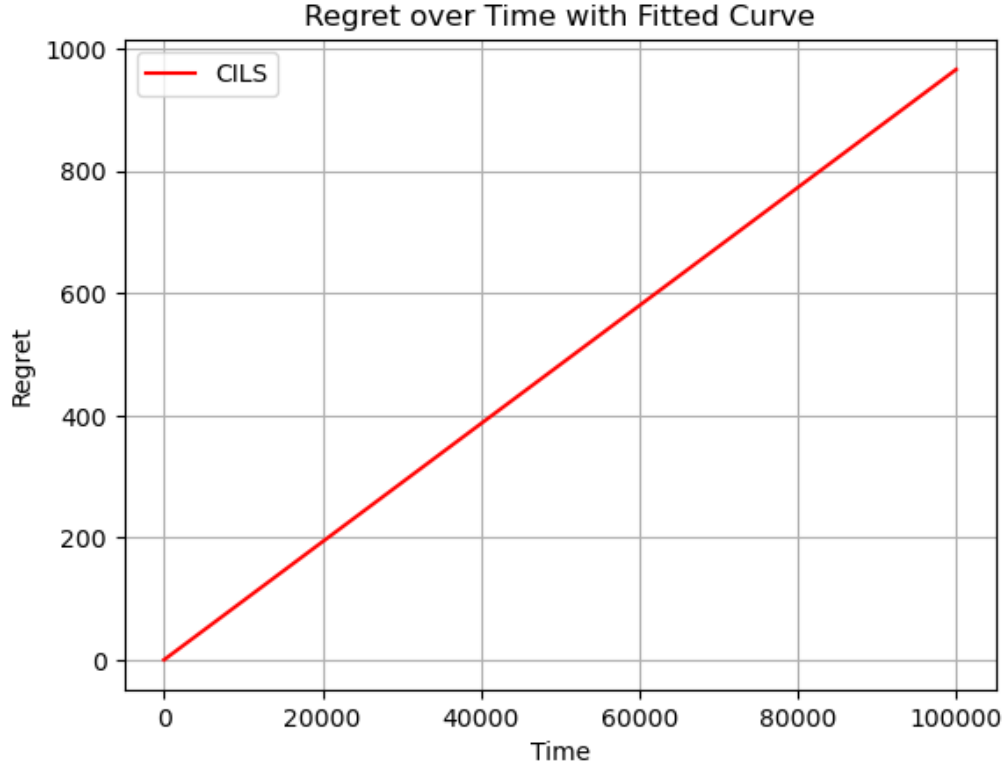
The prices over time is given as



And the expected regret is

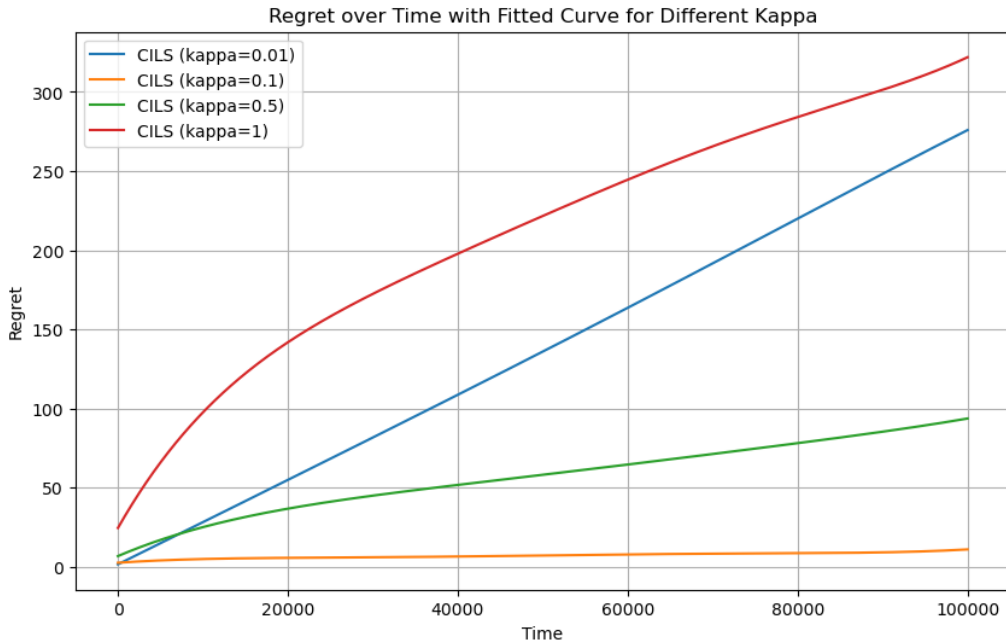


(ii) Let $\kappa = 0$, and the expected regret is given as



Thus, we verify that $\kappa = 0$ would lead to linear regret.

- (iii) For different values of $T \in [0, 100000]$, we implement the CILS algorithms with parameters $\kappa \in \{0.01, 0.1, 0.5, 1\}$ respectively. And the results are given as follows:



From the graph, we can find that when $T > 10000$, $\kappa = 0.1$ is best and has the lowest expected regret. Based on this, if κ becomes too large or too small, our algorithms perform worse in both cases, which means theoretically we can always find an 'optimal' value of κ to minimise the expected regret.

The reasons might include: If κ is too small, the constrained condition is more strict so that our CILS algorithm becomes closer to the greedy ILS algorithm, which means that our algorithm may not converge to the optimal price value and lead to a large regret. And, if κ is too large, the algorithm may oscillate or diverge, leading to higher regret. So we can consider κ as a parameter which controls the trade-off between exploration and exploitation in the algorithm, and too large or too small κ would cause an imbalance between exploration and exploitation and a relatively large regret.

- (iv) By Lemma 3 in Keskin and Zeevi (2014), we know there exist finite positive constants ρ and k such that, under any pricing policy π ,

$$\mathbb{P} \{ \|\hat{\theta}_t - \theta\| \geq \delta, J_t \geq m \} \leq kt \exp \left(-\rho \left(\delta \wedge \delta^2 \right) m \right),$$

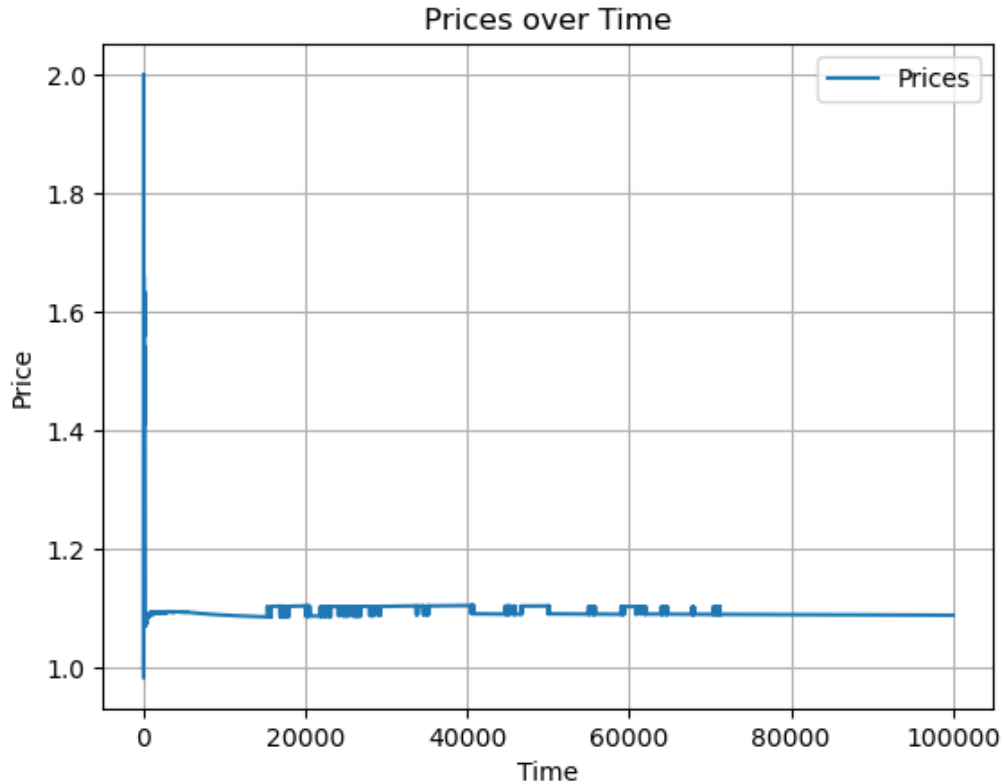
for all $\delta, m > 0$ and all $t \geq 2$. Then we can define

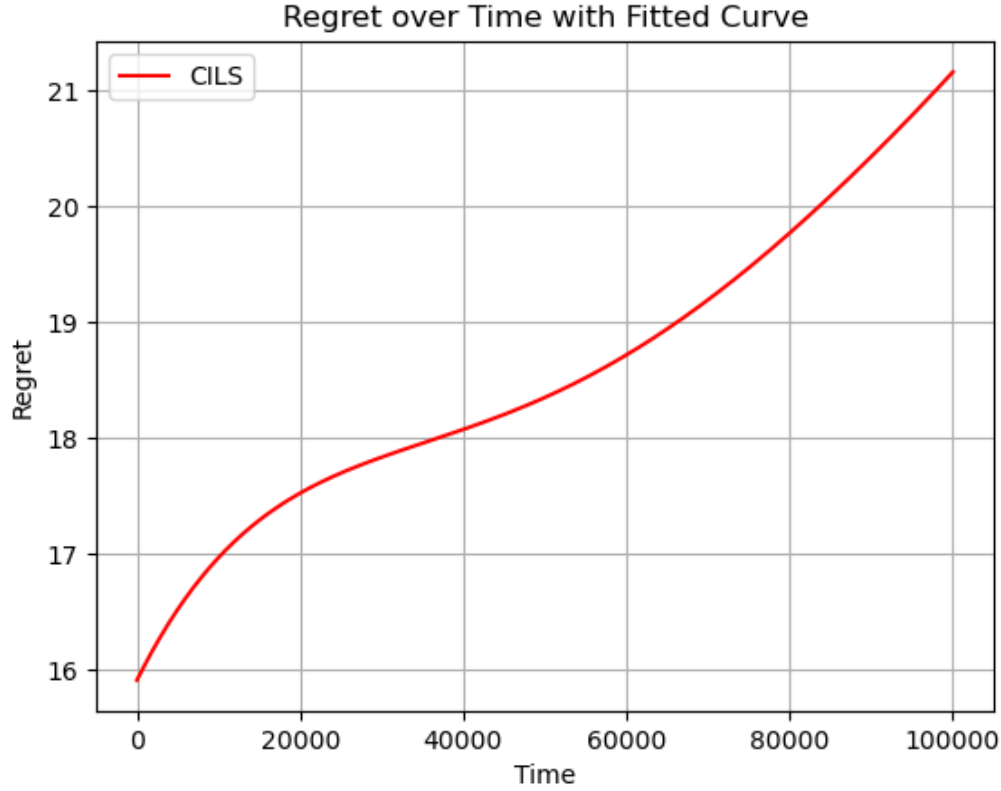
$$\tilde{K}_0 = (1 + 5k) \max_{j=1,2} \left\{ \max_{\theta} \left\{ \left(\frac{\partial \varphi(\theta)}{\partial \theta_j} \right)^2 \right\} \right\},$$

and give the optimal κ^* as $\kappa^* = \left(\frac{8\tilde{K}_0}{\rho} \right)^{\frac{1}{4}}$. And, to adaptively learn the ‘optimal’ κ over time, we let the adapted κ be given as

$$\kappa_t = \frac{(p_t - \bar{p}_{t-1})^2}{t^{-\frac{1}{2}}}.$$

Repeat the same experiment, we obtain the results as follows:





From the above graphs, we can find that the adaptive CILS algorithm converges to the optimal value of price faster, which leads to lower expected regrets.

□

EXERCISE 5.2 (Explore-Then-Exploit for Dynamic Pricing and Learning). Consider the dynamic pricing and learning model from the slides with linear demand $d_t = \theta_1 + \theta_2 p_t + \epsilon_t$, where ϵ_t are *i.i.d.* $N(0, 1)$, $(\theta_1, \theta_2) \in \Theta$, and Θ is a compact rectangle in $(0, \infty) \times (-\infty, 0)$. We assume that $\frac{\theta_1}{-2\theta_2} \in (\underline{p}, \bar{p})$ for all $(\theta_1, \theta_2) \in \Theta$. In addition, we assume that the time horizon $T \in \mathbb{N}$ is known in advance by the decision maker.

- (i) Design an *explore-then-commit* type policy.
- (ii) Prove an (instance-dependent) $O(\sqrt{T})$ regret bound.

SOLUTION. (i) We have to break the time horizon T into two phases: the exploration phase and the exploitation phase.

Exploration Phase. Let the first $\tau = \sqrt{T}$ periods for exploration. Choose selling prices p_t that are well-spread over the interval $[p_{\min}, p_{\max}]$ (for instance, we can select prices randomly from a uniform distribution over $[p_{\min}, p_{\max}]$). Collect data (p_t, d_t) for $t = 1, 2, \dots, \tau$.

Exploitation Phase. At the end of the exploration phase, use the collected data to estimate the parameters θ_1 and θ_2 . This can be done using ordinary least squares:

$$\hat{\theta}(t) := \mathcal{P}_{\Theta} \left\{ \arg \min_{\vartheta \in \mathbb{R}^2} \sum_{i=1}^t (d_i - \vartheta_1 - \vartheta_2 p_i)^2 \right\},$$

where $\mathcal{P}_\Theta$ is the projection operator on Θ . Using the estimated parameters $\hat{\theta}_1$ and $\hat{\theta}_2$, compute the optimal price

$$p^* = -\frac{\hat{\theta}_1}{2\hat{\theta}_2}$$

that maximises the expected revenue. For the remaining $T - \tau$ periods, set the selling price to p^* .

- (ii) We calculate the regret in two phases: exploration and exploitation, and then put them together.

Regret in the Exploration Phase. The regret for period t is

$$R(t) = p^*(\theta_1 + \theta_2 p^*) - p_t(\theta_1 + \theta_2 p_t),$$

where $p^* = -\frac{\theta_1}{2\theta_2}$ is the optimal price. Since $\tau = \sqrt{T}$, the total regret is

$$\sum_{t=1}^{\tau} R(t) \leq \sqrt{T} \cdot \max_{p \in [p_{\min}, p_{\max}]} (p^*(\theta_1 + \theta_2 p^*) - p_t(\theta_1 + \theta_2 p_t)),$$

which is of order $O(\sqrt{T})$.

Regret in the Exploitation Phase. Using ordinary least squares during the exploration phase, we can estimate the parameters θ_1 and θ_2 with the standard error decreasing as the number of samples increases. Given that $\tau = \sqrt{T}$, the estimation error is

$$\text{Var}(\hat{\theta}_i) = O\left(\frac{1}{\tau}\right) = O(\sqrt{T}).$$

Using the delta method, the variance of the estimated optimal price can be approximated as $O(\frac{1}{\tau}) = O(\sqrt{T})$. Then the expected regret during the exploitation phase is bounded by

$$(T - \tau)O(\sqrt{T}) = (T - \sqrt{T})O(\sqrt{T}) = O(\sqrt{T}).$$

Combining the regrets from both phases completes the proof for the $O(\sqrt{T})$ regret bound of the explore-then-commit policy. $\square$

EXERCISE 5.3. Suppose that a DM is trying to find an ‘input value’ x^* that steers a corresponding ‘output value’ to $c \in \mathbb{R}$ in expectation, using a stochastic approximation scheme. More formally, suppose that $X_1 \in \mathbb{R}$ is non-random and

$$X_{t+1} = X_t - a_t(Y_t - c), \quad \text{for all } t \in \mathbb{N}.$$

Here $(a_t)_{t \in \mathbb{N}}$ is a non-random sequence in $[0, \infty)$, to be determined by the DM, and Y_t is an observation that satisfies

$$Y_t = \alpha X_t + \beta + \epsilon_t,$$

where $(\epsilon_t)_{t \in \mathbb{N}}$ are *i.i.d.* Gaussian random variables with expectation zero and variance $\sigma^2 \in (0, \infty)$, and $\alpha > 0, \beta \in \mathbb{R}$ are unknown parameters that lie in a (known) set $[\alpha_{\min}, \alpha_{\max}] \times [\beta_{\min}, \beta_{\max}] \in (0, \infty) \times \mathbb{R}$. Determine a sequence $(a_t)_{t \in \mathbb{N}}$ and show that $\mathbb{E}[(X_t - x^*)^2] = O(t^{-1})$.

SOLUTION. Let $a_t = \frac{1}{t}$ for all $t \in \mathbb{N}$, then we can calculate:

For this calculation, write

Q_1

$$:= \mathbb{E}[\mathbb{E}[(X_{t-1} - x^*)^2 - 2(X_{t-1} - x^*)(a_{t-1}(Y_{t-1} - c)) + a_{t-1}^2(Y_{t-1} - c)^2 \mid X_{t-1}]]$$

Then

$$\mathbb{E}[(X_t - x^*)^2]$$

$$= \mathbb{E}[\mathbb{E}[(X_t - x^*)^2 \mid X_{t-1}]] = \mathbb{E}\left[\mathbb{E}\left[(X_{t-1} - a_{t-1}(Y_{t-1} - c) - x^*)^2 \mid X_{t-1}\right]\right]$$

$$= Q_1 = \mathbb{E}[(X_{t-1} - x^*)^2] - 2a_{t-1}\mathbb{E}[(X_{t-1} - x^*)(\alpha X_{t-1} + \beta - c)] + a_{t-1}^2\mathbb{E}[\mathbb{E}[(Y_{t-1} - c)^2 \mid X_{t-1}]],$$

and since we have $Y_{t-1} \mid X_{t-1} \sim \mathcal{N}(\alpha X_{t-1} + \beta, \sigma^2)$,

$$\mathbb{E}[\mathbb{E}[(Y_{t-1} - c)^2 \mid X_{t-1}]] = \mathbb{E}[(\alpha X_{t-1} + \beta - c)^2 + \sigma^2] = \mathbb{E}[(\alpha X_{t-1} + \beta - c)^2] + \sigma^2.$$

Thus, we obtain

$$\begin{aligned} & \mathbb{E}[(X_t - x^*)^2] - \mathbb{E}[(X_{t-1} - x^*)^2] \\ &= -2a_{t-1}\mathbb{E}[(X_{t-1} - x^*)(\alpha X_{t-1} + \beta - c)] + a_{t-1}^2\left(\mathbb{E}[(\alpha X_{t-1} + \beta - c)^2] + \sigma^2\right). \end{aligned}$$

By telescoping, we have

$$\mathbb{E}[(X_t - x^*)^2] = (X_1 - x^*)^2 - 2 \sum_{s=1}^{t-1} a_s \mathbb{E}[(X_s - x^*)(\alpha X_s + \beta - c)] + \sum_{s=1}^{t-1} a_s^2 \left(\mathbb{E}[(\alpha X_s + \beta - c)^2] + \sigma^2\right).$$

Now we denote

$$b_t = \mathbb{E}[(X_t - x^*)^2], \quad d_t = \mathbb{E}[(X_t - x^*)(\alpha X_t + \beta - c)], \quad e_t = \mathbb{E}[(\alpha X_t + \beta - c)^2] + \sigma^2$$

such that

$$b_t = b_1 - 2 \sum_{s=1}^{t-1} a_s d_s + \sum_{s=1}^{t-1} a_s^2 e_s.$$

Since $\alpha > 0$ and $\alpha x^* + \beta = c$, we know that $(X_s - x^*)(\alpha X_s + \beta - c) \geq 0$ so that $d_t \geq 0$. Also, e_t is bounded and $\sum_{s=1}^{t-1} a_s^2 e_s$ also converges if we substitute $a_t = \frac{1}{t}$. So we have

$$\sum_{s=1}^{t-1} a_s d_s \leq \frac{1}{2} \left(b_1 + \sum_{s=1}^{t-1} a_s^2 e_s \right) < \infty.$$

Thus, the series $\sum_{s=1}^{t-1} a_s d_s$ must converge, which means the limit $\lim_{t \rightarrow \infty} b_t$ exists. Then by analysis we can find that a sequence k_t such that

$$d_t \geq k_t b_t, \quad \sum_{t=1}^{\infty} a_t k_t b_t = \infty.$$

Since

$$\sum_{t=1}^{\infty} a_t k_t b_t \leq \sum_{t=1}^{\infty} a_t d_t < \infty,$$

we can infer that $\lim_{t \rightarrow \infty} b_t = 0$. Then, applying *Theorem 1* in M. Broadie, D. Cicek and A. Zeevi (2011), there exists a constant $C > 0$ such that

$$\mathbb{E}[(X_t - x^*)^2] \leq C a_t = C t^{-1}.$$

Hence, we conclude that $\mathbb{E}[(X_t - x^*)^2] = O(t^{-1})$. $\square$

Assortment Optimisation and Learning

Discrete Choice Models. Probabilistic models that describe how agents choose from alternatives.

Consider a number of alternatives $i = 1, \dots, n$, from which an agent selects exactly one. To each item, the agent associates a *utility* $U_i \in \mathbb{R}$. Her choice maximises her utility (with some tie-breaking rule). Often $U_i = \theta^\top x_i$, where x_i models *features* of alternative i . Deterministic model.

But: under (seemingly) equal circumstances, people often make different choices. More realistic: *random utility model*:

$$U_i = \underbrace{u_i}_{\text{non-random, 'explainable'}} + \underbrace{\epsilon_i}_{\text{random disturbance, unobservable}},$$

and

$$\mathbb{P}(\text{agent selects item } j) = \mathbb{P}\left(U_j = \arg \max_i U_i\right).$$

Remark. To avoid tie-breaks, convenient to assume ϵ_i are atom-less. $\epsilon_1, \dots, \epsilon_n$ random, but can be correlated. Even for i.i.d. ϵ_i , $\arg \max_i U_i$ rarely has closed-form expression. Exception 1: i.i.d. Gumbel noise with common variance. (If i.i.d.: insensitive to mean or variance of $\{\epsilon_i\}$). Exception 2: negative exponential.

$$\text{pdf } f(x) = \exp(-(x + \exp(-x))), \quad \text{cdf } F(x) = \exp(-\exp(-x))$$

Expectation $\gamma \approx 0.5772$, Euler-Mascheroni constant.

If $(\epsilon_i)_{1 \leq i \leq n}$ i.i.d. and standard Gumbel, then

$$\begin{aligned} \mathbb{P}(U_i \geq U_j, \forall j) &= \mathbb{P}(\epsilon_i + u_i - u_j \geq \epsilon_j, \forall j) = \int_{-\infty}^{\infty} f(x) \prod_{j \neq i} F(x + u_i - u_j) dx \\ &= \int_{-\infty}^{\infty} e^{-(x + e^{-x})} \prod_{j \neq i} e^{-e^{-(x + u_i - u_j)}} dx = \int_{-\infty}^{\infty} e^{-x} e^{-\sum_{j=1}^n e^{-(x + u_i - u_j)}} dx \\ &= \int_0^{\infty} e^{-\sum_{j=1}^n s} e^{-(u_i - u_j)} ds \quad \text{where } s = e^{-x} \\ &= \frac{1}{\sum_{j=1}^n e^{-(u_i - u_j)}} = \frac{e^{u_i}}{\sum_{j=1}^n e^{u_j}}. \end{aligned}$$

Closed-form expression, probability proportional to e^{u_i} . *Multinomial logit model* (MNL). If only a subset $S \subset \{1, \dots, n\}$ of alternatives is available,

$$\mathbb{P}(\text{choice } i \text{ from } S) = \frac{e^{u_i}}{\sum_{j \in S} e^{u_j}}, \quad i \in S.$$

Remark. Merely n parameters. Without parametric model, too many parameters: N products

$$\sum_{k=1}^N \binom{N}{k} k = 2^{N-1} \cdot N \text{ parameters.}$$

5120 for $N = 10$, 10,485,760 for $N = 20$. Estimation with MLE is tractable (concave log-likelihood).

EXAMPLE 6.1 (Limited Substitution Behaviour). Suppose {car, red bus, blue bus}, each with probability 1/3 (all u_i equal). Now suppose the red bus is not available. We want

$$\mathbb{P}(\text{car}) = 1/3, \quad \mathbb{P}(\text{blue bus}) = 2/3.$$

Under the MNL model:

$$\mathbb{P}(\text{car}) = 1/2, \quad \mathbb{P}(\text{blue bus}) = 1/2.$$

MNL model has ‘Independence of irrelevant alternatives’ property.

Consider

$$U_i = u_i - \epsilon_i, \quad i = 1, \dots, n, \quad \epsilon_i \text{ i.i.d. Exp}(1)$$

W.l.o.g. $u_1 \leq \dots \leq u_n$.

$$\begin{aligned} \mathbb{P}(U_i \geq U_j \forall j) &= \mathbb{P}(u_i - \epsilon_i \geq u_j - \epsilon_j \forall j) = \mathbb{P}(\epsilon_j \geq u_j - u_i - \epsilon_i \forall j) \\ &= \int_0^\infty e^{-x} \prod_{j \neq i} \bar{F}(u_j - u_i - x) dx \\ &= \int_0^\infty e^{-x} \prod_{j \neq i} e^{-(u_j - u_i - x)^+} dx = \frac{\exp\left(-\sum_{j=i}^n (u_j - u_i)\right)}{n - i + 1} - \sum_{k=1}^{i-1} \frac{\exp\left(-\sum_{j=k}^n (u_j - u_k)\right)}{(n - k)(n - k + 1)}. \end{aligned}$$

This model (Daganzo 1979, Alptekinoglu and Semple 2016) is also being analysed in the literature: more complicated than MNL, sometimes better performance in practice (Aouad, Feldman, Segev 2022; Berbeglia, Garassino, Vulcano G 2021).

Remark. (i) *More advanced choice models: nested logit, mixed logit, and many more (how rich should a ‘good’ model be?)*

(ii) *Many applications: transport, social sciences, business optimisation.*

(iii) *‘How does a particular property affect what people choose?’*

SECTION 6.1

| Assortment Optimisation

A seller has $N \in \mathbb{N}$ products available, denoted as $[N] := \{1, \dots, N\}$ with *valuations* $v_1, \dots, v_N > 0$, and *marginal profits* $0 < w_1 \leq \dots \leq w_N$. The seller offers an assortment $S \subset [N]$ to a customer, who chooses according to a MNL model:

$$\mathbb{P}(Y = i \mid S) = \frac{v_i}{1 + \sum_{j \in S} v_j}, \quad i \in S.$$

Expected revenue function of the seller:

$$r(S) := \sum_{i \in S} w_i \cdot \frac{v_i}{1 + \sum_{j \in S} v_j}, \quad S \subset [N].$$

Remark. There is also the outside option 0, with $v_0 = 1$ and $w_0 = 0$. And

$$\mathbb{P}(Y = 0 \mid S) = \frac{1}{1 + \sum_{j \in S} v_j}.$$

Unconstrained Assortment Optimisation Problem

$$\begin{aligned} & \max_{S \subset [N]} r(S) \\ &= \max \left\{ \rho \geq 0 : \exists S \subset [N] : \sum_{i \in S} \frac{w_i v_i}{1 + \sum_{j \in S} v_j} \geq \rho \right\} \\ &= \max \left\{ \rho \geq 0 : \exists S \subset [N] : \sum_{i \in S} (w_i - \rho) v_i \geq \rho \right\} = \max \left\{ \rho \geq 0 : \max_{S \subset [N]} \sum_{i \in S} (w_i - \rho) v_i \geq \rho \right\}. \end{aligned}$$

The inner maximisation problem is solved by $S_\rho := \{i \in [N] : w_i \geq \rho\}$ and $r(S_\rho^*, v) = \rho^*$ (from first equation).

The optimal assortment is of the form ‘offer the k most expensive products’. In a learning setting where $v_1, \dots, v_n$ are unknown, simple translation to a MAB with n arms yields $\Omega(\sqrt{NT})$ regret.

Suppose $v = (v_1, \dots, v_N)$ is *unknown*. Assumed to lie in

$$\mathcal{V} = \left\{ v : \frac{1}{1 + \frac{1}{N} \sum_{i=1}^N v_i} \geq p_0 \right\}$$

for some known $p_0 \in (0, 1)$. Assume *w.l.o.g.* $0 < w_1 < \dots < w_N = 1$. Policy: select $S_t \subset \{1, \dots, N\}$, for each $t \in \mathbb{N}$, based on available data.

$$R_v^\pi(T) = T \cdot \max_{S \subset \{1, \dots, N\}} r(S, v) - \mathbb{E} \left[\sum_{t=1}^T r(S_t, v) \right], \quad R^\pi(T) = \sup_{v \in \mathcal{V}} \Delta^\pi(T).$$

regret; worst-case regret.

Optimal revenue ρ^* satisfies

$$\sum_{i:w_i \geq \rho} (w_i - \rho)v_i = \rho.$$

Find root of function $r(\{i : w_i \geq \rho\}) - \rho$.

EXAMPLE 6.2 (Simple Policy Based on Stochastic Approximation). Let Y_t denote the purchased product, and $w_0 := 0$. Let $\rho_1 \geq 0$ and $\alpha > 0, \beta \geq 0$. For each t , set

$$S_t = \{i \in \{1, \dots, N\} : w_i \geq \rho_t\}, \quad \rho_{t+1} = \rho_t + \frac{\alpha}{t + \beta} \cdot (w_{Y_t} - \rho_t).$$

THEOREM 6.3 (Regret for Stochastic Approximation).

If $\alpha \geq \frac{1}{p_0}$ and $\beta \geq \alpha - 1$ then there is a $C > 0$ such that, for all $T \in \mathbb{N}$, the regret $R(T) \leq C\sqrt{T}$.

Regret $\Delta^\pi(T) = O(\sqrt{T})$. Also, for any $0 < w_1 < \dots < w_N$, regret $\Delta^\pi(T) = \Omega\sqrt{T}$. Simple policy is asymptotically optimal. Combinatorial problem reduces to learning a single number.

Constrained Assortment Optimisation Problem: LP Formulation

Let $\mathcal{S}$ be a collection of subsets of $[N]$, then

$$\max_{S \in \mathcal{S}} r(S) = \max \left\{ \rho \geq 0 : \max_{S \in \mathcal{S}} \sum_{i \in S} (w_i - \rho)v_i \geq \rho \right\}.$$

The inner maximisation problem is solved by $S_\rho := \{i \in [N] : w_i \geq \rho\}$ provided this assortment is in $\mathcal{S}$. If not, $(w_i - \rho)v_i > (w_j - \rho)v_j \geq 0$ while $w_i < w_j$ is possible (unconstrained: simply offer both i and j). Constraints: optimal assortments not always ‘offer the k most expensive products’. Consider $\mathcal{S} := \{S \subset [N] : |S| \leq K\}$ for some K .

THEOREM 6.4.

The following two optimisation problems have equal optimal values:

$$z^* = \max \left\{ \frac{\sum_{i \in S} w_i v_i}{1 + \sum_{j \in S} v_j} : S \subset [N], |S| \leq K \right\} \quad (\text{OP})$$

$$z_{LP} = \max \left\{ \sum_{i=1}^n w_i p_i \mid \sum_{i=0}^n p_i = 1, \sum_{i=1}^n \frac{p_i}{v_i} \leq K p_0, 0 \leq p_i \leq v_i p_0, \forall i \right\} \quad (\text{LP})$$

Proof. For all solutions S to (OP),

$$p_0 = \frac{1}{1 + \sum_{j \in S} v_j} \quad \text{and} \quad p_i = \frac{v_i}{1 + \sum_{j \in S} v_j} \mathbf{1}_{\{i \in S\}}$$

gives a feasible solution to (LP) with same objective, hence $z^* \leq z_{LP}$.

Now let p^* be a *basic* optimal solution to (LP) (corner solution, at least $\dim(p^*)$ constraints active). LP has $(n+1)$ variables, 1 equality constraint $\Rightarrow$ at least n inequalities tight $\Rightarrow$ at least $n-1$ variables $p_i^* \in \{0, v_i p_0\}$. Suppose, say, $0 < p_1^* < v_1 p_0$.

$$K p_0^* = \sum_{i=1}^n \frac{p_i^*}{v_i} = \frac{p_1^*}{v_1} + \sum_{i=2}^n \frac{p_i^*}{v_i} \quad \text{Contradiction.}$$

Hence $p_j^* \in \{0, v_j p_0^*\}$ for all j ; then $S^* = \{j : p_j^* = v_j p_0^*\}$ is feasible for (OP).

$$\sum_{j=0}^n p_j^* = p_0^* + \sum_{j \in S^*} v_j p_0^* = 1 \implies p_0^* = \frac{1}{1 + \sum_{j \in S^*} v_j}$$

and $p_j^* = v_j^*/(1 + \sum_{j \in S^*} v_j) \implies \text{same objective} \implies z_{LP} \leq z^*$. □

SECTION 6.2

| Elegant Observation

Suppose we fix a feasible assortment $S_t = S$ for all $t \in \mathbb{N}$, let $i \in S$,

$$\begin{aligned} \mathbb{P}(X_t = 0 \mid X_t \in \{0, i\}) &= \frac{\mathbb{P}(X_t = 0)}{\mathbb{P}(X_t = 0) + \mathbb{P}(X_t = i)} \\ &= \frac{1}{1 + \sum_{j \in S} v_j} \left(\frac{1}{1 + \sum_{j \in S} v_j} + \frac{v_i}{1 + \sum_{j \in S} v_j} \right) = \frac{1}{1 + v_i}. \end{aligned}$$

Number of sales of product $i \in S$ until the first no-purchase is geometrically distributed with expectation v_i . Unbiased estimator of v_i , independent of other products in S . Having estimates of v_i , how are we going to set our assortments?

Principle of Optimism. Create *confidence bounds* for rewards of your decisions (assortments), ‘be optimistic’ and act according to upper confidence bound (UCB). If you were ‘too optimistic’, you will learn and decrease the UCB.

Let t_l be the time of the l -th no-purchase, $t_0 := 0$. Call $\mathcal{E}_l := \{t_{l-1} + 1, \dots, t_l\}$ the l -th cycle. Keep assortment *fixed* during cycles. Let $\bar{v}_{i,l}$ average # purchases product i when offered, in first l cycles

$$v_{i,l}^{\text{UCB}} := \bar{v}_{i,l} + 4\sqrt{\bar{v}_{i,l}(1 + \bar{v}_{i,l}) \frac{\log(1 + T_i(l))}{1 + T_i(l)}} \quad \text{if } T_i(l) \geq 1, \quad v_{i,l}^{\text{UCB}} := v_{\max} \quad \text{if } T_i(l) = 0,$$

where $T_i(l)$ counts how often i was offered in the first l cycles. In cycle l , offer $S(l) := S^*(v_{1,l-1}^{\text{UCB}}, \dots, v_{N,l-1}^{\text{UCB}})$.

THEOREM 6.5.

The worst-case regret is $O(\sqrt{NT} \cdot \sqrt{\log(NT)})$.

Proof.

LEMMA 6.6.

If $v_i \leq v_{i,l-1}^{\text{UCB}}$ then

$$r(S, v_{(l-1)}^{\text{UCB}}) - r(S, v) \leq \frac{\sum_{i \in S} v_{i,l-1}^{\text{UCB}} - v_i}{1 + \sum_{j \in S} v_j}$$

and

$$r(S^*(v), v) - r(S^*(v), w) \leq 0 \quad \text{if } v \leq w.$$

Suppose that $v \leq v_{(l-1)}^{\text{UCB}}$.

$$r(S^*, v) - r(S(l), v) = r(S^*, v) - r(S^*, v_{(l-1)}^{\text{UCB}}) + r(S^*, v_{(l-1)}^{\text{UCB}}) - r(S(l), v_{(l-1)}^{\text{UCB}}) \leq r(S(l), v_{(l-1)}^{\text{UCB}}) - r(S(l), v).$$

Hence, on this event, the regret in a cycle satisfies

$$(1 + \sum_{j \in S(l)} v_j)(r(S^*, v) - r(S(l), v)) \leq \sum_{j \in S(l)} v_{j,l-1}^{\text{UCB}} - v_j.$$

Proof.

□

□

Regret Lower Bound

Let $0 < w_1 \leq \dots \leq w_N$ arbitrary. Let

$$v_i^{(0)} := \frac{\gamma}{K(w_i - \gamma)}$$

for some $0 < \gamma < w_1$. Then

$$r(S, v^{(0)}) = \frac{\sum_{i \in S} w_i v_i^{(0)}}{1 + \sum_{i \in S} v_i^{(0)}} = \frac{\gamma \cdot \sum_{i \in S} \frac{w_i}{w_i - \gamma}}{K + \sum_{i \in S} \frac{\gamma}{w_i - \gamma}} = \frac{\gamma \cdot \sum_{i \in S} \frac{w_i}{w_i - \gamma}}{K - |S| + \sum_{i \in S} \frac{w_i}{w_i - \gamma}}.$$

Increasing in the sum in the numerator and in $|S|$. Optimal if $|S| = K$. But then $K - |S| = 0$ and $r(S, v^{(0)}) = \gamma$. Any assortment of size K is optimal. Idea: slightly perturb this base vector $v^{(0)}$:

$$v_i^{(S)} := \begin{cases} v_i^{(0)}(1 + \epsilon) & \text{if } i \in S \\ v_i^{(0)} & \text{if } i \notin S \end{cases} \quad S \text{ is optimal for } v^{(S)}.$$

Intuitive idea:

- (i) Nature chooses a random $v^{(S)}$.
- (ii) If ϵ is small: *difficult* to learn from data $\Rightarrow$ takes many samples. If ϵ is large: wrong assortment gives a *large loss*.
- (iii) ‘Difficult’ ϵ is small but not *too* small.
- (iv) KL-divergence inequalities lead to regret lower bound $\Omega(\sqrt{NT})$ assuming $K < N/2$; for details see *Chen and Wang 2018* or *Peeters and den Boer 2021b*.

Remark. (i) There are other policies, e.g. based on MLE or Thompson Sampling.

- (ii) Regret rate only gives ‘first order’ performance. Optimal ‘constant’ before the $\sqrt{NT}$?
- (iii) Many papers on MNL learning with features (of products and/or of customers). E.g. purchase probability

$$\frac{\exp(\theta^\top x_{i,t})}{1 + \sum_{j \in S} \exp(\theta^\top x_{j,t})}$$

where $\{x_{i,t} : t \in \mathbb{N}\}$ are i.i.d. adversarial, or something in between.

- (iv) Related: contextual multi-armed bandit. Ongoing research topic.
- (v) Typical issue: sparsity.
- (vi) Other choice models: estimation + optimisation is more involved. Model selection question again relevant.

SECTION 6.3

| Model Selection in Data-Driven Decision Problems

Models in Data-Driven Optimisation

$$\max_{x \in \mathcal{X}} \mathbb{E}[r(x, Y(x))] = \max_{x \in \mathcal{X}} \int_{\mathcal{Y}} r(x, y) dF_{Y(x)}(y),$$

where x decision variable, $\mathcal{X}$ admissible decisions. $Y(x)$ random variable, supported on $\mathcal{Y}$. $F_{Y(x)}$ unknown cdf of $Y(x)$. $r : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ reward function.

A model M describes the unknown cdf's $F_{Y(x)}$ for all x :

$$M = \{F_{x,\theta} \text{ cdf on } \mathcal{Y} \mid x \in \mathcal{X}, \theta \in \Theta\},$$

where Θ may be infinite dimensional. If $F_{x,\theta^*} \equiv F_{Y(x)}$ for some $\theta^* \in \Theta$, then M is correctly specified.

Data is a set $d \in (\mathcal{X} \times \mathcal{Y})^n$, for some $n \in \mathbb{N}$. An estimator maps data to parameters:

$$\tau : (\mathcal{X} \times \mathcal{Y})^n \rightarrow \Theta.$$

An optimisation algorithm maps parameters to decisions: $\chi : \Theta \rightarrow \mathcal{X}$. Typically, $\chi(\theta)$ maximises the reward as function of θ . Sometimes χ is a heuristic.

Model M + data $d \rightarrow$ estimate $\tau(d) \rightarrow$ decision $\chi(\tau(d))$.

Now we have $K + 1$ models

$$M^{(k)} = \{F_{x,\theta}^{(k)} \text{ cdf on } \mathcal{Y} \mid x \in \mathcal{X}, \theta \in \Theta^{(k)}\}, \quad k = 0, 1, \dots, K.$$

‘complex’ model $M^{(0)}$ with estimator $\tau^{(0)}$ and algorithm $\chi^{(0)}$, ‘simplified’ models $M^{(k)}$ with estimator $\tau^{(k)}$ and algorithm $\chi^{(k)}$, $k = 1, \dots, K$, and a data set d_0 . Which model should we use?

Assume $M^{(0)}$ is correctly specified, with true but unknown parameter θ^* . Write

$$r(x, \theta) = \int_{\mathcal{Y}} r(x, y) dF_{x,\theta}^{(0)}(y)$$

for the reward function under $M^{(0)}$. Model k suggests decision $x^{(k)} := \chi^{(k)}(\tau^{(k)}(d_0))$, for each k . Goal: estimate

$$\arg \max_{k \in \{0, 1, \dots, K\}} r(x^{(k)}, \theta^*).$$

Naive estimator

$$\arg \max_{k \in \{0, 1, \dots, K\}} r(x^{(k)}, \tau^{(0)}(d_0))$$

does not work, since typically $x^{(0)}$ maximises $r(\cdot, \tau^{(0)}(d_0))$.

Bootstrap. Draw new $\{y_i\}$, according to $\tau^{(0)}(d_0)$; re-estimate θ^* based on the resampled data set d_r ; replace θ^* by the re-estimated parameter $\tau^{(0)}(d_r)$.

Write

$$\theta_0 = \tau^{(0)}(x_1, y_1, \dots, x_n, y_n).$$

Replace each y_i by $y_i^r \sim F_{x_i, \theta_0}^{(0)}$. Let

$$\theta_r = \tau^{(0)}(x_1, y_1^r, \dots, x_n, y_n^r).$$

Select model

$$\arg \max_{k \in \{0,1,\dots,K\}} r(x^{(k)}, \theta_r)$$

(smallest k in case of tie).

Cross-validation. Divide the data into two equal parts. Estimate parameters based on one half and evaluate the reward based on the other half.

EXAMPLE 6.7 (Assortment Optimisation).

Exercises

EXERCISE 6.1. In assortment optimisation under the MNL model *without* capacity constraint, the optimal assortment is of the form ‘offer the k most expensive products’, for some k . If there is a capacity constraint (of the form ‘you cannot offer more than l products in an assortment’) this structure is lost. Explain why.

SOLUTION. In the unconstrained case, the optimal assortment involves offering the k most expensive products, because for the MNL model, the probability of choosing a product is based on its valuation relative to the sum of valuations of all offered products, i.e.,

$$\mathbb{P}(Y = i \mid S) = \frac{v_i}{1 + \sum_{j \in S} v_j}, \quad i \in S,$$

and we can write the expected revenue function of the seller as:

$$r(S) := \sum_{i \in S} \frac{w_i v_i}{1 + \sum_{j \in S} v_j}, \quad S \subset [N] := \{1, \dots, N\},$$

where $v_1, \dots, v_N > 0$ are the valuations and $0 < w_1 \leq \dots \leq w_N$ are the marginal profits. And

$$\max_{S \subset [N]} r(S) = \max \left\{ \rho \geq 0 : \max_{S \subset [N]} \sum_{i \in S} (w_i - \rho) v_i \geq \rho \right\},$$

where the inner maximisation problem is solved by $S_\rho := \{i \in [N] : w_i \geq \rho\}$ and $r(S_\rho^*, v) = \rho^*$. Therefore, products with higher marginal profits w_i contribute more to $r(S)$.

When a capacity constraint l is introduced, this structure is lost. Because adding a product to an assortment also increases the denominator $1 + \sum_{j \in S} v_j$, making the influence of every single product on the overall probability decrease. Adding more products does not always increase the revenue linearly or even beneficially. Instead, we have to find the best combination of products, given the limited slots, to balance choice probabilities and marginal profits effectively. $\square$

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